Module 4: Chapters

- Compound Angles
- Trigonometric Equations
- Fundamentals of Geometry
- Similarity of Triangles
- Heights and Distances

TRIGONOMETRIC RATIOS & IDENTITIES

1. INTRODUCTION TO TRIGONOMETRY :

The word 'trigonometry' is derived from the Greek words 'trigon' and 'metron' and it means 'measuring the sides of a triangle'. The subject was originally developed to solve geometric problems involving triangles. It was studied by sea captains for navigation, surveyor to map out the new lands, by engineers and others. Currently, trigonometry is used in many areas such as the science of seismology, designing electric circuits, describing the state of an atom, predicting the heights of tides in the ocean, analysing a musical tone and in many other areas.

(a) **Measurement of angles :** Commonly two systems of measurement of angles are used.

(i) **Sexagesimal or English System :** Here 1 right angle = 90° (degrees)

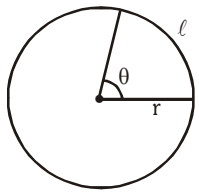
1° = 60' (minutes)

1' = 60" (seconds)

(ii) **Circular system :** Here an angle is measured in radians. One radian corresponds to the angle subtended by an arc of length 'r' at the centre of the circle of radius r. It is a constant quantity and does not depend upon the radius of the circle.

(b) Relation between the these systems : $\frac{D}{90} = \frac{R}{\pi/2}$

(c) If θ is the angle subtended at the centre of a circle of radius 'r', by an arc of length 'ℓ' then $\frac{\ell}{r} = \theta$.



Note that here ℓ, r are in the same units and θ is always in radians.

Illustration 1 : If the arcs of same length in two circles subtend angles of 60° and 75° at their centres. Find the ratio of their radii.

Solution : Let r_1 and r_2 be the radii of the given circles and let their arcs of same length 's' subtend angles of 60° and 75° at their centres.

Now, $60° = \left(60 \times \frac{\pi}{180}\right)^c = \left(\frac{\pi}{3}\right)^c$ and $75° = \left(75 \times \frac{\pi}{180}\right)^c = \left(\frac{5\pi}{12}\right)^c$

$\therefore \quad \frac{\pi}{3} = \frac{s}{r_1}$ and $\frac{5\pi}{12} = \frac{s}{r_2}$

$\Rightarrow \quad \frac{\pi}{3} r_1 = s$ and $\frac{5\pi}{12} r_2 = s \Rightarrow \frac{\pi}{3} r_1 = \frac{5\pi}{12} r_2 \Rightarrow 4r_1 = 5r_2 \Rightarrow r_1 : r_2 = 5 : 4$ **Ans.**

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Do yourself - 1 :

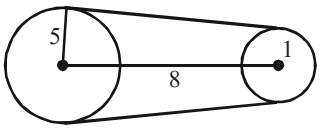
1. The radius of a circle is 30 cm. Find the length of smaller arc of the circle if the length of the chord of the arc is 30 cm.

2. A man is running around a regular hexagonal field of side length 6m, so that he always at a distance 3 metre from the nearest boundary. Find the length of path travelled by him in one round.

3. Convert the following measurement into radians :

(a) 25º 30' 30" (b) 10º 42' 30" (c) 9º 18' 42"

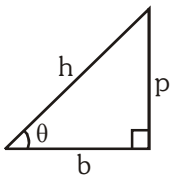
4. A belt is tied up across two circular pulleys of radii 5m and 1m respectively whose centres are seperated at a distance 8m. (as shown). Find the length of the belt required.



5. Find the number of degrees, minutes and seconds in the angle at the centre of a circle, whose radius is 5m, which is subtended by an arc of length 6 m. (Consider π = 22/7)

2. T-RATIOS (or Trigonometric functions) :

In a right angle triangle



$\sin\theta = \frac{p}{h}$; $\cos\theta = \frac{b}{h}$; $\tan\theta = \frac{p}{b}$; $\operatorname{cosec}\theta = \frac{h}{p}$; $\sec\theta = \frac{h}{b}$ and $\cot\theta = \frac{b}{p}$

'p' is perpendicular ; 'b' is base and 'h' is hypotenuse.

3. BASIC TRIGONOMETRIC IDENTITIES :

(1) $\sin\theta . \operatorname{cosec}\theta = 1$

(2) $\cos\theta . \sec\theta = 1$

(3) $\tan\theta . \cot\theta = 1$

(4) $\tan\theta = \frac{\sin\theta}{\cos\theta}$ & $\cot\theta = \frac{\cos\theta}{\sin\theta}$

(5) $\sin^2\theta + \cos^2\theta = 1$ or $\sin^2\theta = 1 - \cos^2\theta$ or $\cos^2\theta = 1 - \sin^2\theta$

(6) $\sec^2\theta - \tan^2\theta = 1$ or $\sec^2\theta = 1 + \tan^2\theta$ or $\tan^2\theta = \sec^2\theta - 1$

(7) $\sec\theta + \tan\theta = \frac{1}{\sec\theta - \tan\theta}$

(8) $\operatorname{cosec}^2\theta - \cot^2\theta = 1$ or $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$ or $\cot^2\theta = \operatorname{cosec}^2\theta - 1$

(9) $\operatorname{cosec}\theta + \cot\theta = \frac{1}{\operatorname{cosec}\theta - \cot\theta}$

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(10) Expressing trigonometrical ratio in terms of each other :

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\cot\theta$	$\sec\theta$	$\text{cosec}\,\theta$
$\sin\theta$	$\sin\theta$	$\sqrt{1-\cos^2\theta}$	$\frac{\tan\theta}{\sqrt{1+\tan^2\theta}}$	$\frac{1}{\sqrt{1+\cot^2\theta}}$	$\frac{\sqrt{\sec^2\theta-1}}{\sec\theta}$	$\frac{1}{\text{cosec}\,\theta}$
$\cos\theta$	$\sqrt{1-\sin^2\theta}$	$\cos\theta$	$\frac{1}{\sqrt{1+\tan^2\theta}}$	$\frac{\cot\theta}{\sqrt{1+\cot^2\theta}}$	$\frac{1}{\sec\theta}$	$\frac{\sqrt{\text{cosec}^2\theta-1}}{\text{cosec}\,\theta}$
$\tan\theta$	$\frac{\sin\theta}{\sqrt{1-\sin^2\theta}}$	$\frac{\sqrt{1-\cos^2\theta}}{\cos\theta}$	$\tan\theta$	$\frac{1}{\cot\theta}$	$\sqrt{\sec^2\theta-1}$	$\frac{1}{\sqrt{\text{cosec}^2\theta-1}}$
$\cot\theta$	$\frac{\sqrt{1-\sin^2\theta}}{\sin\theta}$	$\frac{\cos\theta}{\sqrt{1-\cos^2\theta}}$	$\frac{1}{\tan\theta}$	$\cot\theta$	$\frac{1}{\sqrt{\sec^2\theta-1}}$	$\sqrt{\text{cosec}^2\theta-1}$
$\sec\theta$	$\frac{1}{\sqrt{1-\sin^2\theta}}$	$\frac{1}{\cos\theta}$	$\sqrt{1+\tan^2\theta}$	$\frac{\sqrt{1+\cot^2\theta}}{\cot\theta}$	$\sec\theta$	$\frac{\text{cosec}\theta}{\sqrt{\text{cosec}^2\theta-1}}$
$\text{cosec}\,\theta$	$\frac{1}{\sin\theta}$	$\frac{1}{\sqrt{1-\cos^2\theta}}$	$\frac{\sqrt{1+\tan^2\theta}}{\tan\theta}$	$\sqrt{1+\cot^2\theta}$	$\frac{\sec\theta}{\sqrt{\sec^2\theta-1}}$	$\text{cosec}\,\theta$

Illustration 2 : If $\sin\theta+\sin^2\theta=1$, then prove that $\cos^{12}\theta+3\cos^{10}\theta+3\cos^8\theta+\cos^6\theta-1=0$

Solution : Given that $\sin\theta = 1 - \sin^2\theta = \cos^2\theta$

L.H.S. $= \cos^6\theta(\cos^2\theta + 1)^3 - 1 = \sin^3\theta(1 + \sin\theta)^3 - 1 = (\sin\theta + \sin^2\theta)^3 - 1 = 1 - 1 = 0$

Illustration 3 : $4(\sin^6\theta + \cos^6\theta) - 6(\sin^4\theta + \cos^4\theta)$ is equal to

(A) 0 (B) 1 (C) –2 (D) none of these

Solution : $4[(\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta)] - 6[(\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta]$

$= 4[1 - 3\sin^2\theta\cos^2\theta] - 6[1 - 2\sin^2\theta\cos^2\theta]$

$= 4 - 12\sin^2\theta\cos^2\theta - 6 + 12\sin^2\theta\cos^2\theta = -2$ **Ans.(C)**

Do yourself - 2 :

1. If $\cot\theta = \frac{4}{3}$, then find the value of $\sin\theta$, $\cos\theta$ and $\text{cosec}\,\theta$ in first quadrant.

2. If $\sin\theta + \text{cosec}\,\theta = 2$, then find the value of $\sin^8\theta + \text{cosec}^8\theta$

Prove the following statements in their respective valid domains :

3. $\cos^4A - \sin^4A + 1 = 2\cos^2A$

4. $(\sin A + \cos A)(1 - \sin A\cos A) = \sin^3A + \cos^3A$.

5. $\frac{\sin A}{1+\cos A} + \frac{1+\cos A}{\sin A} = 2\,\text{cosec}\,A$

6. $\cos^6A + \sin^6A = 1 - 3\sin^2A\cos^2A$.

7. $\sqrt{\frac{1-\sin A}{1+\sin A}} = \sec A - \tan A \quad (-90^\circ < A < 90^\circ)$

8. $\dfrac{1}{\cot A + \tan A} = \sin A \cos A$

9. $\dfrac{1-\tan A}{1+\tan A} = \dfrac{\cot A - 1}{\cot A + 1}$

10. $\dfrac{\tan A}{1-\cot A} + \dfrac{\cot A}{1-\tan A} = \sec A \operatorname{cosec} A + 1$

11. $\dfrac{\cos A}{1-\tan A} + \dfrac{\sin A}{1-\cot A} = \sin A + \cos A.$

12. $\sec^4 A - \sec^2 A = \tan^4 A + \tan^2 A.$

13. $\cot^4 A + \cot^2 A = \operatorname{cosec}^4 A - \operatorname{cosec}^2 A.$

14. $\tan^2 A - \sin^2 A = \sin^4 A \sec^2 A = \tan^2 A.\sin^2 A$

15. $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$

16. $\dfrac{1}{\operatorname{cosec} A - \cot A} - \dfrac{1}{\sin A} = \dfrac{1}{\sin A} - \dfrac{1}{\operatorname{cosec} A + \cot A}$

17. $\dfrac{\cot A.\cos A}{\cot A + \cos A} = \dfrac{\cot A - \cos A}{\cot A.\cos A}$

18. $\left(\dfrac{1}{\sec^2\alpha - \cos^2\alpha} + \dfrac{1}{\operatorname{cosec}^2\alpha - \sin^2\alpha}\right)\cos^2\alpha \sin^2\alpha = \dfrac{1-\cos^2\alpha\sin^2\alpha}{2+\cos^2\alpha\sin^2\alpha}$

19. $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A \cos^2 A)$

20. $\dfrac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \dfrac{1+\sin A}{\cos A}$

21. $(\tan\alpha + \operatorname{cosec}\beta)^2 - (\cot\beta - \sec\alpha)^2 = 2\tan\alpha\cot\beta(\operatorname{cosec}\alpha + \sec\beta).$

22. $2\sec^2\alpha - \sec^4\alpha - 2\operatorname{cosec}^2\alpha + \operatorname{cosec}^4\alpha = \cot^4\alpha - \tan^4\alpha.$

23. If $\sin\theta$ equals to $\dfrac{x^2-y^2}{x^2+y^2}$, find the value of $\cos\theta$ and $\cot\theta$, where $\theta \in (0,90^\circ)$.

24. If $\sin\theta = \dfrac{m^2+2mn}{m^2+2mn+2n^2}$, prove that $\tan\theta = \dfrac{m^2+2mn}{2mn+2n^2}$, where $\theta \in (0,90^\circ)$.

25. If $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$, prove that $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$.

26. Prove that $\operatorname{cosec}^6\alpha - \cot^6\alpha = 3\operatorname{cosec}^2\alpha\cot^2\alpha + 1.$

27. Express $2\sec^2 A - \sec^4 A - 2\operatorname{cosec}^2 A + \operatorname{cosec}^4 A$ in terms of $\tan A$.

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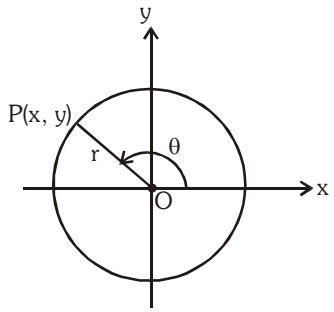
4. NEW DEFINITION OF T-RATIOS :



By using rectangular coordinates the definitions of trigonometric functions can be extended to angles of any size in the following way (see diagram). A point P is taken with coordinates (x, y). The radius vector OP has length r and the angle θ is taken as the directed angle measured anticlockwise from the x-axis. The three main trigonometric functions are then defined in terms of r and the coordinates x and y.

$\sin\theta = y/r$,

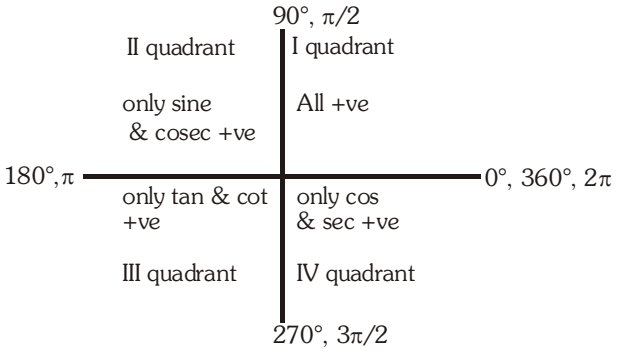
$\cos\theta = x/r$

$\tan\theta = y/x$,

(The other function are reciprocals of these)

This can give negative values of the trigonometric functions.

5. SIGNS OF TRIGONOMETRIC FUNCTIONS IN DIFFERENT QUADRANTS :



6. TRIGONOMETRIC FUNCTIONS OF ALLIED ANGLES :

(a) $\sin(2n\pi + \theta) = \sin\theta$, $\cos(2n\pi + \theta) = \cos\theta$, where $n \in I$

(b)

$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$
$\sin(90° - \theta) = \cos\theta$	$\cos(90° - \theta) = \sin\theta$
$\sin(90° + \theta) = \cos\theta$	$\cos(90° + \theta) = -\sin\theta$
$\sin(180° - \theta) = \sin\theta$	$\cos(180° - \theta) = -\cos\theta$
$\sin(180° + \theta) = -\sin\theta$	$\cos(180° + \theta) = -\cos\theta$
$\sin(270° - \theta) = -\cos\theta$	$\cos(270° - \theta) = -\sin\theta$
$\sin(270° + \theta) = -\cos\theta$	$\cos(270° + \theta) = \sin\theta$
$\sin(360° - \theta) = -\sin\theta$	$\cos(360° - \theta) = \cos\theta$
$\sin(360° + \theta) = \sin\theta$	$\cos(360° + \theta) = \cos\theta$

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Do yourself - 3 :

1. If $\sin A = \frac{11}{61}$, find tanA, cosA, and sec A.

2. If $\cos\theta = \frac{4}{5}$, find sin θ and cot θ.

3. If $\tan\theta = \frac{1}{\sqrt{7}}$, find the value of $\frac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \sec^2\theta}$

4. If $\cot\theta = \frac{15}{8}$, find sin θ and cosec θ.

5. If 2 sin θ = 2 – cos θ, find sin θ.

6. If 8 sin θ = 4 + cos θ, find sin θ.

7. If tan θ + sec θ = 1.5, find sin θ.

8. If cot θ + cosec θ = 5, find cos θ.

9. If 3 $\sec^4\theta + 8 = 10\sec^2\theta$, find the values of tan θ.

10. If $\tan^2\theta + \sec\theta = 5$, find cos θ.

11. If tan θ + cot θ = 2, find sin θ.

12. If $\sec^2\theta = 2 + 2\tan\theta$, find tan θ.

13. If $\tan\theta = \frac{2x(x+1)}{2x+1}$, find sin θ and cos θ.

7. VALUES OF T-RATIOS OF SOME STANDARD ANGLES :

Angles	**0°**	**30°**	**45°**	**60°**	**90°**	**180°**	**270°**
T-ratio	**0**	**π/6**	**π/4**	**π/3**	**π/2**	**π**	**3π/2**
sin θ	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1	0	–1
cos θ	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0	–1	0
tan θ	0	$1/\sqrt{3}$	1	$\sqrt{3}$	N.D.	0	N.D.
cot θ	N.D.	$\sqrt{3}$	1	$1/\sqrt{3}$	0	N.D.	0
sec θ	1	$2/\sqrt{3}$	$\sqrt{2}$	2	N.D.	–1	N.D.
cosecθ	N.D.	2	$\sqrt{2}$	$2/\sqrt{3}$	1	N.D.	–1

N.D. → Not Defined

(a) $\sin n\pi = 0$; $\cos n\pi = (-1)^n$; $\tan n\pi = 0$ where $n \in I$

(b) $\sin(2n+1)\frac{\pi}{2} = (-1)^n$; $\cos(2n+1)\frac{\pi}{2} = 0$ where $n \in I$

Do yourself - 4 :

1. Verify the following identities for A = 30º as well as for A = 45º

(a) $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

(b) $\sin 2A = 2\sin A \cos A$ (c) $\cos 3A = 4\cos^3 A - 3\cos A$

(d) $\sin 3A = 3\sin A - 4\sin^3 A$ (e) $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

2. Find the value of

(a) $\sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ$ (b) $\tan^2 30^\circ + \tan^2 45^\circ + \tan^2 60^\circ$

(c) $\sin 30^\circ \cos 60^\circ + \sin 30^\circ \sin 60^\circ$ (d) $\cos 45^\circ \cos 60^\circ - \sin 45^\circ \sin 60^\circ$

Illustration 4 : If $\sin\theta = -\dfrac{1}{2}$ and $\tan\theta = \dfrac{1}{\sqrt{3}}$ then θ is equal to -

(A) 30° (B) 150° (C) 210° (D) none of these

Solution : Let us first find out θ lying between 0 and 360°.

Since $\sin\theta = -\dfrac{1}{2} \Rightarrow \theta = 210^\circ$ or 330° and $\tan\theta = \dfrac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$ or 210°

Hence , $\theta = 210^\circ$ or $\dfrac{7\pi}{6}$ is the value satisfying both. **Ans. (C)**

Do yourself - 5 :

1. (i) If $\cos\theta = -\dfrac{1}{2}$ and $\pi < \theta < \dfrac{3\pi}{2}$, then find the value of $4\tan^2\theta - 3\operatorname{cosec}^2\theta$.

(ii) Prove that : (a) $\cos 570^\circ \sin 510^\circ + \sin(-330^\circ)\cos(-390^\circ) = 0$

(b) $\tan\dfrac{11\pi}{3} - 2\sin\dfrac{9\pi}{3} - \dfrac{3}{4}\operatorname{cosec}^2\dfrac{\pi}{4} + 4\cos^2\dfrac{17\pi}{6} = \dfrac{3-2\sqrt{3}}{2}$

2. Evaluate :

(a) $\sin 420^\circ \cos 390^\circ + \cos(-300^\circ)\sin(-330^\circ)$ (b) $\tan 225^\circ \cot 405^\circ + \tan 765^\circ \cot 675^\circ$

3. What are the values of $\cos A - \sin A$ and $\tan A + \cot A$ when A has the values

(a) $\dfrac{2\pi}{3}$ (b) $\dfrac{7\pi}{4}$ (c) $\dfrac{11\pi}{3}$ (d) $\dfrac{5\pi}{4}$

4. Express the following quantities in terms of the ratios of a positive angle, which is less than 45º

(a) $\sin(-65^\circ)$ (b) $\cos(-928^\circ)$ (c) $\tan 1145^\circ$ (d) $\cot(-1054^\circ)$

5. What is the sign of $\sin A + \cos A$ for the following values of A ?

(a) 140º (b) –356º (c) –1125º.

6. What is the sign of $\sin A - \cos A$ for the following values of A ?

(a) 215º (b) –634º (c) – 457º

7. Find the sines and cosines of all angles in the first four quadrant whose tangents are equal to cos 135º.

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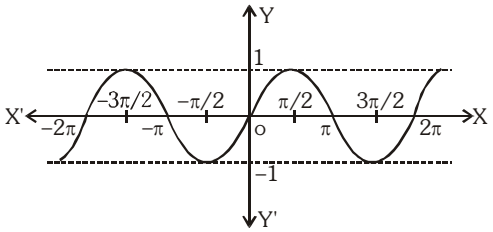
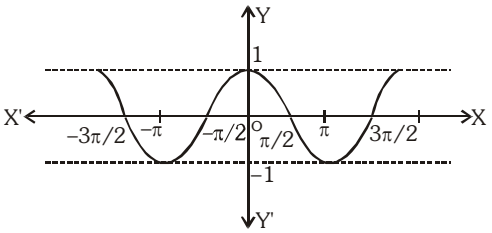
8. GRAPH OF TRIGONOMETRIC FUNCTIONS :

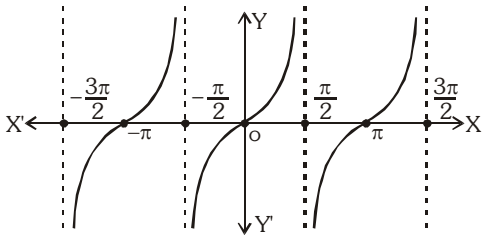
(i) y = sinx



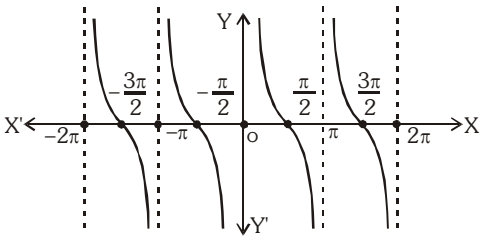
(ii) y = cosx



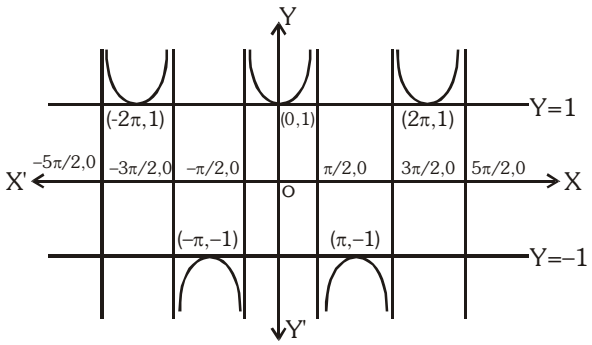
(iii) y = tanx



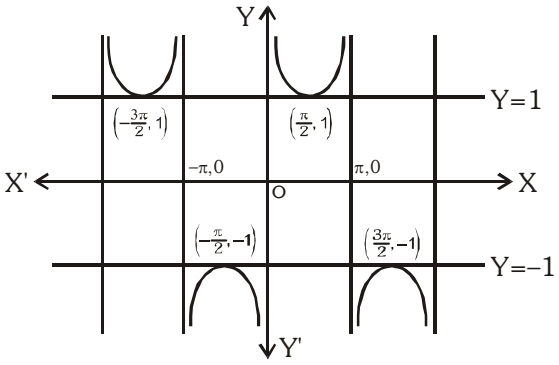
(iv) y = cotx



(v) y = secx



(vi) y = cosecx



9. DOMAINS, RANGES AND PERIODICITY OF TRIGONOMETRIC FUNCTIONS :

T-Ratio	Domain	Range	Period
sin x	R	[–1,1]	2π
cos x	R	[–1,1]	2π
tan x	$R-\{(2n+1)\pi/2 ; n\in I\}$	R	π
cot x	$R-\{n\pi : n \in I\}$	R	π
sec x	$R- \{(2n+1) \pi/2 : n \in I\}$	$(-\infty,-1] \cup[1,\infty)$	2π
cosec x	$R- \{n\pi : n \in I\}$	$(-\infty,-1] \cup[1,\infty)$	2π

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Do yourself - 6 :

1. Prove that the equation $\sin\theta = x + \frac{1}{x}$ is impossible if x be real.
2. Show that the equation $\sec^2\theta = \frac{4xy}{(x+y)^2}$ is only possible when x = y.
3. The number of real solutions of the equation $\sin(e^x) = 2^x + 2^{-x}$ is -
   (A) 1 (B) 0 (C) 2 (D) Infinite
4. Draw graphs of
   (a) y = sin2x (b) y = 2cos3x (c) y = 4 tanx
5. Find number of solutions of the equation $\sin(\pi x) + x^2 + 1 = 2x$

10. TRIGONOMETRIC RATIOS OF THE SUM & DIFFERENCE OF TWO ANGLES :

(i) $\sin(A + B) = \sin A \cos B + \cos A \sin B$.

(ii) $\sin(A - B) = \sin A \cos B - \cos A \sin B$.

(iii) $\cos(A + B) = \cos A \cos B - \sin A \sin B$

(iv) $\cos(A - B) = \cos A \cos B + \sin A \sin B$

(v) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

(vi) $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

(vii) $\cot(A + B) = \frac{\cot B \cot A - 1}{\cot B + \cot A}$

(viii) $\cot(A - B) = \frac{\cot B \cot A + 1}{\cot B - \cot A}$

Some more results :

(i) $\sin^2 A - \sin^2 B = \sin(A + B).\sin(A - B) = \cos^2 B - \cos^2 A$.

(ii) $\cos^2 A - \sin^2 B = \cos(A+B).\cos(A - B)$.

Illustration 5 : Prove that $\sqrt{3}\,\text{cosec}20° - \sec 20° = 4$.

Solution : L.H.S. $= \frac{\sqrt{3}}{\sin 20°} - \frac{1}{\cos 20°} = \frac{\sqrt{3}\cos 20° - \sin 20°}{\sin 20°.\cos 20°}$

$$= \frac{4\left(\frac{\sqrt{3}}{2}\cos 20° - \frac{1}{2}\sin 20°\right)}{2\sin 20° \cos 20°} = \frac{4(\sin 60.\cos 20° - \cos 60°.\sin 20°)}{\sin 40°}$$

$$= 4.\frac{\sin(60° - 20°)}{\sin 40°} = 4.\frac{\sin 40°}{\sin 40°} = 4 = \text{R.H.S.}$$

Illustration 6 : Prove that $\tan 70° = \cot 70° + 2\cot 40°$.

Solution : L.H.S. $= \tan 70° = \tan(20° + 50°) = \frac{\tan 20° + \tan 50°}{1 - \tan 20° \tan 50°}$

or $\tan 70° - \tan 20° \tan 50° \tan 70° = \tan 20° + \tan 50°$

or $\tan 70° = \tan 70° \tan 50° \tan 20° + \tan 20° + \tan 50° = 2\tan 50° + \tan 20°$

$= \cot 70° + 2\cot 40°$ = R.H.S.

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Do yourself - 7 :

1. If $\sin A = \frac{3}{5}$ and $\cos B = \frac{9}{41}$, $0 < A, B < \frac{\pi}{2}$, then find the value of the following :

(a) sin(A + B) (b) sin(A – B) (c) cos(A + B) (d) cos(A – B)

2. If x + y = 45°, then prove that :

(a) (1 + tanx)(1 + tany) = 2 (b) (cotx – 1)(coty – 1) = 2

(Remember these results)

3. If $\sin\alpha = \frac{3}{5}$ and $\cos\beta = \frac{9}{41}$, find the value of sin(α – β) and cos(α + β).

4. If $\sin\alpha = \frac{15}{17}$ and $\cos\beta = \frac{12}{13}$, find the values of sin(α + β), cos(α – β), and tan(α + β)

Prove that

5. cos(45º – A) cos(45º – B) – sin(45º – A) sin(45º – B) = sin(A + B).

6. sin(45º + A) cos(45º – B) + cos(45º + A) sin(45º – B) = cos(A – B).

7. $\frac{\sin(A-B)}{\cos A\cos B} + \frac{\sin(B-C)}{\cos B\cos C} + \frac{\sin(C-A)}{\cos C\cos A} = 0$

8. sin105º + cos105º = cos45º

9. sin75º – sin15º = cos105º + cos15º

10. cos(α + β) cosγ – cos(β + γ)cosα = sinβ sin(γ – α)

11. sin(n + 1)A sin(n – 1) A + cos(n + 1)A cos(n – 1)A = cos2A.

12. sin(n + 1)A sin(n + 2) A + cos(n + 1)A cos(n + 2)A = cosA.

13. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, find the values of tan(2A + B) and tan(2A – B)

14. If $\tan\alpha = \frac{5}{6}$ and $\tan\beta = \frac{1}{11}$, prove that $\alpha + \beta = \frac{\pi}{4}$

Prove that

15. $\tan\left(\frac{\pi}{4}+\theta\right) \times \tan\left(\frac{3\pi}{4}+\theta\right) = -1$

16. $\cot\left(\frac{\pi}{4}+\theta\right)\cot\left(\frac{\pi}{4}-\theta\right) = 1$

17. $1 + \tan A \tan\frac{A}{2} = \tan A \cot\frac{A}{2} - 1 = \sec A$

18. tan3A tan2A tanA = tan3A – tan2A – tanA.

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11. FORMULAE TO TRANSFORM THE PRODUCT INTO SUM OR DIFFERENCE :

(i) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$. **(ii)** $2\cos A\sin B = \sin(A+B) - \sin(A-B)$

(iii) $2\cos A\cos B = \cos(A+B) + \cos(A-B)$ **(iv)** $2\sin A\sin B = \cos(A-B) - \cos(A+B)$

Illustration 7 : If $\sin 2A = \lambda \sin 2B$, then prove that $\dfrac{\tan(A+B)}{\tan(A-B)} = \dfrac{\lambda+1}{\lambda-1}$.

Solution : Given $\sin 2A = \lambda \sin 2B$

$$\Rightarrow \quad \frac{\sin 2A}{\sin 2B} = \frac{\lambda}{1}$$

Applying componendo & dividendo,

$$\frac{\sin 2A + \sin 2B}{\sin 2B - \sin 2A} = \frac{\lambda+1}{1-\lambda}$$

$$\Rightarrow \quad \frac{2\sin\left(\frac{2A+2B}{2}\right)\cos\left(\frac{2A-2B}{2}\right)}{2\cos\left(\frac{2B+2A}{2}\right)\sin\left(\frac{2B-2A}{2}\right)} = \frac{\lambda+1}{1-\lambda}$$

$$\Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\sin\{-(A-B)\}} = \frac{\lambda+1}{1-\lambda} \quad \Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\times -\sin(A-B)} = \frac{\lambda+1}{-(\lambda-1)}$$

$$\Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\sin(A-B)} = \frac{\lambda+1}{\lambda-1} \quad \Rightarrow \quad \tan(A+B)\cot(A-B) = \frac{\lambda+1}{\lambda-1}$$

$$\Rightarrow \quad \frac{\tan(A+B)}{\tan(A-B)} = \frac{\lambda+1}{\lambda-1}$$

Do yourself - 8 :

1. Simplify $\dfrac{\sin 75° - \sin 15°}{\cos 75° + \cos 15°}$

Express follwing as a sum or difference of angles used in arguments :

2. $2\sin 5\theta \sin 7\theta$.

3. $2\sin 54° \sin 66°$

4. $\cos(36° - A)\cos(36° + A) + \cos(54° + A)\cos(54° - A) = \cos 2A$.

5. $\cos A \sin(B - C) + \cos B \sin(C - A) + \cos C \sin(A - B) = 0$.

6. $\sin(45° + A)\sin(45° - A) = \dfrac{1}{2}\cos 2A$

7. $\sin(\beta - \gamma)\cos(\alpha - \delta) + \sin(\gamma - \alpha)\cos(\beta - \delta) + \sin(\alpha - \beta)\cos(\gamma - \delta) = 0$.

8. $2\cos\dfrac{\pi}{13}\cos\dfrac{9\pi}{13} + \cos\dfrac{3\pi}{13} + \cos\dfrac{5\pi}{13} = 0$

12. FORMULAE TO TRANSFORM SUM OR DIFFERENCE INTO PRODUCT :

(i) $\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(ii) $\sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(iii) $\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(iv) $\cos C - \cos D = 2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{D-C}{2}\right)$

Illustration 8 : $\dfrac{\sin 5\theta + \sin 2\theta - \sin\theta}{\cos 5\theta + 2\cos 3\theta + 2\cos^2\theta + \cos\theta}$ is equal to -

(A) $\tan\theta$ (B) $\cos\theta$ (C) $\cot\theta$ (D) none of these

Solution : L.H.S.= $\dfrac{2\sin 2\theta\cos 3\theta + \sin 2\theta}{2\cos 3\theta.\cos 2\theta + 2\cos 3\theta + 2\cos^2\theta} = \dfrac{\sin 2\theta[2\cos 3\theta + 1]}{2\left[\cos 3\theta(\cos 2\theta + 1) + (\cos^2\theta)\right]}$

$= \dfrac{\sin 2\theta[2\cos 3\theta + 1]}{2\left[\cos 3\theta(2\cos^2\theta) + \cos^2\theta\right]} = \dfrac{\sin 2\theta(2\cos 3\theta + 1)}{2\cos^2\theta(2\cos 3\theta + 1)} = \tan\theta$ **Ans. (A)**

Do yourself - 9 :

1. Prove that

(a) $(\sin 3A + \sin A)\sin A + (\cos 3A - \cos A)\cos A = 0$ **(b)** $\dfrac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta} = \tan 2\theta$

Prove that

2. $\dfrac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} = \tan\theta$

3. $\dfrac{\cos 2B + \cos 2A}{\cos 2B - \cos 2A} = \cot(A+B)\cot(A-B)$

4. $\dfrac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B} = \dfrac{\tan(A+B)}{\tan(A-B)}$

5. $\cos(A + B) + \sin(A - B) = 2\sin(45^\circ + A)\cos(45^\circ + B)$

6. $\dfrac{\cos 3A - \cos A}{\sin 3A - \sin A} + \dfrac{\cos 2A - \cos 4A}{\sin 4A - \sin 2A} = \dfrac{\sin A}{\cos 2A\cos 3A}$

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7. $\dfrac{\tan 5\theta + \tan 3\theta}{\tan 5\theta - \tan 3\theta} = 4\cos 2\theta \cos 4\theta$

8. $\dfrac{\cos 3\theta + 2\cos 5\theta + \cos 7\theta}{\cos\theta + 2\cos 3\theta + \cos 5\theta} = \cos 2\theta - \sin 2\theta \tan 3\theta$

9. $\dfrac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$

10. $\dfrac{\sin(A-C) + 2\sin A + \sin(A+C)}{\sin(B-C) + 2\sin B + \sin(B+C)} = \dfrac{\sin A}{\sin B}$

11. $\dfrac{\sin A - \sin 5A + \sin 9A - \sin 13A}{\cos A - \cos 5A - \cos 9A + \cos 13A} = \cot 4A$

12. $\dfrac{\sin A + \sin B}{\sin A - \sin B} = \tan\dfrac{A+B}{2}\cot\dfrac{A-B}{2}$

13. $\dfrac{\sin A + \sin B}{\cos A + \cos B} = \tan\dfrac{A+B}{2}$

14. $\dfrac{\sin A - \sin B}{\cos B - \cos A} = \cot\dfrac{A+B}{2}$

15. $\dfrac{\cos(A+B+C) + \cos(-A+B+C) + \cos(A-B+C) + \cos(A+B-C)}{\sin(A+B+C) + \sin(-A+B+C) - \sin(A-B+C) + \sin(A+B-C)} = \cot B$

16. $\cos\left\{\theta + \left(n - \dfrac{3}{2}\right)\phi\right\} - \cos\left\{\theta + \left(n + \dfrac{3}{2}\right)\phi\right\} = 2\sin\dfrac{3\phi}{2}.\sin(\theta + n\phi)$

17. $\sin\dfrac{\theta}{2}\sin\dfrac{7\theta}{2} + \sin\dfrac{3\theta}{2}\sin\dfrac{11\theta}{2} = \sin 2\theta \sin 5\theta$

18. $\cos 2\theta \cos\dfrac{\theta}{2} - \cos 3\theta \cos\dfrac{9\theta}{2} = \sin 5\theta \sin\dfrac{5\theta}{2}$

19. $\dfrac{2\sin(A-C)\cos C - \sin(A-2C)}{2\sin(B-C)\cos C - \sin(B-2C)} = \dfrac{\sin A}{\sin B}$

20. $\dfrac{\sin A \sin 2A + \sin 3A \sin 6A + \sin 4A \sin 13A}{\sin A \cos 2A + \sin 3A \cos 6A + \sin 4A \cos 13A} = \tan 9A$

21. $\dfrac{\cos 2A \cos 3A - \cos 2A \cos 7A + \cos A \cos 10A}{\sin 4A \sin 3A - \sin 2A \sin 5A + \sin 4A \sin 7A} = \cot 6A \cot 5A$

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13. TRIGONOMETRIC RATIOS OF SUM OF MORE THAN TWO ANGLES :

(i) $\sin(A+B+C) = \sin A\cos B\cos C + \sin B\cos A\cos C + \sin C\cos A\cos B - \sin A\sin B\sin C$

$= \Sigma\sin A\cos B\cos C - \Pi\sin A$

$= \cos A\cos B\cos C\,[\tan A + \tan B + \tan C - \tan A\tan B\tan C]$

(ii) $\cos(A+B+C) = \cos A\cos B\cos C - \sin A\sin B\cos C - \sin A\cos B\sin C - \cos A\sin B\sin C$

$= \Pi\cos A - \Sigma\sin A\sin B\cos C$

$= \cos A\cos B\cos C\,[1 - \tan A\tan B - \tan B\tan C - \tan C\tan A]$

(iii) $\tan(A+B+C) = \dfrac{\tan A + \tan B + \tan C - \tan A\tan B\tan C}{1 - \tan A\tan B - \tan B\tan C - \tan C\tan A} = \dfrac{S_1 - S_3}{1 - S_2}$

14. TRIGONOMETRIC RATIOS OF MULTIPLE ANGLES :

(a) Trigonometrical ratios of an angle 2θ in terms of the angle θ :

(i) $\sin 2\theta = 2\sin\theta\cos\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$

(ii) $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$

(iii) $1 + \cos 2\theta = 2\cos^2\theta$ **(iv)** $1 - \cos 2\theta = 2\sin^2\theta$

(v) $\tan\theta = \dfrac{1-\cos 2\theta}{\sin 2\theta} = \dfrac{\sin 2\theta}{1+\cos 2\theta}$ **(vi)** $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$

Illustration 9: Prove that : $\dfrac{2\cos 2A + 1}{2\cos 2A - 1} = \tan(60° + A)\tan(60° - A)$.

Solution : R.H.S. = $\tan(60° + A)\tan(60° - A)$

$$= \left(\frac{\tan 60° + \tan A}{1 - \tan 60°\tan A}\right)\left(\frac{\tan 60° - \tan A}{1 + \tan 60°\tan A}\right) = \left(\frac{\sqrt{3} + \tan A}{1 - \sqrt{3}\tan A}\right)\left(\frac{\sqrt{3} - \tan A}{1 + \sqrt{3}\tan A}\right)$$

$$= \frac{3 - \tan^2 A}{1 - 3\tan^2 A} = \frac{3 - \dfrac{\sin^2 A}{\cos^2 A}}{1 - 3\dfrac{\sin^2 A}{\cos^2 A}} = \frac{3\cos^2 A - \sin^2 A}{\cos^2 A - 3\sin^2 A} == \frac{2\cos^2 A + \cos^2 A - 2\sin^2 A + \sin^2 A}{2\cos^2 A - 2\sin^2 A - \sin^2 A - \cos^2 A}$$

$$= \frac{2(\cos^2 A - \sin^2 A) + \cos^2 A + \sin^2 A}{2(\cos^2 A - \sin^2 A) - (\sin^2 A + \cos^2 A)} = \frac{2\cos 2A + 1}{2\cos 2A - 1} = \text{L.H.S.}$$

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Do yourself - 10 :

Prove that :

1. $\dfrac{\sin 2A}{1+\cos 2A} = \tan A$ (Remember)

2. $\dfrac{\sin 2A}{1-\cos 2A} = \cot A$ (Remember)

3. tanA – cotA = –2 cot2A. (Remember)

4. $\dfrac{1+\sin 2\theta+\cos 2\theta}{1+\sin 2\theta-\cos 2\theta} = \cot\theta$

5. tanA + cotA = 2 cosec2A

6. $\dfrac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)} = \tan\dfrac{A}{2}\cot\dfrac{B}{2}$

7. $\dfrac{\sec 8A-1}{\sec 4A-1} = \dfrac{\tan 8A}{\tan 2A}$

8. $\dfrac{1+\tan^2(45°-A)}{1-\tan^2(45°-A)} = \text{cosec } 2A$

9. $\dfrac{\sin^2 A-\sin^2 B}{\sin A\cos A-\sin B\cos B} = \tan(A+B)$

10. $\tan\left(\dfrac{\pi}{4}+\theta\right)-\tan\left(\dfrac{\pi}{4}-\theta\right) = 2\tan 2\theta$

11. cot(A + 15º) – tan(A – 15º) $= \dfrac{4\cos 2A}{1+2\sin 2A}$

12. $\dfrac{\sin(n+1)A+2\sin nA+\sin(n-1)A}{\cos(n-1)A-\cos(n+1)A} = \cot\dfrac{A}{2}$

13. sin3A + sin2A – sinA = 4 sin A cos $\dfrac{A}{2}\cos\dfrac{3A}{2}$

14. $\tan 2A = (\sec 2A+1)\sqrt{\sec^2 A-1}$

15. $\cos^3 2\theta + 3\cos 2\theta = 4(\cos^6\theta - \sin^6\theta)$

16. $1+\cos^2 2\theta = 2(\cos^4\theta + \sin^4\theta)$

17. $\sec^2 A(1 + \sec 2A) = 2 \sec 2A$

18. cosecA – 2cot2A cosA = 2sinA

19. $\cot A = \dfrac{1}{2}\left(\cot\dfrac{A}{2}-\tan\dfrac{A}{2}\right)$

20. $\dfrac{2\cos 2^n\theta+1}{2\cos\theta+1} = (2\cos\theta-1)(2\cos 2\theta-1)(2\cos 2^2\theta-1)....(2\cos 2^{n-1}\theta-1)$

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(b) Trigonometrical ratios of an angle 3θ in terms of the angle θ :

(i) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$. **(ii)** $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$.

(iii) $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

Illustration 10: Show that $\sin 12°.\sin 48°.\sin 54° = 1/8$

Solution : L.H.S. $= \frac{1}{2}[\cos 36° - \cos 60°]\sin 54° = \frac{1}{2}\left[\cos 36° \sin 54° - \frac{1}{2}\sin 54°\right]$

$$= \frac{1}{4}[2\cos 36° \sin 54° - \sin 54°] = \frac{1}{4}[\sin 90° + \sin 18° - \sin 54°]$$

$$= \frac{1}{4}[1 - (\sin 54° - \sin 18°)] = \frac{1}{4}[1 - 2\sin 18° \cos 36°]$$

$$= \frac{1}{4}\left[1 - \frac{2\sin 18°}{\cos 18°}\cos 18° \cos 36°\right] = \frac{1}{4}\left[1 - \frac{\sin 36° \cos 36°}{\cos 18°}\right]$$

$$= \frac{1}{4}\left[1 - \frac{2\sin 36° \cos 36°}{2\cos 18°}\right] = \frac{1}{4}\left[1 - \frac{\sin 72°}{2\sin 72°}\right] = \frac{1}{4}\left[1 - \frac{1}{2}\right] = \frac{1}{8} = \text{R.H.S.}$$

Illustration 11: Prove that : $\tan A + \tan(60° + A) + \tan(120° + A) = 3\tan 3A$

Solution : L.H.S. $= \tan A + \tan(60° + A) + \tan(120° + A)$

$= \tan A + \tan(60° + A) + \tan\{180° - (60° - A)\}$

$= \tan A + \tan(60° + A) - \tan(60° - A)$ $[\because \tan(180° - \theta) = -\tan\theta]$

$$= \tan A + \frac{\tan 60° + \tan A}{1 - \tan 60° \tan A} - \frac{\tan 60° - \tan A}{1 + \tan 60° \tan A} = \tan A + \frac{\sqrt{3} + \tan A}{1 - \sqrt{3}\tan A} - \frac{\sqrt{3} - \tan A}{1 + \sqrt{3}\tan A}$$

$$= \tan A + \frac{\sqrt{3} + \tan A + 3\tan A + \sqrt{3}\tan^2 A - \sqrt{3} + \tan A + 3\tan A - \sqrt{3}\tan^2 A}{(1 - \sqrt{3}\tan A)(1 + \sqrt{3}\tan A)}$$

$$= \tan A + \frac{8\tan A}{1 - 3\tan^2 A} = \frac{\tan A - 3\tan^3 A + 8\tan A}{1 - 3\tan^2 A}$$

$$= \frac{9\tan A - 3\tan^3 A}{1 - 3\tan^2 A} = 3\left(\frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}\right) = 3\tan 3A = \text{R.H.S.}$$

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Do yourself - 11 :

1 Prove that :

(a) $\cot\theta \cot(60° - \theta) \cot(60° + \theta) = \cot 3\theta$ (b) $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

(c) $\cos 20° \cos 40° \cos 60° \cos 80° = \frac{1}{16}$

Prove that

2. $\sin\alpha \sin(60° - \alpha) \sin(60° + \alpha) = \frac{1}{4}\sin 3\alpha$

3. $\cos\alpha \cos(60° - \alpha) \cos(60° + \alpha) = \frac{1}{4}\cos 3\alpha$

4. $\cot\alpha + \cot(60° + \alpha) - \cot(60° - \alpha) = 3\cot 3\alpha$.

5. $\cos 4\alpha = 1 - 8\cos^2\alpha + 8\cos^4\alpha$

6. $\sin 4A = 4\sin A\cos^3 A - 4\cos A\sin^3 A$.

7. $\cos 6\alpha = 32\cos^6\alpha - 48\cos^4\alpha + 18\cos^2\alpha - 1$.

8. If $\cos x + \sin x = a, \left(-\frac{\pi}{2} < x < -\frac{\pi}{4}\right)$, then cos2x is equal to

(A) a^2 (B) $a\sqrt{(2-a)}$ (C) $a\sqrt{(2+a)}$ (D) $a\sqrt{(2-a^2)}$

9. If $\cos A = \frac{3}{4}$, then the value of expression $32\sin\frac{A}{2}\sin\frac{5A}{2}$ is equal to

(A) 11 (B) –11 (C) 12 (D) 4

15. TRIGONOMETRIC RATIOS OF SUB MULTIPLE ANGLES :

Since the trigonometric relations are true for all values of angle θ, they will be true if instead of θ be substitute $\frac{\theta}{2}$

(i) $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

(ii) $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = 2\cos^2\frac{\theta}{2} - 1 = 1 - 2\sin^2\frac{\theta}{2} = \frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

(iii) $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ **(iv)** $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$

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(v) $\tan\frac{\theta}{2}=\frac{1-\cos\theta}{\sin\theta}=\frac{\sin\theta}{1+\cos\theta}$

(vi) $\tan\theta=\frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$

(vii) $\sin\frac{\theta}{2}=\pm\sqrt{\frac{1-\cos\theta}{2}}$

(viii) $\cos\frac{\theta}{2}=\pm\sqrt{\frac{1+\cos\theta}{2}}$

(ix) $\tan\frac{\theta}{2}=\pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$

(x) $2\sin\frac{\theta}{2}=\pm\sqrt{1+\sin\theta}\pm\sqrt{1-\sin\theta}$

(xi) $2\cos\frac{\theta}{2}=\pm\sqrt{1+\sin\theta}\mp\sqrt{1-\sin\theta}$

(xii) $\tan\frac{\theta}{2}=\frac{\pm\sqrt{1+\tan^2\theta}-1}{\tan\theta}$

for **(vii)** to **(xii)** , we decide the sign of ratio according to value of θ.

Illustration 12: $\sin 67\frac{1}{2}^\circ+\cos 67\frac{1}{2}^\circ$ is equal to

(A) $\frac{1}{2}\sqrt{4+2\sqrt{2}}$ (B) $\frac{1}{2}\sqrt{4-2\sqrt{2}}$ (C) $\frac{1}{4}\left(\sqrt{4+2\sqrt{2}}\right)$ (D) $\frac{1}{4}\left(\sqrt{4-2\sqrt{2}}\right)$

Solution : $\sin 67\frac{1}{2}^\circ+\cos 67\frac{1}{2}^\circ=\sqrt{1+\sin 135^\circ}=\sqrt{1+\frac{1}{\sqrt{2}}}$ (using cosA + sinA = $\sqrt{1+\sin 2A}$)

$=\frac{1}{2}\sqrt{4+2\sqrt{2}}$ **Ans.(A)**

Do yourself - 12 :

1 Find the value of

(a) $\sin\frac{\pi}{8}$ (b) $\cos\frac{\pi}{8}$ (c) $\tan\frac{\pi}{8}$

2. $\frac{\cos A}{1\mp\sin A}=\tan\left(45^\circ\pm\frac{A}{2}\right)$

3. $\frac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}=\tan\frac{\theta}{2}$

4. If $\sin\theta=\frac{1}{2}$ and $\sin\phi=\frac{1}{3}$, find the values of $\sin(\theta+\phi)$ and $\sin(2\theta+2\phi)$.

5. If $\cos\alpha=\frac{11}{61}$ and $\sin\beta=\frac{4}{5}$, find the values of $\sin^2\frac{\alpha-\beta}{2}$ and $\cos^2\frac{\alpha+\beta}{2}$, the angles α and β being positive acute angles.

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6. If $\cos\alpha = \frac{3}{5}$ and $\cos\beta = \frac{4}{5}$, find the value of $\cos\frac{\alpha-\beta}{2}$, the angles α and β being positive acute angles.

7. Given $\sec\theta = 1\frac{1}{4}$, find $\tan\frac{\theta}{2}$ and $\tan\theta$.

8. Find the values of (a) $\sin 7\frac{1}{2}^\circ$ (b) $\cos 7\frac{1}{2}^\circ$ (c) $\tan 11\frac{1}{4}^\circ$

9. If $\sin\theta + \sin\phi = a$ and $\cos\theta + \cos\phi = b$, find the value of $\tan\frac{\theta-\phi}{2}$.

Prove that

10. $(\cos\alpha+\cos\beta)^2 + (\sin\alpha-\sin\beta)^2 = 4\cos^2\frac{\alpha+\beta}{2}$.

11. $(\cos\alpha+\cos\beta)^2 + (\sin\alpha+\sin\beta)^2 = 4\cos^2\frac{\alpha-\beta}{2}$.

12. $(\cos\alpha-\cos\beta)^2 + (\sin\alpha-\sin\beta)^2 = 4\sin^2\frac{\alpha-\beta}{2}$

13. $\sec\left(\frac{\pi}{4}+\theta\right)\sec\left(\frac{\pi}{4}-\theta\right) = 2\sec 2\theta$

14. $\tan\left(45^\circ+\frac{A}{2}\right) = \sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$

15. $\sin^2\left(\frac{\pi}{8}+\frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8}-\frac{A}{2}\right) = \frac{1}{\sqrt{2}}\sin A$

16. $\cos^2\alpha + \cos^2(\alpha + 120^\circ) + \cos^2(\alpha - 120^\circ) = \frac{3}{2}$

17. $\cos^4\frac{\pi}{8} + \cos^4\frac{3\pi}{8} + \cos^4\frac{5\pi}{8} + \cos^4\frac{7\pi}{8} = \frac{3}{2}$

18. $\sin^4\frac{\pi}{8} + \sin^4\frac{3\pi}{8} + \sin^4\frac{5\pi}{8} + \sin^4\frac{7\pi}{8} = \frac{3}{2}$

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16. TRIGONOMETRIC RATIOS OF SOME STANDARD ANGLES :

(i) $\sin 18° = \sin\frac{\pi}{10} = \frac{\sqrt{5}-1}{4} = \cos 72° = \cos\frac{2\pi}{5}$

(ii) $\cos 36° = \cos\frac{\pi}{5} = \frac{\sqrt{5}+1}{4} = \sin 54° = \sin\frac{3\pi}{10}$

(iii) $\sin 72° = \sin\frac{2\pi}{5} = \frac{\sqrt{10+2\sqrt{5}}}{4} = \cos 18° = \cos\frac{\pi}{10}$

(iv) $\sin 36° = \sin\frac{\pi}{5} = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54° = \cos\frac{3\pi}{10}$

(v) $\sin 15° = \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75° = \cos\frac{5\pi}{12}$

(vi) $\cos 15° = \cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75° = \sin\frac{5\pi}{12}$

(vii) $\tan 15° = \tan\frac{\pi}{12} = 2-\sqrt{3} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \cot 75° = \cot\frac{5\pi}{12}$

(viii) $\tan 75° = \tan\frac{5\pi}{12} = 2+\sqrt{3} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \cot 15° = \cot\frac{\pi}{12}$

(ix) $\tan(22.5°) = \tan\frac{\pi}{8} = \sqrt{2}-1 = \cot(67.5°) = \cot\frac{3\pi}{8}$

(x) $\tan(67.5°) = \tan\frac{3\pi}{8} = \sqrt{2}+1 = \cot(22.5°) = \cot\frac{\pi}{8}$

Illustration 13: Evaluate sin78° – sin66° – sin42° + sin6°.

Solution : The expression $= (\sin 78° - \sin 42°) - (\sin 66° - \sin 6°) = 2\cos(60°)\sin(18°) - 2\cos 36°.\sin 30°$

$$= \sin 18° - \cos 36° = \left(\frac{\sqrt{5}-1}{4}\right) - \left(\frac{\sqrt{5}+1}{4}\right) = -\frac{1}{2}$$

Do yourself - 13 :

1 Find the value of

(a) $\sin\frac{\pi}{10} + \sin\frac{13\pi}{10}$ (b) $\cos^2 48° - \sin^2 12°$

Evaluate :

2. $\sin^2 72° - \sin^2 60°$

3. $\sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}$

4. $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ$

5. $\cos\frac{\pi}{15}\cos\frac{2\pi}{15}\cos\frac{3\pi}{15}\cos\frac{4\pi}{15}\cos\frac{5\pi}{15}\cos\frac{6\pi}{15}\cos\frac{7\pi}{15}$

6. $16\cos\frac{2\pi}{15}\cos\frac{4\pi}{15}\cos\frac{8\pi}{15}\cos\frac{14\pi}{15}$

7. Two parallel chords of a circle, which are on the same side of the centre, subtend angles of 72° and 144° respectively at the centre. Prove that the perpendicular distance between the chords is half the radius of the circle.

8. In any circle prove that the chord which subtends 108° at the centre is equal to the sum of the two chords which subtend angles of 36° and 60°.

9. If $A = \cos\frac{\pi}{9}\cos\frac{4\pi}{9}\cos\frac{16\pi}{9}, B = \cos\frac{2\pi}{9}\cos\frac{8\pi}{9}\cos\frac{32\pi}{9}$, then value of $\left|\frac{1}{16A.B}\right|$ is

10. The value of $\cos\frac{\pi}{10}\cos\frac{2\pi}{10}\cos\frac{4\pi}{10}\cos\frac{8\pi}{10}\cos\frac{16\pi}{10}$ is

(A) $\frac{\sqrt{10+2\sqrt{5}}}{64}$ (B) $-\frac{\cos(\pi/10)}{16}$ (C) $\frac{\cos(\pi/10)}{16}$ (D) $-\frac{\sqrt{10+2\sqrt{5}}}{16}$

17. CONDITIONAL TRIGONOMETRIC IDENTITIES :

If $A + B + C = 180^\circ$, then

(i) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

(ii) $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

(iii) $\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$

(iv) $\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}$

(v) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

(vi) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

(vii) $\sin A + \sin B + \sin C = 4 \cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

(viii) $\cos A + \cos B + \cos C = 1 + 4 \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

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Illustration 14: In any triangle ABC, sin A – cos B = cos C, then angle B is (Where $\angle B > \angle C$)

(A) $\pi/2$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/6$

Solution : We have , sin A – cos B = cos C

sin A = cos B + cos C

$$\Rightarrow \quad 2\sin\frac{A}{2}\cos\frac{A}{2} = 2\cos\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)$$

$$\Rightarrow \quad 2\sin\frac{A}{2}\cos\frac{A}{2} = 2\cos\left(\frac{\pi-A}{2}\right)\cos\left(\frac{B-C}{2}\right) \qquad \because \quad A+B+C=\pi$$

$$\Rightarrow \quad 2\sin\frac{A}{2}\cos\frac{A}{2} = 2\sin\frac{A}{2}\cos\left(\frac{B-C}{2}\right)$$

$$\Rightarrow \quad \cos\frac{A}{2} = \cos\frac{B-C}{2} \text{ or } A = B - C \;\; ; \text{ But } A+B+C=\pi$$

Therefore $2B = \pi \Rightarrow B = \pi/2$ **Ans.(A)**

Illustration 15: If $A + B + C = \frac{3\pi}{2}$, then cos 2A + cos 2B + cos2C is equal to-

(A) 1 – 4cosA cosB cosC (B) 4 sinA sin B sinC

(C) 1 + 2cosA cosB cosC (D) 1 – 4 sinA sinB sinC

Solution : cos 2A + cos 2B + cos 2C = 2 cos (A + B) cos (A – B) + cos 2C

$$= 2\cos\left(\frac{3\pi}{2} - C\right)\cos(A-B) + \cos 2C \quad \because A+B+C = \frac{3\pi}{2}$$

$$= -2\sin C\cos(A-B) + 1 - 2\sin^2 C = 1 - 2\sin C\,[\cos(A-B) + \sin C]$$

$$= 1 - 2\sin C\left[\cos(A-B) + \sin\left(\frac{3\pi}{2} - (A+B)\right)\right]$$

$$= 1 - 2\sin C\,[\cos(A-B) - \cos(A+B)] = 1 - 4\sin A\sin B\sin C$$

Ans.(D)

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Do yourself - 14 :

1. If ABCD is a cyclic quadrilateral, then find the value of sinA + sinB – sinC – sinD

2. If A + B + C = $\frac{\pi}{2}$, then find the value of tanA tanB + tanBtanC + tanC tanA

If A + B + C = 180º, then prove that

3. sin2A + sin2B – sin2C = 4cosA cosB sinC

4. cos2A + cos2B – cos2C = 1 – 4sinA sinB cosC

5. $\sin A + \sin B - \sin C = 4\sin\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2}$

6. $\sin^2 A + \sin^2 B - \sin^2 C = 2\sin A\sin B\cos C$

7. $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A\cos B\cos C$.

8. $\cos^2 A + \cos^2 B - \cos^2 C = 1 - 2\sin A\sin B\cos C$

9. $\sin^2\frac{A}{2} + \sin^2\frac{B}{2} + \sin^2\frac{C}{2} = 1 - 2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

10. $\sin^2\frac{A}{2} + \sin^2\frac{B}{2} - \sin^2\frac{C}{2} = 1 - 2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

11. $\sin(B + 2C) + \sin(C + 2A) + \sin(A + 2B) = 4\sin\frac{B-C}{2}\sin\frac{C-A}{2}\sin\frac{A-B}{2}$.

12. $\sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2} - 1 = 4\sin\frac{\pi - A}{4}\sin\frac{\pi - B}{4}\sin\frac{\pi - C}{4}$

13. $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = 8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

14. sin(B + C – A) + sin(C + A – B) + sin(A + B – C) = 4sinA sinB sinC

15. If x + y + z = xyz prove that

(a) $\frac{3x - x^3}{1 - 3x^2} + \frac{3y - y^3}{1 - 3y^2} + \frac{3z - z^3}{1 - 3z^2} = \frac{3x - x^3}{1 - 3x^2}\cdot\frac{3y - y^3}{1 - 3y^2}\cdot\frac{3z - z^3}{1 - 3z^2}$

(b) $x(1 - y^2)(1 - z^2) + y(1 - z^2)(1 - x^2) + z(1 - x^2)(1 - y^2) = 4xyz$

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18. MAXIMUM & MINIMUM VALUES OF TRIGONOMETRIC EXPRESSIONS :

(i) $a\cos\theta + b\sin\theta$ will always lie in the interval $[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}]$ i.e. the maximum and minimum values are $\sqrt{a^2+b^2}, -\sqrt{a^2+b^2}$ respectively.

(ii) Minimum value of $a^2\tan^2\theta + b^2\cot^2\theta = 2ab$ where a, b > 0

(iii) $-\sqrt{a^2+b^2+2ab\cos(\alpha-\beta)} \le a\cos(\alpha+\theta) + b\cos(\beta+\theta) \le \sqrt{a^2+b^2+2ab\cos(\alpha-\beta)}$ where α and β areknown angles.

(iv) In case a quadratic in $\sin\theta$ & $\cos\theta$ is given then the maximum or minimum values can be obtained by making perfect square.

Illustration 16: Prove that : $-4 \le 5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right) + 3 \le 10$, for all values of θ.

Solution : We have, $5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right) = 5\cos\theta + 3\cos\theta\cos\frac{\pi}{3} - 3\sin\theta\sin\frac{\pi}{3} = \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta$

Since, $-\sqrt{\left(\frac{13}{2}\right)^2 + \left(-\frac{3\sqrt{3}}{2}\right)^2} \le \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta \le \sqrt{\left(\frac{13}{2}\right)^2 + \left(-\frac{3\sqrt{3}}{2}\right)^2}$

$\Rightarrow \quad -7 \le \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta \le 7$

$\Rightarrow \quad -7 \le 5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right) \le 7$ for all θ.

$\Rightarrow \quad -7+3 \le 5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right) + 3 \le 7+3$ for all θ.

$\Rightarrow \quad -4 \le 5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right) + 3 \le 10$ for all θ.

Illustration 17: Find the maximum value of $1 + \sin\left(\frac{\pi}{4}+\theta\right) + 2\cos\left(\frac{\pi}{4}-\theta\right)$ -

(A) 1 (B) 2 (C) 3 (D) 4

Solution : We have $1 + \sin\left(\frac{\pi}{4}+\theta\right) + 2\cos\left(\frac{\pi}{4}-\theta\right)$

$= 1 + \frac{1}{\sqrt{2}}(\cos\theta + \sin\theta) + \sqrt{2}(\cos\theta + \sin\theta) = 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)(\cos\theta + \sin\theta)$

$= 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right) . \sqrt{2}\cos\left(\theta - \frac{\pi}{4}\right)$

$\therefore$ maximum value $= 1 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right) . \sqrt{2} = 4$ **Ans. (D)**

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Do yourself - 15 :

1. Find maximum and minimum value of $5\cos\theta + 3\sin\left(\theta + \frac{\pi}{6}\right)$ for all real values of θ.

2. Find the minimum value of $\cos\theta + \cos2\theta$ for all real values of θ.

3. Find maximum and minimum value of $\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2$.

4. Find the maximum and minimum values of

(i) $\cos2x + \cos^2x$ (ii) $\cos^2\left(\frac{\pi}{4} + x\right) + (\sin x - \cos x)^2$

5. If $\alpha + \beta = 90^\circ$, then find the maximum value of $\sin\alpha.\sin\beta$.

6. Find the maximum and minimum value of $1 + 2\sin x + 3\cos^2x$

7. Find the minimum value of $4\sec^2x + 9\operatorname{cosec}^2x$

8. Find the maximum and minimum value of $9\cos^2x + 48\sin x.\cos x - 5\sin^2x - 2$

9. Find the maximum and minimum value of $2\sin\left(\theta + \frac{\pi}{6}\right) + \sqrt{3}\cos\left(\theta - \frac{\pi}{6}\right)$

10. Find minimum value of (i) $3\sin^2x + 27\operatorname{cosec}^2x$ (ii) $27\sin^2x + 3\operatorname{cosec}^2x$

19. IMPORTANT RESULTS :

(i) $\sin\theta \sin(60^\circ - \theta)\sin(60^\circ + \theta) = \frac{1}{4}\sin3\theta$

(ii) $\cos\theta.\cos(60^\circ - \theta)\cos(60^\circ + \theta) = \frac{1}{4}\cos3\theta$

(iii) $\tan\theta \tan(60^\circ - \theta)\tan(60^\circ + \theta) = \tan3\theta$

(iv) $\cot\theta \cot(60^\circ - \theta)\cot(60^\circ + \theta) = \cot3\theta$

(v) (a) $\sin^2\theta + \sin^2(60^\circ + \theta) + \sin^2(60^\circ - \theta) = \frac{3}{2}$

(b) $\cos^2\theta + \cos^2(60^\circ + \theta) + \cos^2(60^\circ - \theta) = \frac{3}{2}$

(c) $\tan\theta + \tan(60^\circ + \theta) + \tan(120^\circ + \theta) = 3\tan3\theta$

(vi) (a) If $\tan A + \tan B + \tan C = \tan A \tan B \tan C$, then $A + B + C = n\pi, n \in I$

(b) If $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$, then $A + B + C = (2n + 1)\frac{\pi}{2}, n \in I$

(vii) $\cos\theta \cos2\theta \cos4\theta \ldots \cos(2^{n-1}\theta) = \frac{\sin(2^n\theta)}{2^n\sin\theta}$

(viii) (a) $\cot A - \tan A = 2\cot2A$ (b) $\cot A + \tan A = 2\operatorname{cosec}2A$

(ix) $\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots \sin(\alpha + \overline{n-1}\,\beta) = \dfrac{\sin\left\{\alpha + \left(\frac{n-1}{2}\right)\beta\right\}\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$

(x) $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha + 2\beta) + \ldots + \cos(\alpha + \overline{n-1}\,\beta) = \dfrac{\cos\left\{\alpha + \left(\frac{n-1}{2}\right)\beta\right\}\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$

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Do yourself - 16 :

1 Evaluate $\sin\frac{\pi}{n}+\sin\frac{3\pi}{n}+\sin\frac{5\pi}{n}+.....$ to n terms

2. Prove that : $\sin\theta + \sin3\theta + \sin5\theta +...+ \sin(2n-1)\theta = \frac{\sin^2 n\theta}{\sin\theta}$

3. Find the average of sin2º, sin4º,sin6º,..., sin180º

4. Prove that : $\cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}=\frac{1}{2}$

5. Find sum of the following series :

(a) $\cos\frac{\pi}{2n+1}+\cos\frac{3\pi}{2n+1}+\cos\frac{5\pi}{2n+1}+...$ up to n terms.

(b) $\sin2\alpha + \sin3\alpha + \sin4\alpha +...+ \sin n\alpha$, where $(n+2)\alpha = 2\pi$

6. If $S=\cos^2\frac{\pi}{n}+\cos^2\frac{2\pi}{n}+...+\cos^2\frac{(n-1)\pi}{n}$, then S equals (where $n \geq 2$, $n \in N$)

(A) $\frac{n}{2(n+1)}$ (B) $\frac{1}{2(n-1)}$ (C) $\frac{n-2}{2}$ (D) $\frac{n}{2}$

Miscellaneous Illustration :

Illustration 18 : Prove that

$$\tan\alpha + 2\tan2\alpha + 2^2\tan2^2\alpha + ... + 2^{n-1}\tan 2^{n-1}\alpha + 2^n\cot 2^n\alpha = \cot\alpha$$

Solution : We know $\tan\theta = \cot\theta - 2\cot 2\theta$ .....(i)

Putting $\theta = \alpha, 2\alpha, 2^2\alpha$, ...in (i), we get

$\tan\alpha = (\cot\alpha - 2\cot 2\alpha)$

$2(\tan 2\alpha) = 2(\cot 2\alpha - 2\cot 2^2\alpha)$

$2^2(\tan 2^2\alpha) = 2^2(\cot 2^2\alpha - 2\cot 2^3\alpha)$

.....

$2^{n-1}(\tan 2^{n-1}\alpha) = 2^{n-1}(\cot 2^{n-1}\alpha - 2\cot 2^n\alpha)$

Adding,

$$\tan\alpha + 2\tan2\alpha + 2^2\tan^2\alpha + ... + 2^{n-1}\tan 2^{n-1}\alpha = \cot\alpha - 2^n\cot 2^n\alpha$$

$$\therefore \quad \tan\alpha + 2\tan2\alpha + 2^2\tan^2\alpha + ... + 2^{n-1}\tan 2^{n-1}\alpha + 2^n\cot 2^n\alpha = \cot\alpha$$

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Illustration 19: If A,B,C and D are angles of a quadrilateral and $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\sin\frac{D}{2}=\frac{1}{4}$, prove that A = B = C = D = π/2.

Solution : $\left(2\sin\frac{A}{2}\sin\frac{B}{2}\right)\left(2\sin\frac{C}{2}\sin\frac{D}{2}\right)=1$

$$\Rightarrow \quad \left\{\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right\}\left\{\cos\left(\frac{C-D}{2}\right)-\cos\left(\frac{C+D}{2}\right)\right\}=1$$

Since, A + B = 2π – (C + D), the above equation becomes,

$$\Rightarrow \quad \left\{\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right\}\left\{\cos\left(\frac{C-D}{2}\right)+\cos\left(\frac{A+B}{2}\right)\right\}=1$$

$$\Rightarrow \cos^2\left(\frac{A+B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\left\{\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{C-D}{2}\right)\right\}+1-\cos\left(\frac{A-B}{2}\right)\cos\left(\frac{C-D}{2}\right)=0$$

This is a quadratic equation in $\cos\left(\frac{A+B}{2}\right)$ which has real roots.

$$\Rightarrow \quad \left\{\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{C-D}{2}\right)\right\}^2-4\left\{1-\cos\left(\frac{A-B}{2}\right).\cos\left(\frac{C-D}{2}\right)\right\}\geq 0$$

$$\left(\cos\frac{A-B}{2}+\cos\frac{C-D}{2}\right)^2\geq 4$$

$\Rightarrow \quad \cos\frac{A-B}{2}+\cos\frac{C-D}{2}\geq 2$, Now both $\cos\frac{A-B}{2}$ and $\cos\frac{C-D}{2}\leq 1$

$$\Rightarrow \quad \cos\frac{A-B}{2}=1 \;\&\; \cos\frac{C-D}{2}=1$$

$$\Rightarrow \quad \frac{A-B}{2}=0=\frac{C-D}{2}$$

$\Rightarrow$ A = B, C = D.

Similarly A = C, B = D ⇒ A = B = C = D = π/2

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EXERCISE (O-1)

Straight Objective Type

1. If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is -

(A) 0 (B) 2 (C) 1 (D) 3

TR0001

2. $2(\sin^6\theta + \cos^6\theta) - 3(\sin^4\theta + \cos^4\theta) + 1$ is equal to -

(A) 2 (B) 0 (C) 4 (D) 6

TR0002

3. If $x = y\cos\frac{2\pi}{3} = z\cos\frac{4\pi}{3}$, then $xy + yz + zx =$

(A) –1 (B) 0 (C) 1 (D) 2

TR0003

4. The value of $\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + ... + \sin 360^\circ$ is equal to -

(A) 0 (B) 1 (C) $\sqrt{3}$ (D) 2

TR0004

5. If $\cos(\alpha + \beta) + \sin(\alpha - \beta) = 0$ and $2010\tan\beta + 1 = 0$, then $\tan\alpha$ is equal to

(A) 1 (B) –1 (C) 2010 (D) $\frac{1}{2010}$

TR0005

6. If $\cos x + \cos y + \cos\alpha = 0$ and $\sin x + \sin y + \sin\alpha = 0$, then $\cot\left(\frac{x+y}{2}\right) =$

(A) $\sin\alpha$ (B) $\cos\alpha$ (C) $\cot\alpha$ (D) $2\sin\alpha$

TR0006

7. $\frac{\sin(A-C) + 2\sin A + \sin(A+C)}{\sin(B-C) + 2\sin B + \sin(B+C)}$ is equal to -

(A) $\tan A$ (B) $\frac{\sin A}{\sin B}$ (C) $\frac{\cos A}{\cos B}$ (D) $\frac{\sin C}{\cos B}$

TR0007

8. The expression $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta}$ is equals -

(A) $\tan\theta$ (B) $\tan 2\theta$ (C) $\sin 2\theta$ (D) $\cos 2\theta$

TR0008

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9. $\dfrac{1+\sin 2\theta+\cos 2\theta}{1+\sin 2\theta-\cos 2\theta}=$

(A) $\frac{1}{2}\tan\theta$ (B) $\frac{1}{2}\cot\theta$ (C) $\tan\theta$ (D) $\cot\theta$

TR0009

10. If $\tan x + \tan y = 25$ and $\cot x + \cot y = 30$, then the value of $\tan(x + y)$ is

(A) 150 (B) 200 (C) 250 (D) 100

TR0010

11. If $A = \tan 6^\circ \tan 42^\circ$ and $B = \cot 66^\circ \cot 78^\circ$, then -

(A) A = 2B (B) A = 1/3B (C) A = B (D) 3A = 2B

TR0011

12. In a right angled triangle the hypotenuse is $2\sqrt{2}$ times the perpendicular drawn from the opposite vertex. Then the other acute angles of the triangle are

(A) $\frac{\pi}{3}$ and $\frac{\pi}{6}$ (B) $\frac{\pi}{8}$ and $\frac{3\pi}{8}$ (C) $\frac{\pi}{4}$ and $\frac{\pi}{4}$ (D) $\frac{\pi}{5}$ and $\frac{3\pi}{10}$

TR0012

13. If $\tan\alpha = (1+2^{-x})^{-1}$, $\tan\beta = (1+2^{x+1})^{-1}$, then $\alpha + \beta =$

(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $\pi/2$

TR0013

14. If $3\sin\alpha = 5\sin\beta$, then $\dfrac{\tan\frac{\alpha+\beta}{2}}{\tan\frac{\alpha-\beta}{2}}=$

(A) 1 (B) 2 (C) 3 (D) 4

TR0014

15. If $\cos\alpha = \dfrac{2\cos\beta-1}{2-\cos\beta}$ then $\tan\frac{\alpha}{2}\cdot\cot\frac{\beta}{2}$ has the value equal to {where $\alpha, \beta \in (0, \pi)$}

(A) 2 (B) $\sqrt{2}$ (C) 3 (D) $\sqrt{3}$

TR0015

16. If $\tan B = \dfrac{n\sin A\cos A}{1-n\cos^2 A}$ then $\tan(A + B)$ equals

(A) $\dfrac{\sin A}{(1-n)\cos A}$ (B) $\dfrac{(n-1)\cos A}{\sin A}$ (C) $\dfrac{\sin A}{(n-1)\cos A}$ (D) $\dfrac{\sin A}{(n+1)\cos A}$

TR0016

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17. The value of $\operatorname{cosec}\frac{\pi}{18} - \sqrt{3}\sec\frac{\pi}{18}$ is a

(A) surd
(B) rational which is not integral
(C) negative integer
(D) natural number

TR0017

18. The value of $\cot x + \cot(60^\circ + x) + \cot(120^\circ + x)$ is equal to :

(A) $\cot 3x$
(B) $\tan 3x$
(C) $3\tan 3x$
(D) $\frac{3 - 9\tan^2 x}{3\tan x - \tan^3 x}$

TR0018

19. If $\frac{5\pi}{2} < x < 3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}}$ is

(A) $-\cot\frac{x}{2}$
(B) $\cot\frac{x}{2}$
(C) $\tan\frac{x}{2}$
(D) $-\tan\frac{x}{2}$

TR0019

20. The value of $\cot 7\frac{1}{2}^\circ + \tan 67\frac{1}{2}^\circ - \cot 67\frac{1}{2}^\circ - \tan 7\frac{1}{2}^\circ$ is :

(A) a rational number
(B) irrational number
(C) $2(3 + 2\sqrt{3})$
(D) $2(3 - \sqrt{3})$

TR0020

21. If $x + y = 3 - \cos 4\theta$ and $x - y = 4\sin 2\theta$ then

(A) $x^4 + y^4 = 9$
(B) $\sqrt{x} + \sqrt{y} = 16$
(C) $x^3 + y^3 = 2(x^2 + y^2)$
(D) $\sqrt{x} + \sqrt{y} = 2$

TR0021

22. If $A = \sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7}$ and $B = \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7}$ then $\sqrt{A^2 + B^2}$ is equal to

(A) 1
(B) $\sqrt{2}$
(C) 2
(D) $\sqrt{3}$

TR0022

23. If tan A + tan B + tan C = tan A. tan B. tan C, then -

(A) A,B,C must be angles of a triangle
(B) the sum of any two of A,B,C is equal to the third
(C) A + B + C must be n integral multiple of π
(D) None of these

TR0023

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24. If $f(x) = \frac{\sin 3x}{\sin x}$, $x \neq n\pi$, then the range of values of f(x) for real values of x is -

(A) $[-1,3]$ (B) $(-\infty,-1]$ (C) $(3, +\infty)$ (D) $[-1,3)$

TR0024

25. Maximum and minimum value of $2\sin^2\theta - 3\sin\theta + 2$ is -

(A) $\frac{1}{4}, -\frac{7}{4}$ (B) $\frac{1}{4}, \frac{21}{4}$ (C) $\frac{21}{4}, -\frac{3}{4}$ (D) $7, \frac{7}{8}$

TR0025

26. For $\theta \in (0, \pi/2)$, the maximum value of $\sin\left(\theta+\frac{\pi}{6}\right)+\cos\left(\theta+\frac{\pi}{6}\right)$ is attained at $\theta =$

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

TR0026

27. Minimum value of the expression $\cos^2\theta - (6 \sin\theta \cos\theta) + 3 \sin^2\theta + 2$, is -

(A) $4+\sqrt{10}$ (B) $4-\sqrt{10}$ (C) 0 (D) 4

TR0027

28. The value of $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}$ is :-

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{2}$ (D) 1

TR0028

29. The exact value of $\frac{96 \sin 80° \sin 65° \sin 35°}{\sin 20° + \sin 50° + \sin 110°}$ is equal to

(A) 12 (B) 24 (C) –12 (D) 48

TR0029

30. If m and n are positive integers satisfying

$1 + \cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta + \cos 10\theta = \frac{\cos m\theta \cdot \sin n\theta}{\sin\theta}$ then $(m + n)$ is equal to

(A) 9 (B) 10 (C) 11 (D) 12

TR0030

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EXERCISE (O-2)

Multiple Correct Answer Type

1. If cosx = tanx, then which of the following is/are true ?

(A) $\frac{1}{\sin x}+\cos^4 x=1$ (B) $\frac{1}{\sin x}+\cos^4 x=2$ (C) $\cos^4x + \cos^2x = 1$ (D) $\cos^4x + \cos^2x = 2$

TR0031

2. If $\tan A=-\frac{1}{2}$ and $\tan B=-\frac{1}{3}$, then A + B =

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{7\pi}{4}$

TR0032

3. If $\cos(A-B)=\frac{3}{5}$ and tanA tanB = 2, then which of the following is/are correct ?

(A) cosA cosB = $-\frac{1}{5}$ (B) $\sin A\sin B=\frac{2}{5}$ (C) $\cos(A+B)=-\frac{1}{5}$ (D) $\sin A\cos B=\frac{4}{5}$

TR0033

4. If $\sin t+\cos t=\frac{1}{5}$, then $\tan\frac{t}{2}$ is equal to

(A) –1 (B) $-\frac{1}{3}$ (C) 2 (D) $-\frac{1}{6}$

TR0034

5. If $\tan^2\alpha + 2\tan\alpha.\tan2\beta = \tan^2\beta + 2\tan\beta.\tan2\alpha$, then we may have

(A) $\tan^2\alpha + 2\tan\alpha.\tan2\beta = 0$ (B) $\tan\alpha + \tan\beta = 0$

(C) $\tan^2\beta + 2\tan\beta.\tan2\alpha = 1$ (D) $\tan\alpha = \tan\beta$

TR0035

6. If $3\sin\beta = \sin(2\alpha + \beta)$, then $\tan(\alpha + \beta) - 2\tan\alpha$ is

(A) independent of α (B) independent of β

(C) dependent of both α and β (D) independent of α but dependent of β

TR0036

7. If $L = \cos^2 84^\circ + \cos^2 36^\circ + \cos 36^\circ \cos 84^\circ$

$M = \cot 73^\circ \cot 47^\circ \cot 13^\circ$

$N = 4 \sin 156^\circ \sin 84^\circ \sin36^\circ$, then which of the following option(s) is(are) correct ?

(A) L < 1 (B) M > tan 2 (C) $N > \sin\frac{\pi}{4}$ (D) 0 < LMN

TR0037

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8. If $\frac{\cos 3x}{\cos x} = \frac{1}{3}$ for some angle x, $0 \le x \le \frac{\pi}{2}$, which of the following is/are true ?

(A) $\frac{\sin 3x}{\sin x} = \frac{7}{3}$ (B) $\cos 2x = \frac{2}{3}$ (C) $\tan x = \frac{1}{\sqrt{5}}$ (D) $\sin 2x = \frac{2\sqrt{5}}{6}$

TR0038

9. If $\alpha + \beta + \gamma = 2\pi$, then

(A) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = \tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(B) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} + \tan\frac{\beta}{2}\tan\frac{\gamma}{2} + \tan\frac{\gamma}{2}\tan\frac{\alpha}{2} = 1$

(C) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = -\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(D) $\tan\frac{\alpha}{4}\tan\frac{\beta}{4} + \tan\frac{\beta}{4}\tan\frac{\gamma}{4} + \tan\frac{\gamma}{4}\tan\frac{\alpha}{4} = 1$

TR0039

10. If $x + y = z$, then $\cos^2 x + \cos^2 y + \cos^2 z - 2\cos x \cos y \cos z$ is equal to

(A) $\cos^2 z$ (B) $\sin^2 z$ (C) $\cos(x + y - z)$ (D) 1

TR0040

11. If A,B,C are angles of a triangle ABC and tanA tanC = 3; tan B tanC = 6 then, which is (are) correct?

(A) $A = \frac{\pi}{4}$ (B) tanA tanB = 2 (C) $\frac{\tan A}{\tan C} = 3$ (D) tanB = 2 tanA

TR0041

12. Which of the following is/are true ?

(A) $\sin\left(\frac{\pi}{4} + \theta\right)\sin\left(\frac{\pi}{4} - \theta\right) = \frac{\cos 2\theta}{2}$

(B) In a ΔABC, if tanA = 2, tanB = 3, then tanC = 1

(C) Minimum value of $4\tan^2\theta + \cot^2\theta$ is 4 (wherever defined)

(D) $2(\sin^6 x + \cos^6 x) - 3(\sin^4 x + \cos^4 x) = -1$

TR0042

13. Let $f_n(\theta) = \sum_{r=0}^{n} \frac{1}{4^r}\sin^4\left(2^r\theta\right)$. Then which of the following alternative(s) is/are correct ?

(A) $f_2\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ (B) $f_3\left(\frac{\pi}{8}\right) = \frac{2+\sqrt{2}}{4}$ (C) $f_4\left(\frac{3\pi}{2}\right) = 1$ (D) $f_5(\pi) = 0$

TR0043

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14. Let $y = \dfrac{\cos x + \cos 2x + \cos 3x + \cos 4x + \cos 5x + \cos 6x + \cos 7x}{\sin x + \sin 2x + \sin 3x + \sin 4x + \sin 5x + \sin 6x + \sin 7x}$, then which of the following hold good?

(A) The value of y when $x = \pi/8$ is not defined. (B) The value of y when $x = \pi/16$ is 1.

(C) The value of y when $x = \pi/32$ is $\sqrt{2} - 1$. (D) The value of y when $x = \pi/48$ is $2 + \sqrt{3}$.

TR0044

15. The value of $\dfrac{(\cos 11° + \sin 11°)}{(\cos 11° - \sin 11°)}$ is

(A) –tan 304° (B) tan 56° (C) cot 214° (D) cot 34°

TR0111

16. The value of $\dfrac{\sin x + \cos x}{\cos^3 x} =$

(A) $1 + \tan x + \tan^2 x - \tan^3 x$ (B) $1 + \tan x + \tan^2 x + \tan^3 x$

(C) $1 - \tan x + \tan^2 x + \tan^3 x$ (D) $(1 + \tan x) \sec^2 x$

TR0112

17. If (sec A + tan A) (sec B + tan B) (sec C + tan C) = (sec A – tan A) (sec B – tan B) (sec C – tan C) then each side can be

(A) 1 (B) –1 (C) 0 (D) none

TR0113

18. Which of the following is correct ?

(A) sin 1° > sin 1 (B) sin 1° < sin 1 (C) cos 1° > cos 1 (D) cos 1° < cos 1

TR0114

19. If $P_n = \cos^n\theta + \sin^n\theta$ and $Q_n = \cos^n\theta - \sin^n\theta$, then which of the following is/are true.

(A) $P_n - P_{n-2} = -\sin^2\theta \cos^2\theta\, P_{n-4}$ (B) $Q_n - Q_{n-2} = -\sin^2\theta \cos^2\theta\, Q_{n-4}$

(C) $P_4 = 1 - 2 \sin^2\theta \cos^2\theta$ (D) $Q_4 = \cos^2\theta - \sin^2\theta$

TR0115

20. For $0 < \theta < \pi/2$, $\tan\theta + \tan 2\theta + \tan 3\theta = 0$ if

(A) $\tan \theta = 0$ (B) $\tan 2\theta = 0$

(C) $\tan 3\theta = 0$ (D) $\tan\theta \tan 2\theta = 2$

TR0116

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EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question Nos. 1 to 3

Consider the polynomial $P(x) = (x - \cos 36°)(x - \cos 84°)(x - \cos 156°)$

1. The coefficient of x^2 is

(A) 0 (B) 1 (C) $-\frac{1}{2}$ (D) $\frac{\sqrt{5}-1}{2}$

TR0045

2. The coefficient of x is

(A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $-\frac{3}{4}$ (D) zero

TR0045

3. The absolute term in P(x) has the value equal to

(A) $\frac{\sqrt{5}-1}{4}$ (B) $\frac{\sqrt{5}-1}{16}$ (C) $\frac{\sqrt{5}+1}{16}$ (D) $\frac{1}{16}$

TR0045

Paragraph for Question 4 to 6

If $\cos\alpha + \cos\beta = a$ and $\sin\alpha + \sin\beta = b$ and θ is arithmetic mean between α and β, then

$\sin 2\theta + \cos 2\theta = 1 + \frac{nb(a-b)}{a^2+b^2}$ where, $n \in I$

On the basis of above information answer the following :

4. The value of n is -

(A) 0 (B) 1 (C) 2 (D) –2

TR0117

5. If for n obtained in above question, $\sin^n A = x$, then sinA sin2A sin3A sin4A is a polynomial in x of degree -

(A) 5 (B) 6 (C) 7 (D) 8

TR0118

6. If degree of polynomial obtained in previous question is p and $(p - 5) + \sin\theta$, $\cos\theta$, $\tan\theta$ are in G.P., then $\cos^9\theta + \cos^6\theta + 3\cos^5\theta - 1 =$

(A) p (B) 0 (C) 2p (D) –1

TR0119

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Paragraph for Question 7 and 8

Let a,b,c are respectively the sines and p,q,r are respectively the cosines of $\alpha, \alpha + \frac{2\pi}{3}$ and $\alpha + \frac{4\pi}{3}$, then

7. The value of (a + b + c) + (ab + bc + ca) is less than or equal to

(A) 0 (B) $\frac{3}{4}$ (C) 1 (D) $-\frac{3}{4}$

TR0046

8. The value of (qc – rb) is equal to $\frac{\sqrt{\alpha}}{\beta}$; α, β are coprime then -

(A) $\alpha + \beta = 5$ (B) $\alpha - \beta = -1$ (C) $\alpha - \beta = 1$ (D) $\alpha + \beta = 7$

TR0046

Matrix Match Type

9.

	List-I		List-II
(I)	The value of tan 20° . tan 40° . tan 60° . tan 80° is	(P)	Odd natural Number
			TR0054
(II)	The value of $4\left(\sin^4\frac{\pi}{16}+\sin^4\frac{3\pi}{16}+\sin^4\frac{5\pi}{16}+\sin^4\frac{7\pi}{16}\right)$ is	(Q)	Even natural Number
			TR0055
(III)	$\tan\alpha + 2\tan2\alpha + 4\tan4\alpha + 8\cot8\alpha = \lambda\cot\lambda\alpha$ then λ is	(R)	3
(IV)	$(4\cos^2 9° - 3)(4\cos^2 27° - 3) = \tan k°$, then k is	(S)	6
			TR0057
		(T)	1
		(U)	9

Which of the following is **CORRECT** ?

(A) (I) → (P,R) ; (II) → (Q,S) ; (III) → (P,T) ; (IV) → (P, U)

(B) (I) → (P,T) ; (II) → (Q,R) ; (III) → (P,U) ; (IV) → (R, U)

(C) (I) → (R,S) ; (II) → (T,U) ; (III) → (P,T) ; (IV) → (P, Q)

(D) (I) → (P,R) ; (II) → (P,U) ; (III) → (Q,S) ; (IV) → (S, T)

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10. In the following matrix match **Column-I** has some quantities and **Column-II** has some comments or other quantities

Match the each element in **Column-I** with corresponding element(s) in **Column-II**

Column-I	**Column-II**
(A) The value of $\left\|4\left(2\cos^3\frac{\pi}{7}-\cos^2\frac{\pi}{7}-\cos\frac{\pi}{7}\right)\right\|$ is	(P) 4
(B) If A + B + C = π and cosA = cosB. cosC then tanB. tanC has the value equal to	(Q) 8
(C) $4\left(\frac{\cos 20^\circ+8\sin 70^\circ\sin 50^\circ\sin 10^\circ}{\sin^2 80^\circ}\right)$ is equal to	(R) 2
(D) The maximum value of $12\sin\theta - 9\sin^2\theta$ is	(S) 1
	(T) 6

TR0052

EXERCISE (O-4)

Numerical Grid Type

1. If acute angle A = 3B and $\sin A = \frac{4}{5}$ then $\left|\frac{3\sec B - 4\operatorname{cosec} B}{2}\right|$ is

TR0047

2. If the value of $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \cos\frac{7\pi}{7} = -\frac{\ell}{2}$. Find the value of ℓ.

TR0048

3. If $\cot(\theta - \alpha)$, $3\cot\theta$, $\cot(\theta + \alpha)$ are in AP (where, $\theta \neq \frac{n\pi}{2}, \alpha \neq k\pi, n, k \in I$), then $\frac{2\sin^2\theta}{\sin^2\alpha}$ is equal to

TR0049

4. If $k_1 = \tan 27\theta - \tan\theta$ and $k_2 = \frac{\sin\theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta}$ then $\left(\frac{k_1}{k_2}\right)$ is equal to

TR0050

5. The value of the expression $\frac{1 - 4\sin 10^\circ \sin 70^\circ}{2\sin 10^\circ}$ is

TR0051

6. If $\cos(\alpha + \beta) = \frac{4}{5}$; $\sin(\alpha - \beta) = \frac{5}{13}$ & α, β lie between 0 & $\frac{\pi}{4}$, then find the value of $\tan 2\alpha$.

TR0059

7. If the value of the expression $\sin 25^\circ . \sin 35^\circ . \sin 85^\circ$ can be expressed as $\frac{\sqrt{a} + \sqrt{b}}{c}$, where $a, b, c \in N$ and are in their lowest form, find the value of $\left(\frac{a + b + c}{18}\right)$.

TR0060

8. Given that $(1 + \tan 1^\circ)(1 + \tan 2^\circ)....(1 + \tan 45^\circ) = 2^n$, find $\left(\frac{n}{4}\right)$.

TR0067

9. The value of $\frac{1}{100}(\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ)$.

TR0074

10. Calculate : $\frac{2\cos 40^\circ - \cos 20^\circ}{\sin 20^\circ}$

TR0064

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EXERCISE (JM)

1. In a ΔPQR, if $3 \sin P + 4 \cos Q = 6$ and $4 \sin Q + 3 \cos P = 1$, then the angle R is equal to : **[AIEEE-2012]**

(1) $\frac{3\pi}{4}$ (2) $\frac{5\pi}{6}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{4}$

TR0083

2. The expression $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$ can be written as **[JEE-MAIN 2013]**

(1) sinA cosA + 1 (2) secA cosecA + 1 (3) tanA + cotA (4) secA + cosecA

TR0084

3. Let $f_K(x) = \frac{1}{k}\left(\sin^k x+\cos^k x\right)$ where $x \in R$ and $k \geq 1$. Then $f_4(x) - f_6(x)$ equals : **[JEE-MAIN 2014]**

(1) $\frac{1}{6}$ (2) $\frac{1}{3}$ (3) $\frac{1}{4}$ (4) $\frac{1}{12}$

TR0085

4. If $5(\tan^2 x - \cos^2 x) = 2\cos 2x + 9$, then the value of cos4x is :- **[JEE-MAIN 2017]**

(1) $-\frac{7}{9}$ (2) $-\frac{3}{5}$ (3) $\frac{1}{3}$ (4) $\frac{2}{9}$

TR0086

5. For any $\theta \in \left(\frac{\pi}{4},\frac{\pi}{2}\right)$, the expression $3(\sin\theta - \cos\theta)^4 + 6(\sin\theta + \cos\theta)^2 + 4\sin^6\theta$ equals : **[JEE-MAIN 2019]**

(1) $13 - 4\cos^6\theta$ (2) $13 - 4\cos^4\theta + 2\sin^2\theta\cos^2\theta$

(3) $13 - 4\cos^2\theta + 6\cos^4\theta$ (4) $13 - 4\cos^2\theta + 6\sin^2\theta\cos^2\theta$

TR0087

6. The value of $\cos\frac{\pi}{2^2}\cdot\cos\frac{\pi}{2^3}\cdot.....\cdot\cos\frac{\pi}{2^{10}}\cdot\sin\frac{\pi}{2^{10}}$ is : **[JEE-MAIN 2019]**

(1) $\frac{1}{256}$ (2) $\frac{1}{2}$ (3) $\frac{1}{512}$ (4) $\frac{1}{1024}$

TR0088

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7. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ for k = 1, 2, 3, ... Then for all $x \in R$, the value of $f_4(x) - f_6(x)$ is equal to :- **[JEE-MAIN 2019]**

(1) $\frac{5}{12}$ (2) $\frac{-1}{12}$ (3) $\frac{1}{4}$ (4) $\frac{1}{12}$

TR0089

8. The maximum value of $3\cos\theta+5\sin\left(\theta-\frac{\pi}{6}\right)$ for any real value of θ is : **[JEE-MAIN 2019]**

(1) $\sqrt{19}$ (2) $\frac{\sqrt{79}}{2}$ (3) $\sqrt{31}$ (4) $\sqrt{34}$

TR0090

9. Let α and β be two real roots of the equation $(k + 1)\tan^2x - \sqrt{2}\,\lambda\tan x = (1 - k)$, where $k(\neq -1)$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50$, then a value of λ is ; **[JEE-MAIN 2020]**

(1) 5 (2) 10 (3) $5\sqrt{2}$ (4) $10\sqrt{2}$

TR0091

10. The value of $\cos^3\left(\frac{\pi}{8}\right)\cdot\cos\left(\frac{3\pi}{8}\right)+\sin^3\left(\frac{\pi}{8}\right)\cdot\sin\left(\frac{3\pi}{8}\right)$ is : **[JEE-MAIN 2020]**

(1) $\frac{1}{4}$ (2) $\frac{1}{\sqrt{2}}$ (3) $\frac{1}{2\sqrt{2}}$ (4) $\frac{1}{2}$

TR0092

11. If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of $27\sec^6\alpha + 8\operatorname{cosec}^6\alpha$ is equal to :

(1) 350 (2) 500 (3) 400 (4) 250

[JEE-MAIN 2021]

TR0120

12. 16sin(20°) sin(40°) sin(80°) is equal to :

(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) 3 (D) $4\sqrt{3}$

[JEE-MAIN 2022]

TR0121

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EXERCISE (JA)

1. Let $\theta \in \left(0, \frac{\pi}{4}\right)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then -

[JEE 06, 3M,–1M]

(A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$ (C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

TR0093

2. **(a)** If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then **[JEE 2009, 4 + 4]**

(A) $\tan^2 x = \frac{2}{3}$ (B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$ (D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

TR0094

(b) For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right)\operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is (are) -

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

TR0095

3. **(a)** The maximum value of the expression $\frac{1}{\sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta}$ is

TR0096

(b) Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3}+1$ apart. If the chords subtend at the center, angles of $\frac{\pi}{k}$ and $\frac{2\pi}{k}$ where $k > 0$, then the value of [k] is -

[**Note :** [k] denotes the largest integer less than or equal to k] **[JEE 2010, 3+3]**

TR0097

4. Let $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$ and $Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$ be two sets. Then

(A) $P \subset Q$ and $Q - P \neq \varnothing$ (B) $Q \not\subset P$

(C) $P \not\subset Q$ (D) $P = Q$ **[JEE 2011,3]**

TR0098

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5. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4}+\frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{k\pi}{6}\right)}$ is equal to **[JEE(Advanced)-2016, 3(–1)]**

(A) $3-\sqrt{3}$ (B) $2(3-\sqrt{3})$ (C) $2(\sqrt{3}-1)$ (D) $2(2+\sqrt{3})$

TR0099

6. Let α and β be nonzero real numbers such that 2(cosβ – cosα) + cosα cosβ = 1. Then which of the following is/are true ? (where α,β ≠ (2k + 1)π) **[JEE(Advanced)-2017, 4]**

(A) $\tan\left(\frac{\alpha}{2}\right)-\sqrt{3}\tan\left(\frac{\beta}{2}\right)=0$ (B) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right)-\tan\left(\frac{\beta}{2}\right)=0$

(C) $\tan\left(\frac{\alpha}{2}\right)+\sqrt{3}\tan\left(\frac{\beta}{2}\right)=0$ (D) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right)+\tan\left(\frac{\beta}{2}\right)=0$

TR0100

7. Let α and β be real numbers such that $-\frac{\pi}{4}<\beta<0<\alpha<\frac{\pi}{4}$. If $\sin(\alpha+\beta)=\frac{1}{3}$ and $\cos(\alpha-\beta)=\frac{2}{3}$, then the greatest integer less than or equal to

$$\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\cos\alpha}{\sin\beta}+\frac{\sin\beta}{\cos\alpha}\right)^2$$

is _____. **[JEE(Advanced)-2022]**

TR0122

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ANSWER KEY

Do yourself-1

1. 10π cm **2.** $36 + 6\pi$ **3.** (a) $\left(\frac{3061}{21600}\pi\right)$ (b) $\left(\frac{257}{4320}\pi\right)$ (c) $\frac{5587}{108000}\pi$

4. $\frac{22\pi}{3}+8\sqrt{3}$ **5.** 68º 43' 38"

Do yourself-2

1. $\frac{3}{5},\frac{4}{5},\frac{5}{3}$ **2.** 2 **23.** $\frac{2xy}{x^2+y^2},\frac{2xy}{x^2-y^2}$ **27.** $\left(\frac{1}{\tan^4 A}-\tan^4 A\right)$

Do yourself-3

1. $\left\{\frac{11}{60},\frac{60}{61},\frac{61}{60}\right\};\left\{\frac{-11}{60},\frac{-60}{61},\frac{-61}{60}\right\}$ **2.** $\left\{\frac{3}{5},\frac{4}{3}\right\};\left\{\frac{-3}{5},\frac{-4}{3}\right\}$ **3.** $\left\{\frac{3}{4}\right\}$

4. $\left\{\frac{8}{17},\frac{17}{8}\right\};\left\{\frac{-8}{17},\frac{-17}{8}\right\}$ **5.** $1,\frac{3}{5}$ **6.** $\left\{\frac{3}{5},\frac{5}{13}\right\}$ **7.** $\frac{5}{13}$

8. $\left\{\frac{12}{13}\right\}$ **9.** $\left\{\pm1,\pm\frac{1}{\sqrt{3}}\right\}$ **10.** $\left\{\frac{1}{2},-\frac{1}{3}\right\}$ **11.** $\pm\frac{1}{\sqrt{2}}$

12. $\left\{1\pm\sqrt{2}\right\}$ **13.** $\pm\frac{2x(x+1)}{2x^2+2x+1};\pm\frac{2x+1}{2x^2+2x+1}$

Do yourself-4

2. (a) $\frac{3}{2}$ (b) $4\frac{1}{3}$ (c) $\frac{1+\sqrt{3}}{4}$ (d) $-\frac{\sqrt{3}-1}{2\sqrt{2}}$

Do yourself-5

1. **(i)** 8 **2.** (a) 1 (b) 0

3. (a) $-\left(\frac{\sqrt{3}+1}{2}\right),-\frac{4}{\sqrt{3}}$ (b) $\left(\sqrt{2},-2\right)$ (c) $\frac{\sqrt{3}+1}{2},-\frac{4}{\sqrt{3}}$ (d) 0,2

4. (a) –cos(25º) (b) –cos(28º) (c) cot 25º (d) cot(26º)

5. (a) negative (b) positive (c) zero

6. (a) positive (b) positive (c) negative **7.** $-\frac{1}{\sqrt{3}}$ and $\frac{\sqrt{2}}{\sqrt{3}}$; $\frac{1}{\sqrt{3}}$ and $-\frac{\sqrt{2}}{\sqrt{3}}$

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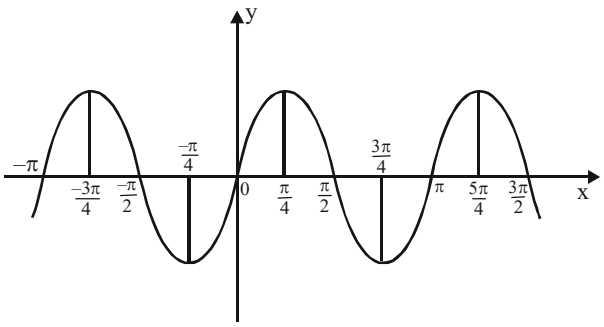
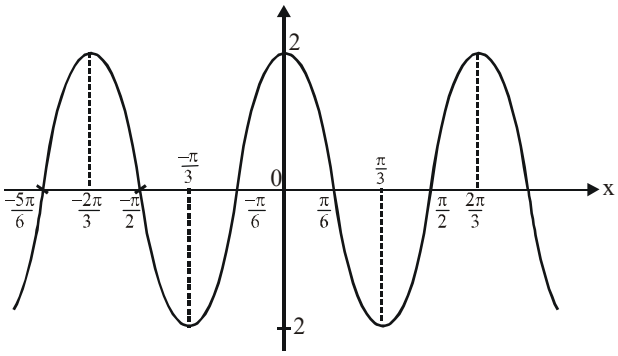
Do yourself-6

3. B

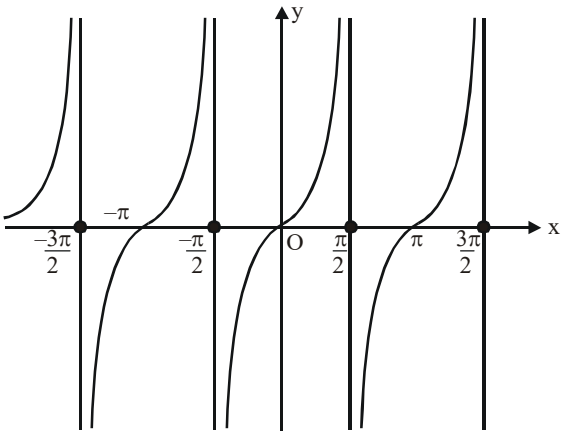
4. (a)



(b)



(c)



5. 2

Do yourself-7

1. (a) $\frac{187}{205}$ (b) $\frac{-133}{205}$ (c) $\frac{-84}{205}$ (d) $\frac{156}{205}$

3. $\sin(\alpha-\beta)=-\frac{133}{205}$ or $\frac{187}{205}$; $\cos(\alpha+\beta)=-\frac{84}{205}$ or $\frac{156}{205}$

4. $\sin(\alpha+\beta)=\frac{220}{221}$ or $\frac{140}{221}$; $\cos(\alpha-\beta)=\pm\frac{171}{221}$ or $\pm\frac{21}{221}$; $\sin(\alpha+\beta)=\frac{220}{221}$ or $\frac{140}{221}$; $\tan(\alpha+\beta)=\pm\frac{220}{21}$ or $\pm\frac{140}{171}$

13. 3 and $\frac{9}{13}$

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Do yourself-8

1. $\frac{1}{\sqrt{3}}$ **2.** $\cos 2\theta - \cos 12\theta$ **3.** $\cos 12^\circ - \cos 120^\circ$

Do yourself-11

8. D **9.** A

Do yourself-12

1 (a) $\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$ (b) $\sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$ (c) $\sqrt{2}-1$ **4.** $\frac{\pm 2\sqrt{2} \pm \sqrt{3}}{6}; \frac{\pm 7\sqrt{3} \pm 4\sqrt{2}}{18}$

5. $\frac{16}{305}; \frac{49}{305}$ **6.** $\frac{7}{5\sqrt{2}}$ **7.** $\pm\frac{1}{3}; \pm\frac{3}{4}$

8. (a) $\frac{\sqrt{4-\sqrt{2}-\sqrt{6}}}{2\sqrt{2}}$; (b) $\frac{\sqrt{4+\sqrt{2}+\sqrt{6}}}{2\sqrt{2}}$; (c) $\sqrt{4+2\sqrt{2}} - \left(\sqrt{2}+1\right)$ **9.** $\pm\sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$

Do yourself-13

1. (a) $-\frac{1}{2}$ (b) $\frac{\sqrt{5}+1}{8}$ **2.** $\frac{\sqrt{5}-1}{8}$ **3.** $\frac{5}{16}$ **4.** 1 **5.** $\frac{1}{2^7}$

6. 1 **9.** 4 **10.** **B**

Do yourself-14

1. 0 **2.** 1

Do yourself-15

1. 7 & –7 **2.** $-\frac{9}{8}$ **3.** $4+\sqrt{10}$ & $4-\sqrt{10}$ **4.** (i) 2, – 1 (ii) 3,0 **5.** $\frac{1}{2}$

6. $\left\{\frac{13}{3}, -1\right\}$ **7.** 25 **8.** {25,–25} **9.** $\left\{\sqrt{13}, -\sqrt{13}\right\}$ **10.** (i) 30 (ii) 18

Do yourself-16

1. 0 **3.** $\frac{\cot 1^\circ}{90}$ **5.** (a) $\frac{1}{2}$ (b) 0 **6.** C

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EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	B	A	B	C	B	B	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	B	D	D	A	D	D	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	D	D	A	B	B	B	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C	B,D	B,C	B,C	B,C,D	A,B	A,B,C,D	A,B,C,D	A,D	C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,D	A,B,C,D	C,D	B,D	A,B,C,D	B,D	A,B	B,C	A,B,C,D	C,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	A	C	B	C	A	B	A,B,C,D	A,C	A	
Q.10	A	B	C	D						
	S	R	Q	P						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5	3	3	2	1	1.69 or 1.70	1.33	5.75	0.04	1.73

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	4	1	1	3	4	1	2	3
Que.	11	12								
Ans.	4	B								

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	
Ans.	B	(a) A,B; (b) C,D	(a) 2; (b) k = 3	D	C	Bonus	1	

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TRIGONOMETRIC EQUATION

1. TRIGONOMETRIC EQUATION :

An equation involving one or more trigonometrical ratios of unknown angles is called a trigonometrical equation.

2. SOLUTION OF TRIGONOMETRIC EQUATION :

A value of the unknown angle which satisfies the given equation is called a solution of the trigonometric equation.

(a) **Principal solution :-** The solution of the trigonometric equation lying in the interval $[0, 2\pi)$.

(b) **General solution :-** Since all the trigonometric functions are many one & periodic, hence there are infinite values of θ for which trigonometric functions have the same value. All such possible values of θ for which the given trigonometric function is satisfied is given by a general formula. Such a general formula is called general solution of trigonometric equation.

(c) **Particular solution :-** The solution of the trigonometric equation lying in the given interval.

3. GENERAL SOLUTIONS OF SOME TRIGONOMETRIC EQUATIONS (TO BE REMEMBERED) :

(a) If $\sin\theta = 0$, then $\theta = n\pi$, $n \in I$ (set of integers)

(b) If $\cos\theta = 0$, then $\theta = (2n+1)\frac{\pi}{2}$, $n \in I$

(c) If $\tan\theta = 0$, then $\theta = n\pi$, $n \in I$

(d) If $\sin\theta = \sin\alpha$, then $\theta = n\pi + (-1)^n\alpha$ where $\alpha \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$, $n \in I$

(e) If $\cos\theta = \cos\alpha$, then $\theta = 2n\pi \pm \alpha$, $n \in I$, $\alpha \in [0,\pi]$

(f) If $\tan\theta = \tan\alpha$, then $\theta = n\pi + \alpha$, $n \in I$, $\alpha \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

(g) If $\sin\theta = 1$, then $\theta = 2n\pi + \frac{\pi}{2} = (4n+1)\frac{\pi}{2}$, $n \in I$

(h) If $\cos\theta = 1$ then $\theta = 2n\pi$, $n \in I$

(i) If $\sin^2\theta = \sin^2\alpha$ or $\cos^2\theta = \cos^2\alpha$ or $\tan^2\theta = \tan^2\alpha$, then $\theta = n\pi \pm \alpha$, $n \in I$

(j) For $n \in I$, $\sin n\pi = 0$ and $\cos n\pi = (-1)^n$, $n \in I$
$\sin(n\pi + \theta) = (-1)^n \sin\theta$ $\cos(n\pi + \theta) = (-1)^n \cos\theta$

(k) $\cos n\pi = (-1)^n$, $n \in I$

If n is an odd integer, then $\sin\frac{n\pi}{2} = (-1)^{\frac{n-1}{2}}, \cos\frac{n\pi}{2} = 0,$

$$\sin\left(\frac{n\pi}{2} + \theta\right) = (-1)^{\frac{n-1}{2}}\cos\theta$$

$$\cos\left(\frac{n\pi}{2} + \theta\right) = (-1)^{\frac{n+1}{2}}\sin\theta$$

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Illustration 1 : Find the set of values of x for which $\frac{\tan 3x - \tan 2x}{1+\tan 3x.\tan 2x} = 1$.

Solution : We have, $\frac{\tan 3x - \tan 2x}{1+\tan 3x.\tan 2x} = 1 \Rightarrow \tan(3x - 2x) = 1 \Rightarrow \tan x = 1$

$\Rightarrow \tan x = \tan\frac{\pi}{4} \Rightarrow x = n\pi + \frac{\pi}{4}, n \in I$ {using $\tan\theta = \tan\alpha \Leftrightarrow \theta = n\pi + \alpha$)

But for this value of x, tan 2x is not defined.

Hence the solution set for x is ϕ. **Ans.**

Do yourself-1 :

1. Find general solutions of the following equations :

(a) $\sin\theta = \frac{1}{2}$ (b) $\cos\left(\frac{3\theta}{2}\right) = 0$ (c) $\tan\left(\frac{3\theta}{4}\right) = 0$

(d) $\cos^2 2\theta = 1$ (e) $\sqrt{3}\sec 2\theta = 2$ (f) $\operatorname{cosec}\left(\frac{\theta}{2}\right) = -1$

2. cos 15 x = sin 5x if

(A) $x = -\frac{\pi}{20} + \frac{n\pi}{5}, n \in I$ (B) $x = \frac{\pi}{40} + \frac{n\pi}{10}, n \in I$

(C) $x = \frac{3\pi}{20} + \frac{n\pi}{5}, n \in I$ (D) $x = -\frac{3\pi}{40} + \frac{n\pi}{10}, n \in I$

3. $\tan(p\pi/4) = \cot(q\pi/4)$ if :

(A) p + q = 0 (B) p + q = 2n + 1

(C) p + q = 2n (D) p + q = 2 (2n + 1)

4. Solve $\tan 2\theta = \tan\frac{2}{\theta}$.

5. What are the most general values θ of which satisfy the equations :

(i) $\sin\theta = \frac{1}{\sqrt{2}}$ (ii) $\tan(x-1) = \sqrt{3}$ (iii) $\tan\theta = -1$

(iv) $\operatorname{cosec}\theta = \frac{2}{\sqrt{3}}$ (v) $2\cot^2\theta = \operatorname{cosec}^2\theta$

6. Solve :

(i) $\sin 9\theta = \sin\theta$ (ii) $\cot\theta + \tan\theta = 2\operatorname{cosec}\theta$ (iii) $\sin 2\theta = \cos 3\theta$

(iv) $\cot\theta = \tan 8\theta$ (v) $\cot\theta - \tan\theta = 2$ (vi) $\operatorname{cosec}\theta = \cot\theta + \sqrt{3}$

(vii) $\tan 2\theta \tan\theta = 1$ (viii) $\tan\theta + \tan 2\theta + \sqrt{3}\tan\theta\tan 2\theta = \sqrt{3}$

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4. IMPORTANT POINTS TO BE REMEMBERED WHILE SOLVING TRIGONOMETRIC EQUATIONS :

(a) For equations of the type $\sin\theta = k$ or $\cos\theta = k$, one must check that $|k| \le 1$.

(b) Avoid squaring the equations, if possible, because it may lead to extraneous solutions. Reject extra solutions if they do not satisfy the given equation.

(c) Do not cancel the common variable factor from the two sides of the equations which are in a product because we may loose some solutions.

(d) The answer should not contain such values of θ, which make any of the terms undefined or infinite.

(i) Check that denominator is not zero at any stage while solving equations.

(ii) If $\tan\theta$ or $\sec\theta$ is involved in the equations, θ should not be odd multiple of $\frac{\pi}{2}$.

(iii) If $\cot\theta$ or $\text{cosec}\,\theta$ is involved in the equation, θ should not be multiple of π or 0.

5. DIFFERENT STRATEGIES FOR SOLVING TRIGONOMETRIC EQUATIONS :

(a) Solving trigonometric equations by factorisation.

e.g. $(2\sin x - \cos x)(1 + \cos x) = \sin^2 x$

$\therefore$ $(2\sin x - \cos x)(1 + \cos x) - (1 - \cos^2 x) = 0$

$\therefore$ $(1 + \cos x)(2\sin x - \cos x - 1 + \cos x) = 0$

$\therefore$ $(1 + \cos x)(2\sin x - 1) = 0$

$\Rightarrow$ $\cos x = -1$ or $\sin x = \frac{1}{2}$

$\Rightarrow$ $\cos x = -1 = \cos\pi$ $\Rightarrow$ $x = 2n\pi + \pi = (2n+1)\pi,\ n \in I$

or $\sin x = \frac{1}{2} = \sin\frac{\pi}{6}$ $\Rightarrow$ $x = k\pi + (-1)^k \frac{\pi}{6},\ k \in I$

Illustration 2 : If $\frac{1}{6}\sin\theta$, $\cos\theta$ and $\tan\theta$ are in G.P. then the general solution for θ is -

(A) $2n\pi \pm \frac{\pi}{3}$ (B) $2n\pi \pm \frac{\pi}{6}$ (C) $n\pi \pm \frac{\pi}{3}$ (D) none of these

Solution : Since, $\frac{1}{6}\sin\theta$, $\cos\theta$, $\tan\theta$ are in G.P.

$\Rightarrow$ $\cos^2\theta = \frac{1}{6}\sin\theta \,.\, \tan\theta$ $\Rightarrow$ $6\cos^3\theta + \cos^2\theta - 1 = 0$

$\therefore$ $(2\cos\theta - 1)(3\cos^2\theta + 2\cos\theta + 1) = 0$

$\Rightarrow$ $\cos\theta = \frac{1}{2}$ (other values of $\cos\theta$ are imaginary)

$\Rightarrow$ $\cos\theta = \cos\frac{\pi}{3}$ $\Rightarrow$ $\theta = 2n\pi \pm \frac{\pi}{3},\ n \in I.$ **Ans. (A)**

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(b) Solving of trigonometric equation by reducing it to a quadratic equation.

e.g. $6 - 10\cos x = 3\sin^2 x$

$\therefore \quad 6 - 10\cos x = 3 - 3\cos^2 x \quad \Rightarrow \quad 3\cos^2 x - 10\cos x + 3 = 0$

$\Rightarrow \quad (3\cos x - 1)(\cos x - 3) = 0 \quad \Rightarrow \quad \cos x = \frac{1}{3}$ or $\cos x = 3$

Since $\cos x = 3$ is not possible as $-1 \le \cos x \le 1$

$\therefore \quad \cos x = \frac{1}{3} = \cos\left(\cos^{-1}\frac{1}{3}\right) \quad \Rightarrow \quad x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right),\ n \in I$

Illustration 3 : Solve $\sin^2\theta - \cos\theta = \frac{1}{4}$ for θ and write the values of θ in the interval $0 \le \theta \le 2\pi$.

Solution : The given equation can be written as

$1 - \cos^2\theta - \cos\theta = \frac{1}{4} \quad \Rightarrow \quad \cos^2\theta + \cos\theta - 3/4 = 0$

$\Rightarrow \quad 4\cos^2\theta + 4\cos\theta - 3 = 0 \quad \Rightarrow \quad (2\cos\theta - 1)(2\cos\theta + 3) = 0$

$\Rightarrow \quad \cos\theta = \frac{1}{2}, -\frac{3}{2}$

Since, $\cos\theta = -3/2$ is not possible as $-1 \le \cos\theta \le 1$

$\therefore \quad \cos\theta = \frac{1}{2} \quad \Rightarrow \quad \cos\theta = \cos\frac{\pi}{3} \quad \Rightarrow \quad \theta = 2n\pi \pm \frac{\pi}{3},\ n \in I$

For the given interval, n = 0 and n = 1.

$\Rightarrow \quad \theta = \frac{\pi}{3}, \frac{5\pi}{3}$ **Ans.**

Illustration 4 : Find the number of solutions of $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$.

Solution : Here, $\tan x + \sec x = 2\cos x \quad \Rightarrow \quad \sin x + 1 = 2\cos^2 x$

$\Rightarrow \quad 2\sin^2 x + \sin x - 1 = 0 \quad \Rightarrow \quad \sin x = \frac{1}{2}, -1$

But $\sin x = -1 \Rightarrow x = \frac{3\pi}{2}$ for which $\tan x + \sec x = 2\cos x$ is not defined.

Thus $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$

$\Rightarrow$ number of solutions of $\tan x + \sec x = 2\cos x$ is 2. **Ans.**

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Illustration 5 : Solve the equation $5\sin^2x - 7\sin x \cos x + 16\cos^2 x = 4$

Solution : To solve this equation we use the fundamental formula of trigonometric identities,

$$\sin^2x + \cos^2x = 1$$

writing the equation in the form,

$$5\sin^2x - 7\sin x \,.\, \cos x + 16\cos^2x = 4(\sin^2x + \cos^2x)$$

$\Rightarrow$ $\sin^2x - 7\sin x \cos x + 12\cos^2 x = 0$

dividing by $\cos^2x$ on both side we get,

$$\tan^2x - 7\tan x + 12 = 0$$

Now it can be factorized as :

$$(\tan x - 3)(\tan x - 4) = 0$$

$\Rightarrow$ $\tan x = 3, 4$

i.e., $\tan x = \tan(\tan^{-1}3)$ or $\tan x = \tan(\tan^{-1}4)$

$\Rightarrow$ $x = n\pi + \tan^{-1} 3$ or $x = n\pi + \tan^{-1} 4$, $n \in I$. **Ans.**

Illustration 6 : If $x \neq \frac{n\pi}{2}$, $n \in I$ and $(\cos x)^{\sin^2 x - 3\sin x + 2} = 1$, then find the general solutions of x.

Solution : As $x \neq \frac{n\pi}{2}$ $\Rightarrow$ $\cos x \neq 0, 1, -1$

So, $(\cos x)^{\sin^2 x - 3\sin x + 2} = 1$ $\Rightarrow$ $\sin^2x - 3\sin x + 2 = 0$

$\therefore$ $(\sin x - 2)(\sin x - 1) = 0 \Rightarrow$ $\sin x = 1, 2$

where $\sin x = 2$ is not possible and $\sin x = 1$ which is also not possible as $x \neq \frac{n\pi}{2}$

$\therefore$ no general solution is possible. **Ans.**

Illustration 7 : Solve the equation $\sin^4x + \cos^4 x = \frac{7}{2}\sin x \,.\, \cos x$.

Solution : $\sin^4x + \cos^4x = \frac{7}{2}\sin x \,.\, \cos x$ $\Rightarrow$ $(\sin^2x + \cos^2x)^2 - 2\sin^2x \cos^2x = \frac{7}{2}\sin x \,.\, \cos x$

$\Rightarrow$ $1 - \frac{1}{2}(\sin 2x)^2 = \frac{7}{4}(\sin 2x)$ $\Rightarrow 2\sin^2 2x + 7\sin 2x - 4 = 0$

$\Rightarrow$ $(2\sin 2x - 1)(\sin 2x + 4) = 0$ $\Rightarrow$ $\sin 2x = \frac{1}{2}$ or $\sin 2x = -4$ (which is not possible)

$\Rightarrow$ $2x = n\pi + (-1)^n \frac{\pi}{6}$, $n \in I$

i.e., $x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}$, $n \in I$ **Ans.**

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Do yourself-2 :

1. Solve the following equations :

(a) $3\sin x + 2\cos^2 x = 0$ (b) $\sec^2 2\alpha = 1 - \tan 2\alpha$

(c) $7\cos^2\theta + 3\sin^2\theta = 4$ (d) $4\cos\theta - 3\sec\theta = \tan\theta$

2. Solve the equation : $2\sin^2\theta + \sin^2 2\theta = 2$ for $\theta \in (-\pi, \pi)$.

3. Solve $\cos^3 x + \cos^2 x - 4\cos^2\frac{x}{2} = 0$

4. Solve $\cot^2\theta + 3\operatorname{cosec}\theta + 3 = 0$

5. The general solution of the equation, $\dfrac{1-\sin x + ... + (-1)^{n-1}\sin^{n-1} x + ...\infty}{1+\sin x + ... + \sin^{n-1} x + ...\infty} = \dfrac{1-\cos 2x}{1+\cos 2x}$ is

(A) $(-1)^n \pi/3 + n\pi$ (B) $(-1)^n \pi/6 + n\pi$

(C) $(-1)^{n+1} \pi/6 + n\pi$ (D) $(-1)^{n-1} \pi/3 + n\pi$

6. If sum of all solutions of the equation $6\sin\frac{x}{2} = \sec\frac{x}{2}$ in $[-2\pi, 4\pi]$ is $k\pi$, then k is equal to

7. If sum of all possible values of $x \in (0, 2\pi)$ satisfying the equation

$2\cos x \cdot \operatorname{cosec} x - 4\cos x - \operatorname{cosec} x = -2$, is $\frac{k\pi}{4}$, then k is equal to

(A) 9 (B) 12 (C) 16 (D) 32

8. Solve

(i) $\tan^2\theta - (1+\sqrt{3})\tan\theta + \sqrt{3} = 0$

(ii) $4\cos\theta - 3\sec\theta = 2\tan\theta$

(iii) $\tan x \,.\, \tan\left(x + \frac{\pi}{3}\right) \,.\, \tan\left(x + \frac{2\pi}{3}\right) = \sqrt{3}$

(c) Solving trigonometric equations by introducing an auxilliary argument.

Consider, $a\sin\theta + b\cos\theta = c$ ..... (i)

$$\therefore \quad \frac{a}{\sqrt{a^2+b^2}}\sin\theta + \frac{b}{\sqrt{a^2+b^2}}\cos\theta = \frac{c}{\sqrt{a^2+b^2}}$$

equation (i) has a solution only if $|c| \le \sqrt{a^2+b^2}$

let $\dfrac{a}{\sqrt{a^2+b^2}} = \cos\phi$, $\dfrac{b}{\sqrt{a^2+b^2}} = \sin\phi$ & $\phi = \tan^{-1}\dfrac{b}{a}$

by introducing this auxillary argument ϕ, equation (i) reduces to

$\sin(\theta + \phi) = \dfrac{c}{\sqrt{a^2+b^2}}$ Now this equation can be solved easily.

Illustration 8 : Find the number of distinct solutions of secx + tanx = $\sqrt{3}$, where $0 \le x \le 3\pi$.

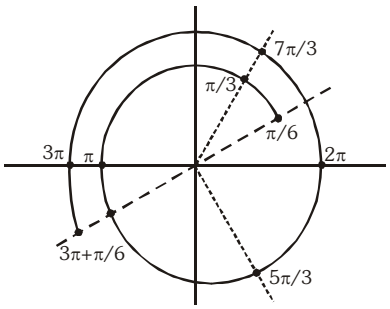
Solution : Here, $\sec x + \tan x = \sqrt{3} \quad \Rightarrow \quad 1 + \sin x = \sqrt{3}\cos x$

or $\sqrt{3}\cos x - \sin x = 1$

dividing both sides by $\sqrt{a^2+b^2}$ i.e. $\sqrt{4} = 2$, we get

$$\Rightarrow \frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x = \frac{1}{2}$$

$$\Rightarrow \cos\frac{\pi}{6}\cos x - \sin\frac{\pi}{6}\sin x = \frac{1}{2} \Rightarrow \cos\left(x+\frac{\pi}{6}\right) = \frac{1}{2}$$



As $0 \le x \le 3\pi$

$$\frac{\pi}{6} \le x + \frac{\pi}{6} \le 3\pi + \frac{\pi}{6}$$

$$\Rightarrow x + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3} \quad \Rightarrow \quad x = \frac{\pi}{6}, \frac{3\pi}{2}, \frac{13\pi}{6}$$

But at $x = \frac{3\pi}{2}$, tanx and secx is not defined.

$\therefore$ Total number of solutions are 2. **Ans.**

Illustration 9 : Prove that the equation kcosx – 3sinx = k + 1 possess a solution iff $k \in (-\infty, 4]$.

Solution : Here, k cosx – 3sinx = k + 1, could be re-written as :

$$\frac{k}{\sqrt{k^2+9}}\cos x - \frac{3}{\sqrt{k^2+9}}\sin x = \frac{k+1}{\sqrt{k^2+9}}$$

or $\cos(x+\phi) = \frac{k+1}{\sqrt{k^2+9}}$, where $\tan\phi = \frac{3}{k}$

which possess a solution only if $-1 \le \frac{k+1}{\sqrt{k^2+9}} \le 1$

i.e., $\left|\frac{k+1}{\sqrt{k^2+9}}\right| \le 1$

i.e., $(k+1)^2 \le k^2 + 9$

i.e., $k^2 + 2k + 1 \le k^2 + 9$

or $k \le 4$

$\Rightarrow$ The interval of k for which the equation (kcosx – 3sinx = k + 1) has a solution is $(-\infty, 4]$.

Ans.

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Do yourself-3 :

1 Solve the following equations :

(a) $\sin x + \sqrt{2} = \cos x$ (b) $\operatorname{cosec}\theta = 1 + \cot\theta$

2. Solve : $\cos\theta + \sin\theta = \cos 2\theta + \sin 2\theta$

3. Solve : $8 \sin x = \dfrac{\sqrt{3}}{\cos x} + \dfrac{1}{\sin x}$

4. Solve

(i) $\sqrt{3}\sin\theta - \cos\theta = \sqrt{2}$ (ii) $5\sin\theta + 2\cos\theta = 5$

(d) Solving trigonometric equations by transforming sum of trigonometric functions into product.

e.g. $\cos 3x + \sin 2x - \sin 4x = 0$

$\cos 3x - 2\sin x \cos 3x = 0$

$\Rightarrow$ $(\cos 3x)(1 - 2\sin x) = 0$

$\Rightarrow$ $\cos 3x = 0$ or $\sin x = \dfrac{1}{2}$

$\Rightarrow$ $\cos 3x = 0 = \cos\dfrac{\pi}{2}$ or $\sin x = \dfrac{1}{2} = \sin\dfrac{\pi}{6}$

$\Rightarrow$ $3x = 2n\pi \pm \dfrac{\pi}{2}$ or $x = m\pi + (-1)^m \dfrac{\pi}{6}$

$\Rightarrow$ $x = \dfrac{2n\pi}{3} \pm \dfrac{\pi}{6}$ or $x = m\pi + (-1)^m \dfrac{\pi}{6}$; $(n, m \in I)$

Illustration 10 : Solve : $\cos\theta + \cos 3\theta + \cos 5\theta + \cos 7\theta = 0$

Solution : We have $\cos\theta + \cos 7\theta + \cos 3\theta + \cos 5\theta = 0$

$\Rightarrow$ $2\cos 4\theta\cos 3\theta + 2\cos 4\theta\cos\theta = 0 \Rightarrow \cos 4\theta(\cos 3\theta + \cos\theta) = 0$

$\Rightarrow$ $\cos 4\theta(2\cos 2\theta\cos\theta) = 0$

$\Rightarrow$ Either $\cos\theta = 0 \Rightarrow \theta = (2n_1 + 1)\pi/2,\ n_1 \in I$

or $\cos 2\theta = 0 \Rightarrow \theta = (2n_2 + 1)\dfrac{\pi}{4},\ n_2 \in I$

or $\cos 4\theta = 0 \Rightarrow \theta = (2n_3 + 1)\dfrac{\pi}{8},\ n_3 \in I$ **Ans.**

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(e) Solving trigonometric equations by transforming a product into sum.

e.g. $\sin 5x.\cos 3x = \sin 6x.\cos 2x$

$\sin 8x + \sin 2x = \sin 8x + \sin 4x$

$\therefore \quad 2\sin 2x\,.\cos 2x - \sin 2x = 0$

$\Rightarrow \quad \sin 2x(2\cos 2x - 1) = 0$

$\Rightarrow \quad \sin 2x = 0 \quad$ or $\quad \cos 2x = \frac{1}{2}$

$\Rightarrow \quad \sin 2x = 0 = \sin 0 \quad$ or $\quad \cos 2x = \frac{1}{2} = \cos\frac{\pi}{3}$

$\Rightarrow \quad 2x = n\pi + (-1)^n \times 0,\ n \in I$ or $\quad 2x = 2m\pi \pm \frac{\pi}{3},\ m \in I$

$\Rightarrow \quad x = \frac{n\pi}{2},\ n \in I \quad$ or $\quad x = m\pi \pm \frac{\pi}{6},\ m \in I$

Illustration 11 : Solve : $\cos\theta\cos 2\theta\cos 3\theta = \frac{1}{4}$; where $0 \le \theta \le \pi$.

Solution : $\frac{1}{2}(2\cos\theta\cos 3\theta)\cos 2\theta = \frac{1}{4} \Rightarrow (\cos 2\theta + \cos 4\theta)\cos 2\theta = \frac{1}{2}$

$\Rightarrow \quad \frac{1}{2}[2\cos^2 2\theta + 2\cos 4\theta\cos 2\theta] = \frac{1}{2} \Rightarrow 1 + \cos 4\theta + 2\cos 4\theta\cos 2\theta = 1$

$\therefore \quad \cos 4\theta\,(1 + 2\cos 2\theta) = 0$

$\cos 4\theta = 0 \quad$ or $\quad (1 + 2\cos 2\theta) = 0$

Now from the first equation : $2\cos 4\theta = 0 = \cos(\pi/2)$

$\therefore \quad 4\theta = \left(n + \frac{1}{2}\right)\pi \Rightarrow \theta = (2n+1)\frac{\pi}{8},\ n \in I$

for $\quad n = 0,\ \theta = \frac{\pi}{8};\ n = 1,\ \theta = \frac{3\pi}{8};\ n = 2,\ \theta = \frac{5\pi}{8};\ n = 3,\ \theta = \frac{7\pi}{8} \qquad (\because\ 0 \le \theta \le \pi)$

and from the second equation :

$\cos 2\theta = -\frac{1}{2} = -\cos(\pi/3) = \cos(\pi - \pi/3) = \cos(2\pi/3)$

$\therefore \quad 2\theta = 2k\pi \pm 2\pi/3 \quad \therefore\ \theta = k\pi \pm \pi/3,\ k \in I$

again for $\quad k = 0,\ \theta = \frac{\pi}{3};\ k = 1,\ \theta = \frac{2\pi}{3} \qquad (\because\ 0 \le \theta \le \pi)$

$\therefore \quad \theta = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$ **Ans.**

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Illustration 12 : The general solution of the trigonometric equation $\sin x \cos 2x + \sin 2x \cos 5x = \sin 3x \cos 5x$, is

(A) $\frac{n\pi}{3}$ (B) $\frac{2n\pi}{9}$ (C) $2n\pi$ (D) $\frac{n\pi}{3} \cup \frac{2n\pi}{9}$

(where $n \in I$)

Solution : $2 \sin x \cos 2x + 2 \sin 2x \cos 5x = 2 \sin 3x \cos 5x$

$\sin 3x - \sin x + \sin 7x - \sin 3x = \sin 8x - \sin 2x$

$\sin 7x - \sin x = \sin 8x - \sin 2x$

$2 \cos 4x \sin 3x = 2 \cos 5x \sin 3x$

$2 \sin 3x [\cos 5x - \cos 4x] = 0$

$\sin 3x = 0 \Rightarrow x = \frac{n\pi}{3}$

if $\cos 5x - \cos 4x = 0$

$2 \sin \frac{9x}{2} \sin \frac{x}{2} = 0$

$\therefore \quad x = \frac{2n\pi}{9}$ or $2n\pi$

Hence general solution is $\frac{n\pi}{3} \cup \frac{2n\pi}{9}, n \in I$

Do yourself-4 :

1. Solve : $4\sin\theta \sin 2\theta \sin 4\theta = \sin 3\theta$.

2. Solve for x : $\sin x + \sin 3x + \sin 5x = 0$.

3. Solve : $\sin 2x + 5\sin x + 1 + 5\cos x = 0$

4. Solve : $3\cos x + 3\sin x + \sin 3x - \cos 3x = 0$

5. Solve : $(1 - \sin 2x)(\cos x - \sin x) = 1 - 2\sin^2 x$.

6. The number of integral values of a for which the equation $\cos 2x + a \sin x = 2a - 7$ possesses a solution is

(A) 2 (B) 3 (C) 4 (D) 5

7. Solve

(i) $\cos\theta + \sin\theta = \cos 2\theta + \sin 2\theta$

(ii) $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$

(iii) $\sin^2 n\theta - \sin^2 (n-1)\theta = \sin^2\theta$, where n is constant and $n \neq 0, 1$

(f) Solving equations by a change of variable :

(i) Equations of the form $P(\sin x \pm \cos x, \sin x . \cos x) = 0$, where $P(y,z)$ is a polynomial, can be solved by the substitution :

$\cos x \pm \sin x = t \quad \Rightarrow \quad 1 \pm 2 \sin x . \cos x = t^2$.

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Illustration 13 : Solve : sin x + cos x = 1 + sin x. cos x.

Solution : put sinx + cosx = t

$\Rightarrow$ $\sin^2x + \cos^2x + 2\sin x \,.\, \cos x = t^2$

$\Rightarrow$ $2\sin x \cos x = t^2 - 1$ $(\because \sin^2x + \cos^2x = 1)$

$\Rightarrow$ $\sin x.\cos x = \left(\dfrac{t^2-1}{2}\right)$

Substituting above result in given equation, we get :

$t = 1 + \dfrac{t^2-1}{2}$

$\Rightarrow$ $2t = t^2 + 1 \Rightarrow t^2 - 2t + 1 = 0$

$\Rightarrow$ $(t-1)^2 = 0$ $\Rightarrow$ $t = 1$

$\Rightarrow$ $\sin x + \cos x = 1$

Dividing both sides by $\sqrt{1^2+1^2}$ i.e. $\sqrt{2}$, we get

$\Rightarrow$ $\dfrac{1}{\sqrt{2}}\sin x + \dfrac{1}{\sqrt{2}}\cos x = \dfrac{1}{\sqrt{2}}$ $\Rightarrow$ $\cos x \cos\dfrac{\pi}{4} + \sin x.\sin\dfrac{\pi}{4} = \dfrac{1}{\sqrt{2}}$

$\Rightarrow$ $\cos\left(x - \dfrac{\pi}{4}\right) = \cos\dfrac{\pi}{4}$ $\Rightarrow$ $x - \dfrac{\pi}{4} = 2n\pi \pm \dfrac{\pi}{4}$

$\Rightarrow$ $x = 2n\pi$ or $x = 2n\pi + \dfrac{\pi}{2} = (4n+1)\dfrac{\pi}{2}$, $n \in I$

(ii) Equations of the form of asinx + bcosx + d = 0, where a, b & d are real numbers can be solved by changing sin x & cos x into their corresponding tangent of half the angle.

Illustration 14 : Solve : 3 cos x + 4 sin x = 5

Solution : $\Rightarrow$ $3\left(\dfrac{1-\tan^2 x/2}{1+\tan^2 x/2}\right) + 4\left(\dfrac{2\tan x/2}{1+\tan^2 x/2}\right) = 5$

$\Rightarrow$ $\dfrac{3-3\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}} + \dfrac{8\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}} = 5$

$\Rightarrow$ $3 - 3\tan^2\dfrac{x}{2} + 8\tan\dfrac{x}{2} = 5 + 5\tan^2\dfrac{x}{2}$ $\Rightarrow$ $8\tan^2\dfrac{x}{2} - 8\tan\dfrac{x}{2} + 2 = 0$

$\Rightarrow$ $4\tan^2\dfrac{x}{2} - 4\tan\dfrac{x}{2} + 1 = 0$ $\Rightarrow$ $\left(2\tan\dfrac{x}{2} - 1\right)^2 = 0$

$\Rightarrow$ $2\tan\dfrac{x}{2} - 1 = 0 \Rightarrow \tan\dfrac{x}{2} = \dfrac{1}{2} = \tan\left(\tan^{-1}\dfrac{1}{2}\right)$

$\Rightarrow$ $\dfrac{x}{2} = n\pi + \tan^{-1}\left(\dfrac{1}{2}\right)$, $n \in I$ $\Rightarrow$ $x = 2n\pi + 2\tan^{-1}\dfrac{1}{2}$, $n \in I$

(g) Solving trigonometric equations with the use of the boundness of the functions involved.

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Illustration 15 : Solve the equation $(\sin x + \cos x)^{1+\sin 2x} = 2$, when $0 \le x \le \pi$.

Solution : We know, $-\sqrt{a^2+b^2} \le a\sin\theta + b\cos\theta \le \sqrt{a^2+b^2}$ and $-1 \le \sin\theta \le 1$.

$\therefore$ $(\sin x + \cos x)$ admits the maximum value as $\sqrt{2}$

and $(1 + \sin 2x)$ admits the maximum value as 2.

Also $\left(\sqrt{2}\right)^2 = 2$.

$\therefore$ the equation could hold only when, $\sin x + \cos x = \sqrt{2}$ and $1 + \sin 2x = 2$

Now, $\sin x + \cos x = \sqrt{2}$ $\Rightarrow$ $\cos\left(x - \frac{\pi}{4}\right) = 1$

$\Rightarrow$ $x = 2n\pi + \pi/4,\ n \in I$ ..... (i)

and $1 + \sin 2x = 2$ $\Rightarrow$ $\sin 2x = 1 = \sin\frac{\pi}{2}$

$\Rightarrow$ $2x = m\pi + (-1)^m\frac{\pi}{2},\ m \in I$ $\Rightarrow$ $x = \frac{m\pi}{2} + (-1)^m\frac{\pi}{4}$ ..... (ii)

The value of x in $[0, \pi]$ satisfying equations (i) and (ii) is $x = \frac{\pi}{4}$ (when $n = 0$ & $m = 0$)

Ans.

Note : $\sin x + \cos x = -\sqrt{2}$ and $1 + \sin 2x = 2$ also satisfies but as $x \ge 0$, this solution is not in domain.

Illustration 16 : Solve for x and y : $2^{\frac{1}{\cos^2 x}}\sqrt{y^2 - y + 1/2} \le 1$

Solution : $2^{\frac{1}{\cos^2 x}}\sqrt{y^2 - y + 1/2} \le 1$ ..... (i)

$$2^{\frac{1}{\cos^2 x}}\sqrt{\left(y - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} \le 1$$

Minimum value of $2^{\frac{1}{\cos^2 x}} = 2$

Minimum value of $\sqrt{\left(y - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{1}{2}$

$\Rightarrow$ Minimum value of $2^{\frac{1}{\cos^2 x}}\sqrt{y^2 - y + \frac{1}{2}}$ is 1

$\Rightarrow$ (i) is possible when $2^{\frac{1}{\cos^2 x}}\sqrt{\left(y - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = 1$

$\Rightarrow$ $\cos^2 x = 1$ and $y = 1/2$ $\Rightarrow \cos x = \pm 1 \Rightarrow x = n\pi$, where $n \in I$.

Hence $x = n\pi,\ n \in I$ and $y = 1/2$. **Ans.**

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Illustration 17 : The number of solution(s) of $2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2+\frac{1}{x^2}$, $0 \le x \le \pi/2$, is/are -

(A) 0 (B) 1 (C) infinite (D) none of these

Solution : Let $y = 2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2+\frac{1}{x^2} \Rightarrow y = (1 + \cos x)\sin^2 x$ and $y = x^2 + \frac{1}{x^2}$

when $y = (1 + \cos x)\sin^2 x = (\text{a number} < 2)(\text{a number} \le 1) \Rightarrow y < 2$ ..... (i)

and when $y = x^2 + \frac{1}{x^2} = \left(x - \frac{1}{x}\right)^2 + 2 \ge 2 \Rightarrow y \ge 2$ ..... (ii)

No value of y can be obtained satisfying (i) and (ii), simultaneously

$\Rightarrow$ No real solution of the equation exists. **Ans. (A)**

Note: If L.H.S. of the given trigonometric equation is always less than or equal to k and RHS is always greater than k, then no solution exists. If both the sides are equal to k for same value of θ, then solution exists and if they are equal for different values of θ, then solution does not exist.

Illustration 18 : The number of ordered pairs (x, y) satisfying $|x| + |y| = 3$ and $\sin\left(\frac{\pi x^2}{3}\right) = 1$ is less than or equal to

(A) 7 (B) 8 (C) 9 (D) 10

Solution : $\sin\left(\frac{\pi}{3}x^2\right) = 1$

$\frac{\pi}{3}x^2 = 2n\pi + \frac{\pi}{2}; n \in I$

$\frac{\pi}{3}x^2 = (4n + 1)\frac{\pi}{2}$

$x^2 = \frac{3}{2}(4n + 1); n \in I$

only n = 0 and n = 1 is possible.

$x^2 = \frac{3}{2}$ or $x^2 = \frac{15}{2}$

$\therefore x = \pm\sqrt{\frac{3}{2}}$ or $x = \pm\sqrt{\frac{15}{2}}$.

for each value of x will get 2 values of y, Hence 8 ordered pairs.

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Do yourself-5 :

1. If $x^2 - 4x + 5 - \sin y = 0$, $y \in [0, 2\pi)$, then -

(A) $x = 1, y = 0$ (B) $x = 1, y = \pi/2$ (C) $x = 2, y = 0$ (D) $x = 2, y = \pi/2$

2. If $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$, $y > 0$, $x \in [0, \pi]$, then find the least positive value of x satisfying the given condition.

3. Solve $\sin 3x + \cos 2x = -2$

4. Solve $\sqrt{3\sin 5x - \cos^2 x - 3} = 1 - \sin x$

5. The number of real solutions of the equation $\sin(e^x) = 5^x + 5^{-x}$ is -

(A) 0 (B) 1 (C) 2 (D) infinitely many

6. If $\sin^3\alpha + \cos^3\beta + 6\sin\alpha \cdot \cos\beta = 8$, where $\alpha, \beta \in [0, 2\pi]$ then number of ordered pairs (α, β) is equal to

(A) 0 (B) 1 (C) 2 (D) 3

6. TRIGONOMETRIC INEQUALITIES :

There is no general rule to solve trigonometric inequations and the same rules of algebra are valid provided the domain and range of trigonometric functions should be kept in mind.

Illustration 19 : Find the solution set of inequality $\sin x > 1/2$.

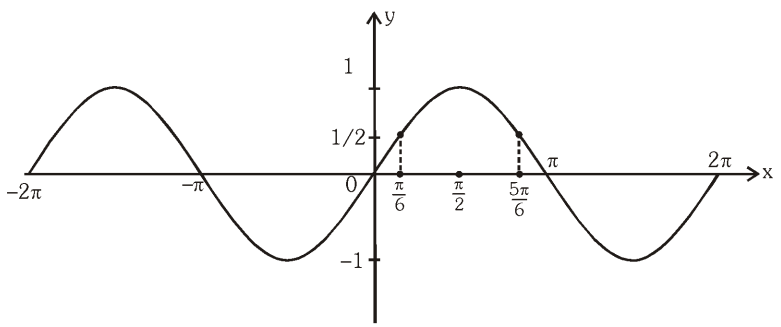
Solution : When $\sin x = \frac{1}{2}$, the two values of x between 0 and 2π are $\pi/6$ and $5\pi/6$.

From the graph of $y = \sin x$, it is obvious that between 0 and 2π,

$\sin x > \frac{1}{2}$ for $\pi/6 < x < 5\pi/6$

Hence, $\sin x > 1/2$

$\Rightarrow \quad 2n\pi + \pi/6 < x < 2n\pi + 5\pi/6, n \in I$



Thus, the required solution set is $\bigcup_{n \in I}\left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right)$ **Ans.**

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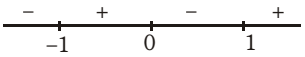
Illustration 20 : Find the values of α lying between 0 and π for which the inequality : $\tan\alpha > \tan^3\alpha$ is valid.

Solution : We have : $\tan\alpha - \tan^3\alpha > 0 \Rightarrow \tan\alpha\,(1 - \tan^2\alpha) > 0$

$\Rightarrow$ $(\tan\alpha)(\tan\alpha + 1)(\tan\alpha - 1) < 0$



So $\tan\alpha < -1,\ 0 < \tan\alpha < 1$

$\therefore$ Given inequality holds for $\alpha \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$ **Ans.**

Do yourself - 6 :

1. Find the solution set of the inequality : $\cos x \geq -1/2$.

2. Find the values of x in the interval $[0, 2\pi]$ for which $4\sin^2 x - 8\sin x + 3 \leq 0$.

3. Solve $\tan^2 x \leq 1$

4. Solve $2\sin^2 x - \sin x - 1 > 0$

5. Solve $\sqrt{\sqrt{3}\cot\theta} < 1$

<u>Miscellaneous Illustration :</u>

Illustration 21 : Solve the following equation : $\tan^2\theta + \sec^2\theta + 3 = 2(\sqrt{2}\sec\theta + \tan\theta)$

Solution : We have $\tan^2\theta + \sec^2\theta + 3 = 2\sqrt{2}\sec\theta + 2\tan\theta$

$\Rightarrow$ $\tan^2\theta - 2\tan\theta + \sec^2\theta - 2\sqrt{2}\sec\theta + 3 = 0$

$\Rightarrow$ $\tan^2\theta + 1 - 2\tan\theta + \sec^2\theta - 2\sqrt{2}\sec\theta + 2 = 0$

$\Rightarrow$ $(\tan\theta - 1)^2 + (\sec\theta - \sqrt{2})^2 = 0$ $\Rightarrow$ $\tan\theta = 1$ and $\sec\theta = \sqrt{2}$

As the periodicity of $\tan\theta$ and $\sec\theta$ are not same, we get

$\theta = 2n\pi + \frac{\pi}{4},\ n \in I$ **Ans.**

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Illustration 22 : Find the solution set of equation $5^{(1+\log_5 \cos x)} = 5/2$.

Solution : Taking log to base 5 on both sides in given equation :

$(1 + \log_5 \cos x).\ \log_5 5 = \log_5(5/2) \quad \Rightarrow \log_5 5 + \log_5 \cos x = \log_5 5 - \log_5 2$

$\Rightarrow \log_5 \cos x = -\log_5 2 \Rightarrow \cos x = 1/2 \qquad \Rightarrow x = 2n\pi \pm \pi/3,\ n \in I$

Ans.

Illustration 23 : If the set of all values of x in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying $|4\sin x + \sqrt{2}| < \sqrt{6}$ is $\left(\frac{a\pi}{24}, \frac{b\pi}{24}\right)$ then find the value of $\left|\frac{a-b}{3}\right|$.

Solution : $|4\sin x + \sqrt{2}| < \sqrt{6}$

$\Rightarrow \quad -\sqrt{6} < 4\sin x + \sqrt{2} < \sqrt{6} \qquad \Rightarrow \quad -\sqrt{6} - \sqrt{2} < 4\sin x < \sqrt{6} - \sqrt{2}$

$\Rightarrow \quad \frac{-(\sqrt{6}+\sqrt{2})}{4} < \sin x < \frac{\sqrt{6}-\sqrt{2}}{4} \qquad \Rightarrow \quad -\frac{5\pi}{12} < x < \frac{\pi}{12}$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Comparing with $\frac{a\pi}{24} < x < \frac{b\pi}{24}$, we get, $a = -10, b = 2$

$\therefore \left|\frac{a-b}{3}\right| = \left|\frac{-10-2}{3}\right| = 4$ **Ans.**

Illustration 24 : The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is -

(A) 0 (B) 5 (C) 6 (D) 10

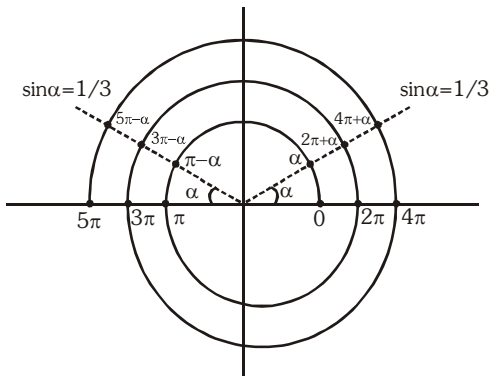
Solution : $3\sin^2 x - 7\sin x + 2 = 0$

$\Rightarrow \quad (3\sin x - 1)(\sin x - 2) = 0$

$\because \quad \sin x \neq 2$

$\Rightarrow \quad \sin x = \frac{1}{3} = \sin\alpha$ (say)

where α is the least positive value of x such that $\sin\alpha = \frac{1}{3}$.



Clearly $0 < \alpha < \frac{\pi}{2}$. We get the solution,

$x = \alpha,\ \pi - \alpha,\ 2\pi + \alpha,\ 3\pi - \alpha,\ 4\pi + \alpha$ and $5\pi - \alpha$.

Hence total six values in $[0, 5\pi]$ **Ans. (C)**

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EXERCISE (O-1)

Straight Objective Type

1. If $2\tan^2\theta = \sec^2\theta$, then the general solution of θ -

(A) $n\pi + \frac{\pi}{4}$ $(n \in I)$ (B) $n\pi - \frac{\pi}{4}$ $(n \in I)$ (C) $n\pi \pm \frac{\pi}{4}$ $(n \in I)$ (D) $2n\pi \pm \frac{\pi}{4}$ $(n \in I)$

TE0001

2. If $\frac{1-\cos 2\theta}{1+\cos 2\theta} = 3$, then the general solution of θ is -

(A) $2n\pi \pm \pi/6$ (B) $n\pi \pm \pi/6$ (C) $2n\pi \pm \pi/3$ (D) $n\pi \pm \pi/3$

where $n \in I$

TE0002

3. The general value of θ satisfying $\sin^2\theta + \sin\theta = 2$ is -

(A) $n\pi + (-1)^n\frac{\pi}{6}$; $n \in \mathbb{Z}$ (B) $2n\pi + \frac{\pi}{4}$; $n \in \mathbb{Z}$

(C) $n\pi + (-1)^n\frac{\pi}{2}$; $n \in \mathbb{Z}$ (D) $n\pi + (-1)^n\frac{\pi}{3}$; $n \in \mathbb{Z}$

TE0003

4. Let $A = \{\theta : \sin(\theta) = \tan(\theta)\}$ and $B = \{\theta : \cos(\theta) = 1\}$ be two sets. Then -

(A) $A = B$ (B) $A \subset B$ and $B - A \neq \phi$

(C) $A \not\subset B$ (D) $B \not\subset A$

TE0004

5. The solution set of $(5 + 4\cos\theta)(2\cos\theta + 1) = 0$ in the interval $[0, 2\pi]$ is :

(A) $\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$ (B) $\left\{\frac{\pi}{3}, \pi\right\}$ (C) $\left\{\frac{2\pi}{3}, \frac{4\pi}{3}\right\}$ (D) $\left\{\frac{2\pi}{3}, \frac{5\pi}{3}\right\}$

TE0005

6. The general solution of equation $4\cos^2 x + 6\sin^2 x = 5$ is -

(A) $x = n\pi \pm \frac{\pi}{2}$ $(n \in I)$ (B) $x = n\pi \pm \frac{\pi}{4}$ $(n \in I)$

(C) $x = n\pi \pm \frac{3\pi}{2}$ $(n \in I)$ (D) None of these

TE0006

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7. If $\tan^2\theta - (1+\sqrt{3})\tan\theta + \sqrt{3} = 0$, then the general value of θ is :

(A) $n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{3}$ (B) $n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{3}$ (C) $n\pi + \frac{\pi}{4}, n\pi - \frac{\pi}{3}$ (D) $n\pi - \frac{\pi}{4}, n\pi - \frac{\pi}{3}$

where $n \in I$

TE0007

8. The general solution of the equation $\tan^2\alpha + 2\sqrt{3}\tan\alpha = 1$ is given by -

(A) $\alpha = \frac{n\pi}{2}\ (n \in I)$ (B) $\alpha = (2n+1)\frac{\pi}{2}\ (n \in I)$

(C) $\alpha = (6n+1)\frac{\pi}{12}\ (n \in I)$ (D) $\alpha = \frac{n\pi}{12}\ (n \in I)$

TE0008

9. The number of solutions of the equation $\sin 2x - 2\cos x + 4\sin x = 4$ in the interval $[0, 5\pi]$ is -

(A) 6 (B) 4 (C) 3 (D) 5

TE0009

10. The set of angles between 0 and 2π satisfying the equation $4\cos^2\theta - 2\sqrt{2}\cos\theta - 1 = 0$ is

(A) $\left\{\frac{\pi}{12}, \frac{5\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}\right\}$ (B) $\left\{\frac{\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{23\pi}{12}\right\}$

(C) $\left\{\frac{5\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}\right\}$ (D) $\left\{\frac{\pi}{12}, \frac{7\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}\right\}$

TE0010

11. If $\tan\theta + \tan 4\theta + \tan 7\theta = \tan\theta \tan 4\theta \tan 7\theta$, then $\theta =$

(A) $\frac{n\pi}{4}$ (B) $\frac{n\pi}{7}$

(C) $\frac{n\pi}{12}, n \neq (12m-6), n, m \in I$ (D) $n\pi$

where $n \in I$

TE0011

12. If $\frac{\tan 2\theta + \tan\theta}{1 - \tan\theta\tan 2\theta} = 0$, then the general value of θ is -

(A) $n\pi$; $n \in I$ (B) $\frac{n\pi}{3}$; $n \in I$ (C) $\frac{n\pi}{4}$ (D) $\frac{n\pi}{6}$; $n \in I$

where $n \in I$

TE0012

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13. The smallest positive angle satisfying the equation $1 + \cos 3x - 2\cos 2x = 0$, is equal to

(A) 15° (B) 22.5° (C) 30° (D) 45°

TE0013

14. If $\tan\theta - \sqrt{2}\sec\theta = \sqrt{3}$, then the general solution of θ is -

(A) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}$ (B) $n\pi + (-1)^n \frac{\pi}{3} - \frac{\pi}{4}$

(C) $n\pi + (-1)^n \frac{\pi}{3} + \frac{\pi}{4}$ (D) $n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{3}$

where $n \in I$

TE0014

15. Number of principal solution(s) of the equation $4 \cdot 16^{\sin^2 x} = 2^{6\sin x}$ is

(A) 1 (B) 2 (C) 3 (D) 4

TE0015

16. The most general values of x for which $\sin x + \cos x = \min\limits_{a \in R} \{1, a^2 - 4a + 6\}$ is given by -

(A) $2n\pi$ (B) $2n\pi + \frac{\pi}{2}$ (C) $n\pi + (-1)^n . \frac{\pi}{4} - \frac{\pi}{4}$ (D) None of these

where $n \in I$

TE0016

17. If the equation $\sin^4 x - (k + 2)\sin^2 x - (k + 3) = 0$ has a solution then k must lie in the interval :

(A) $(-4, -2)$ (B) $[-3, 2)$ (C) $(-4, -3)$ (D) $[-3, -2]$

TE0017

18. Number of values of x satisfying the equation $\log_2(\sin x) + \log_{1/2}(-\cos x) = 0$ in the interval $(-\pi, \pi]$ is equal to -

(A) 0 (B) 1 (C) 2 (D) 3

TE0018

19. The equation $\sin x \cos x = 2$ has :

(A) one solution (B) two solutions (C) infinite solutions (D) no solution

TE0019

20. The number of solutions of the equation $\tan^2 x - \sec^{10} x + 1 = 0$ in (0, 10) is -

(A) 3 (B) 6 (C) 10 (D) 11

TE0020

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21. If $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$, then the greatest positive solution of $1 + \sin^4 x = \cos^2 3x$ is -

(A) π (B) 2π (C) $\frac{5\pi}{2}$ (D) none of these

TE0021

22. The number of solutions of the equation $\sin x = x^2 + x + 1$ is-

(A) 0 (B) 1 (C) 2 (D) None

TE0022

23. **Statement-1:** If $\sin\frac{3x}{2}\cos\frac{5y}{3} = k^8 - 4k^4 + 5$, where $x, y \in R$ then exactly four distinct real values of k are possible.

because

Statement-2: $\sin\frac{3x}{2}$ and $\cos\frac{5y}{3}$ both are less than or equal to one and greater than or equal to – 1.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

TE0023

24. The number of solutions of the equation $2\cos\left(\frac{x}{2}\right) = 3^x + 3^{-x}$ is -

(A) 1 (B) 2 (C) 3 (D) None

TE0024

25. The equation $2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2 + x^{-2}$, $0 < x \le \frac{\pi}{2}$ has

(A) one real solutions (B) more than one real solutions

(C) no real solution (D) none of the above

TE0025

26. If $0 \le x \le 3\pi$, $0 \le y \le 3\pi$ and $\cos x . \sin y = 1$, then the possible number of values of the ordered pair (x, y) is -

(A) 6 (B) 12 (C) 8 (D) 15

TE0026

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27. Given $a^2 + 2a + \text{cosec}^2\left(\frac{\pi}{2}(a+x)\right) = 0$ then, which of the following holds good ?

(A) $a = 1$; $\frac{x}{2} \in I$
(B) $a = -1$; $\frac{x}{2} \in I$
(C) $a \in R$; $x \in \phi$
(D) a , x are finite but not possible to find

TE0027

28. If the equation $\cot^4 x - 2\,\text{cosec}^2 x + a^2 = 0$ has atleast one solution then, sum of all possible integral values of 'a' is equal to

(A) 4
(B) 3
(C) 2
(D) 0

TE0028

29. The complete solution set of the inequality $\tan^2 x - 2\sqrt{2}\tan x + 1 \le 0$ is -

(A) $n\pi + \frac{\pi}{8} \le x \le \frac{3\pi}{8} + n\pi,\ n \in I$
(B) $n\pi + \frac{\pi}{4} \le x \le \frac{3\pi}{4} + n\pi,\ n \in I$
(C) $n\pi + \frac{\pi}{16} \le x \le \frac{3\pi}{8} + n\pi,\ n \in I$
(D) $n\pi + \frac{\pi}{3} \le x \le \frac{2\pi}{3} + n\pi,\ n \in I$

TE0029

30. Number of integral solution(s) of the inequality $2\sin^2 x - 5\sin x + 2 > 0$ in $x \in [0, 2\pi]$, is -

(A) 3
(B) 4
(C) 5
(D) 6

TE0030

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EXERCISE (O-2)

Multiple Correct Answer Type

1. The equation $2\sin\frac{x}{2}.\cos^2x + \sin^2x = 2\sin\frac{x}{2}.\sin^2x + \cos^2x$ has a root for which

(A) $\sin 2x = 1$ (B) $\sin 2x = -1$ (C) $\cos x = \frac{1}{2}$ (D) $\cos 2x = -\frac{1}{2}$

TE0031

2. $\sin^2x - \cos 2x = 2 - \sin 2x$ if

(A) $x = n\pi/2, n \in I$ (B) $\tan x = 3/2$

(C) $x = (2n + 1)\pi/2, n \in I$ (D) $x = n\pi + (-1)^n \sin^{-1}(2/3), n \in I$

TE0032

3. $4\sin^4x + \cos^4x = 1$ if

(A) $x = n\pi$ (B) $x = n\pi \pm \frac{1}{2}\cos^{-1}\left(\frac{1}{5}\right)$

(C) $x = \frac{n\pi}{2}$ (D) none of these, $(n \in I)$

TE0033

4. Which of the following set of values of x satisfies the equation

$2^{(2\sin^2 x - 3\sin x + 1)} + 2^{(2 - 2\sin^2 x + 3\sin x)} = 9$ is -

(A) $x = x\pi, n \in I$ (B) $x = 2n\pi + \frac{\pi}{2}, n \in I$

(C) $x = n\pi \pm \frac{\pi}{6}, n \in I$ (D) $x = n\pi \pm \frac{\pi}{3}, n \in I$

TE0034

5. $5\sin^2 x + \sqrt{3}\sin x\cos x + 6\cos^2x = 5$ if

(A) $\tan x = -1/\sqrt{3}$ (B) $\sin x = 0$

(C) $x = n\pi + \pi/2, n \in I$ (D) $x = n\pi + \pi/6, n \in I$

TE0035

6. If sum of all the solution of the equation $\cot x + \operatorname{cosec} x + \sec x = \tan x$ in $[0, 2\pi]$ is $\frac{k\pi}{2}$, then the value of k is greater than

(A) 1 (B) 2 (C) 3 (D) 4

TE0036

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7. The equation $\sin x + \cos(k + x) + \cos(k - x) = 2$ has real solution(s), then $\sin k$ can be

(A) $\frac{-3}{4}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

TE0037

8. Which of the following equations have no solution ?

(A) $2^{|x|} = \sin x^2$ (B) $3^{|\sin\sqrt{x}|} = |\cos x|$ (C) $x^2 = -\cos x$ (D) $3x^2 = 1 - 2\cos x$

TE0038

9. If $m > 0, n < 5, 0 < m + n < 10$ and $\frac{x^2+5}{2} = x - 2\cos(m + nx)$ has atleast one real root, then

(A) the greatest value of $(m + n)$ is 3π. (B) the greatest value of $(m + n)$ is 2π.
(C) the least value of $(m + n)$ is π. (D) the least value of $(m + n)$ is 2π.

TE0039

10. If $x^4 + 3\cos(ax^2 + bx + c) = 2(x^2 - 2)$ has two solutions with $a, b, c \in (2, 5)$ then
(A) $a + b + c = \pi$ (B) $a - b + c = \pi$ (C) $a + b + c = 3\pi$ (D) $a - b + c = 3\pi$

TE0040

11. The equation $\sin^6 x + \cos^6 x = a^2$ has real solution if

(A) $a \in (-1, 1)$ (B) $a \in \left(-1, -\frac{1}{2}\right)$ (C) $a \in \left(-1, \frac{1}{2}\right)$ (D) $a \in \left(\frac{1}{2}, 1\right)$

TE0101

12. If $\sin(x - y) = \cos(x + y) = 1/2$ then the values of x & y lying between 0 and π are given by :
(A) $x = \pi/4, y = 3\pi/4$ (B) $x = \pi/4, y = \pi/12$
(C) $x = 5\pi/4, y = 5\pi/12$ (D) $x = 11\pi/12, y = 3\pi/4$

TE0102

13. sinx, sin2x, sin3x are in A.P if
(A) $x = n\pi/2, n \in I$ (B) $x = n\pi, n \in I$ (C) $x = 2n\pi, n \in I$ (D) $x = (2n + 1)\pi, n \in I$

TE0103

14. $\cos 4x \cos 8x - \cos 5x \cos 9x = 0$ if
(A) $\cos 12x = \cos 14x$ (B) $\sin 13x = 0$
(C) $\sin x = 0$ (D) $\cos x = 0$

TE0104

15. $\sin x - \cos^2 x - 1$ assumes the least value for the set of values of x given by :
(A) $x = n\pi + (-1)^{n+1}(\pi/6), n \in I$ (B) $x = n\pi + (-1)^n(\pi/6), n \in I$
(C) $x = n\pi + (-1)^n(\pi/3), n \in I$ (D) $x = n\pi - (-1)^n(\pi/6), n \in I$

TE0105

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EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 and 2

Let $f(x) = \cos x + \sin x - 1$ and $g(x) = \sin 2x - 2$.

1. Number of solutions of the equation $f(x) = g(x)$ in $x \in \left[-\pi, \frac{7\pi}{2}\right]$ is equal to

(A) 4 (B) 5 (C) 6 (D) 8

TE0041

2. If the equation $f(x) = k$ has atleast one real solution then the number of possible integral values of k is equal to

(A) 1 (B) 2 (C) 3 (D) 4

TE0041

Paragraph for Question 3 to 5

Let $f(x) = \sin^2 x - (a - 1)\sin x + 2(a - 3)$

On the basis of above information, answer the following questions :

3. If $x \in [0, \pi]$ and $f(x) = 0$ has exactly one real root, then 'a' lies in

(A) (3, 5) (B) (2, 4) (C) (4, 5) (D) none of these

TE0042

4. If $f(x) = 0$ have two real roots in $(0, \pi)$, then $a \in$

(A) (1, 2) (B) (3, 4) (C) $(3, 4) \cup \{5\}$ (D) (3, 5)

TE0042

5. If $f(x) \geq 0\ \forall\ x \in R$ then range of 'a' is

(A) $[2, \infty)$ (B) $[4, \infty)$ (C) $(4, \infty)$ (D) none of these

TE0042

Paragraph for Questions 6 to 8

On the basis of the given polynomial equation

$x^4 - (\sin\theta + \cos\theta)x^3 + (\sin\theta\cos\theta - 1)x^2 + (\sin\theta + \cos\theta)x - \sin\theta\cos\theta = 0.$

Answer the following :

6. Sum of any 2 roots of the equation can never exceed -

(A) –1 (B) 0 (C) 1 (D) 2

TE0106

7. Product of the roots of the equation will always be more than -

(A) –1 (B) 0 (C) 1 (D) 2

TE0107

8. Number of real roots of the equation is/are -

(A) 1 (B) 2 (C) 3 (D) 4

TE0108

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Matrix Match Type

9. **List-I** **List-II**

(I) Number of common solutions of the equations (P) 2

$2\cos^2 x - 3\cos x + 1 = 0$ and $\tan\left(\frac{3x}{4}\right) + 1 = 0$,

where $-\pi < x \le 3\pi$, is equal to

TE0043

(II) Number of solutions of the equation (Q) 12

$\sin^2 x + 2\sin x - 4\cos x - \sin 2x = 0$ in $(0, 2\pi)$ is equal to

TE0044

(III) If the sum of all possible values of $x \in (0, 2\pi)$ satisfying (R) 4

the equation $2\cos x \operatorname{cosec} x - 4\cos x - \operatorname{cosec} x = -2$

is equal to $\frac{k\pi}{4}$ $(k \in N)$, then the value of k is (S) 3

TE0045

(IV) Let $x - y = \frac{1}{4}$ and $\cos(\pi x) \cdot \cos(\pi y) = \frac{1}{2\sqrt{2}}$ (T) 1

where $x, y \in (0, 2)$ then the number of ordered pair(s) (x, y)

is equal to

TE0046

(A) I → P; II → P; III → Q; IV → R (B) I → T; II → P; III → S; IV → R

(C) I → R; II → P; III → S; IV → T (D) I → Q; II → P; III → S; IV → T

10. **Column I** **Column II**

(A) The number of real roots of the equation (P) 1

$\cos^7 x + \sin^4 x = 1$ in $(-\pi, \pi)$ is

TE0047

(B) The number of solutions of the equation (Q) 2

$|\cot x| = \cot x + \frac{1}{\sin x}$ $(0 < x < \pi)$ is

TE0048

(C) The number of solution(s) of equation in $(0, 3\pi)$ (R) 0

$\sin^2 x + 3\sec^2 x = -(\tan^2 x + \cot^2 x)$ is (S) 3

(D) The number of values of $x \in [-2\pi, 2\pi]$, which (T) 5

satisfy $\operatorname{cosec} x = 1 + \cot x$ **TE0050**

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EXERCISE (O-4)

Numerical Grid Type

1. If $\sin \pi x + \cos \pi x = 0$, then the sum of all solutions in [0, 100] is

TE0055

2. If α, β ($\alpha - \beta \neq 2n\pi$, $n \in I$) are different values of θ satisfying the equation $5\cos\theta - 12\sin\theta = 11$. If the value of $\sin(\alpha + \beta) = -\frac{5k}{169}$, then the value of k is

TE0109

3. If $x \in \left(\pi, \frac{3\pi}{2}\right)$ then $4\cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right) + \sqrt{4\sin^4 x + \sin^2 2x}$ is always equal to

TE0110

4. The number of integral values of λ, for which equation $4\cos x + 3\sin x = 2\lambda + 1$ has a solution is

TE0111

5. If $2\cos x + \sin x = 1$, then find the sum of all possible values of $7\cos x + 6\sin x$.

TE0112

6. If $2\tan^2 x - 5\sec x - 1 = 0$ has 7 different roots in $\left[0, \frac{n\pi}{2}\right]$, $n \in N$, then find the greatest value of n.

TE0113

7. If the arithmetic mean of the roots of the equation $4\cos^3 x - 4\cos^2 x - \cos(\pi + x) - 1 = 0$ in the interval [0, 315] is equal to $k\pi$, then find the value of k

TE0114

8. In $(0, 6\pi)$, find the number of solutions of the equation $\tan\theta + \tan 2\theta + \tan 3\theta = \tan\theta.\tan 2\theta.\tan 3\theta$

TE0115

9. Find the number of solutions of the equation $\cos 6x + \tan^2 x + \cos 6x \,.\, \tan^2 x = 1$ in the interval $[0, 2\pi]$.

TE0116

10. Find the number of solutions of $\sin\theta + 2\sin 2\theta + 3\sin 3\theta + 4\sin 4\theta = 10$ in $(0, \pi)$.

TE0117

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EXERCISE (JM)

1. Let A and B denote the statements

A : $\cos\alpha + \cos\beta + \cos\gamma = 0$

B : $\sin\alpha + \sin\beta + \sin\gamma = 0$

If $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) = -\frac{3}{2}$, then - **[AIEEE 2009]**

(1) Both **A** and **B** are true
(2) Both **A** and **B** are false
(3) **A** is true and **B** is false
(4) **A** is false and **B** is true

TE0071

2. The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are: **[AIEEE 2011]**

(1) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$
(2) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$
(3) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$
(4) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

TE0072

3. In a ΔPQR, if $3\sin P + 4\cos Q = 6$ and $4\sin Q + 3\cos P = 1$, then the angle R is equal to : **[AIEEE-2012]**

(1) $\frac{3\pi}{4}$
(2) $\frac{5\pi}{6}$
(3) $\frac{\pi}{6}$
(4) $\frac{\pi}{4}$

TE0073

4. If $0 \le x < 2\pi$, then the number of real values of x, which satisfy the equation $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$, is :- **[JEE(Main) 2016]**

(1) 9
(2) 3
(3) 5
(4) 7

TE0074

5. If sum of all the solutions of the equation $8\cos x \cdot \left(\cos\left(\frac{\pi}{6}+x\right).\cos\left(\frac{\pi}{6}-x\right)-\frac{1}{2}\right) = 1$ in $[0, \pi]$ is $k\pi$, then k is equal to : **[JEE(Main) 2018]**

(1) $\frac{13}{9}$
(2) $\frac{8}{9}$
(3) $\frac{20}{9}$
(4) $\frac{2}{3}$

TE0075

6. If $0 \le x < \frac{\pi}{2}$, then the number of values of x for which $\sin x - \sin 2x + \sin 3x = 0$, is **[JEE(Main) 19]**

(1) 2
(2) 1
(3) 3
(4) 4

TE0076

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7. The sum of all values of $\theta \in \left(0, \frac{\pi}{2}\right)$ satisfying $\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}$ is : **[JEE(Main) 19]**

(1) $\frac{\pi}{2}$ (2) π (3) $\frac{3\pi}{8}$ (4) $\frac{5\pi}{4}$

TE0077

8. Let S = $\{\theta \in [-2\pi, 2\pi] : 2\cos^2\theta + 3\sin\theta = 0\}$. Then the sum of the elements of S is

[JEE(Main) 19]

(1) $\frac{13\pi}{6}$ (2) π (3) 2π (4) $\frac{5\pi}{3}$

TE0078

9. All the pairs (x, y) that satisfy the inequality $2\sqrt{\sin^2 x - 2\sin x + 5} \cdot \frac{1}{4^{\sin^2 y}} \le 1$ also satisfy the eauation.

[JEE(Main) 19]

(1) sin x = |siny| (2) sin x = 2 sin y (3) 2|sinx| = 3siny (4) 2sin x = siny

TE0079

10. The number of solutions of the equation $1 + \sin^4 x = \cos^2 3x,\ x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is : **[JEE(Main) 19]**

(1) 5 (2) 4 (3) 7 (4) 3

TE0080

11. Let S be the set of all $\alpha \in R$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to : **[JEE(Main) 19]**

(1) [2, 6] (2) [3,7] (3) R (4) [1,4]

TE0081

12. If $\sin^4\alpha + 4\cos^4\beta + 2 = 4\sqrt{2}\sin\alpha\cos\beta$; $\alpha, \beta \in [0, \pi]$, then $\cos(\alpha + \beta) - \cos(\alpha - \beta)$ is equal to :

[JEE(Main) 2019]

(1) 0 (2) $-\sqrt{2}$ (3) –1 (4) $\sqrt{2}$

TE0082

13. The number of distinct solutions of the equation $\log_{\frac{1}{2}}|\sin x| = 2 - \log_{\frac{1}{2}}|\cos x|$ in the interval [0, 2π], is ———. **[JEE(Main) 20]**

TE0083

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14. All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in : **[JEE(Main) 2021]**

(1) $\left(0,\frac{\pi}{2}\right)\cup\left(\pi,\frac{3\pi}{2}\right)$

(2) $\left(0,\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\frac{3\pi}{4}\right)\cup\left(\pi,\frac{7\pi}{6}\right)$

(3) $\left(0,\frac{\pi}{4}\right)\cup\left(\frac{\pi}{2},\frac{3\pi}{4}\right)\cup\left(\frac{3\pi}{2},\frac{11\pi}{6}\right)$

(4) $\left(0,\frac{\pi}{4}\right)\cup\left(\frac{\pi}{2},\frac{3\pi}{4}\right)\cup\left(\pi,\frac{5\pi}{4}\right)\cup\left(\frac{3\pi}{2},\frac{7\pi}{4}\right)$

TE0118

15. The number of solutions of the equation $\cos\left(x+\frac{\pi}{3}\right)\cos\left(\frac{\pi}{3}-x\right)=\frac{1}{4}\cos^2 2x$, $x \in [-3\pi, 3\pi]$ is :

(A) 8 (B) 5 (C) 6 (D) 7

[JEE(Main) 2022]

TE0119

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EXERCISE (JA)

1. For $x \in (0, \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has **[JEE(Advanced)-2014, 3(–1)]**

(A) infinitely many solutions (B) three solutions

(C) one solution (D) no solution

TE0087

2. The number of distinct solutions of equation $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is **[JEE 2015, 4M, –0M]**

TE0088

3. Let $S = \left\{x \in (-\pi, \pi) : x \neq 0, \pm\frac{\pi}{2}\right\}$. The sum of all distinct solution of the equation $\sqrt{3}\sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$ in the set S is equal to -

[JEE(Advanced)-2016, 3(–1)]

(A) $-\frac{7\pi}{9}$ (B) $-\frac{2\pi}{9}$ (C) 0 (D) $\frac{5\pi}{9}$

TE0089

4. Let a, b, c be three non-zero real numbers such that the equation

$$\sqrt{3}a\cos x + 2b\sin x = c, \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{b}{a}$ is _____

[JEE(Advanced)-2018, 3(0)]

TE0090

Answer the following by appropriately matching the lists based on the information given in the paragraph

Let $f(x) = \sin(\pi\cos x)$ and $g(x) = \cos(2\pi\sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order :

$X = \{x : f(x) = 0\}$, $Y = \{x : f'(x) = 0\}$

$Z = \{x : g(x) = 0\}$, $W = \{x : g'(x) = 0\}$.

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List-I contains the sets X,Y,Z and W. List -II contains some information regarding these sets.

List-I		**List-II**	
(I)	X	(P)	$\supseteq\left\{\frac{\pi}{2},\frac{3\pi}{2},4\pi,7\pi\right\}$
(II)	Y	(Q)	an arithmetic progression
(III)	Z	(R)	NOT an arithmetic progression
(IV)	W	(S)	$\supseteq\left\{\frac{\pi}{6},\frac{7\pi}{6},\frac{13\pi}{6}\right\}$
		(T)	$\supseteq\left\{\frac{\pi}{3},\frac{2\pi}{3},\pi\right\}$
		(U)	$\supseteq\left\{\frac{\pi}{6},\frac{3\pi}{4}\right\}$

5. Which of the following is the only CORRECT combination ? **[JEE(Advanced)-2019, 3(–1)]**

(1) (II), (R), (S) (2) (I), (P), (R) (3) (II), (Q), (T) (4) (I), (Q), (U)

TE0091

6. Which of the following is the only CORRECT combination ? **[JEE(Advanced)-2019, 3(–1)]**

(1) (IV), (Q), (T) (2) (IV), (P), (R), (S) (3) (III), (R), (U) (4) (III), (P), (Q), (U)

TE0091

7. Let $f : [0, 2] \to \mathbb{R}$ be the function defined by

$f(x) = (3 - \sin(2\pi x)) \sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right)$ **[JEE(Advanced)-2020]**

If $\alpha, \beta \in [0, 2]$ are such that $\{x \in [0, 2] : f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is _____

TE0100

8. Consider the following lists :

List-I		**List-II**	
(I)	$\left\{x \in \left[-\frac{2\pi}{3},\frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$	(P)	has two elements
(II)	$\left\{x \in \left[-\frac{5\pi}{18},\frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$	(Q)	has three elements
(III)	$\left\{x \in \left[-\frac{6\pi}{5},\frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$	(R)	has four elements
(IV)	$\left\{x \in \left[-\frac{7\pi}{4},\frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$	(S)	has five elements
		(T)	has six elements

The correct option is : **[JEE(Advanced)-2022]**

(A) (I) → (P); (II) → (S); (III) → (P); (IV) → (S)
(B) (I) → (P); (II) → (P); (III) → (T); (IV) → (R)
(C) (I) → (Q); (II) → (P); (III) → (T); (IV) → (S)
(D) (I) → (Q); (II) → (S); (III) → (P); (IV) → (R)

TE0120

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ANSWER KEY

Do yourself-1

1. **(a)** $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$ **(b)** $\theta = (2n+1)\frac{\pi}{3}, n \in I$ **(c)** $\theta = \frac{4n\pi}{3}, n \in I$

(d) $\theta = \frac{n\pi}{2}, n \in I$ **(e)** $\theta = n\pi \pm \frac{\pi}{12}, n \in I$ **(f)** $\theta = 2n\pi + (-1)^{n+1}\pi, n \in I$

2. A,B,C,D **3.** D **4.** $\frac{n\pi}{4} \pm \sqrt{1+\frac{n^2\pi^2}{16}}, n \in I$

5. (i) $n\pi + (-1)^n \frac{\pi}{4}, n \in I$ (ii) $n\pi + \frac{\pi}{3} + 1, n \in I$ (iii) $n\pi - \frac{\pi}{4}, n \in I$

(iv) $n\pi + (-1)^n \frac{\pi}{3}, n \in I$ (v) $n\pi \pm \frac{\pi}{4}, n \in I$

6. (i) $\frac{m\pi}{4}, m \in I$ or $\frac{(2m+1)\pi}{10}, m \in I$ (ii) $2n\pi \pm \frac{\pi}{3}, \ n \in I$

(iii) $\left(2n+\frac{1}{2}\right)\frac{\pi}{5}, n \in I$ or $2n\pi - \frac{\pi}{2}, n \in I$ (iv) $\left(n+\frac{1}{2}\right)\frac{\pi}{9}, n \in I$

(v) $\left(n+\frac{1}{4}\right)\frac{\pi}{2}, n \in I$ (vi) $2n\pi + \frac{2\pi}{3}, n \in I$

(vii) $n\pi \pm \frac{\pi}{6}$ (viii) $\left(n+\frac{1}{3}\right)\frac{\pi}{3}, n \in I$

Do yourself-2

1. **(a)** $x = n\pi + (-1)^{n+1}\frac{\pi}{6}, n \in I$ **(b)** $\alpha = \frac{n\pi}{2}$ or $\alpha = \frac{k\pi}{2} + \frac{3\pi}{8}, \ n, k \in I$

(c) $\theta = n\pi \pm \frac{\pi}{3}, n \in I$

(d) $\theta = n\pi + (-1)^n\alpha$, where $\alpha = \sin^{-1}\left(\frac{\sqrt{17}-1}{8}\right)$ or $\sin^{-1}\left(\frac{-1-\sqrt{17}}{8}\right), n \in I$

2. $\theta = \left\{-\frac{\pi}{4}, -\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{2}\right\}$

3. $(2n+1)\pi, n \in I$ **4.** $2n\pi - \frac{\pi}{2}, n \in I$ or $n\pi + (-1)^{n+1}\frac{\pi}{6}, n \in I$

5. B **6.** 3 **7.** B

8. (i) $n\pi + \frac{\pi}{3}, n \in I$ or $n\pi + \frac{\pi}{4}, n \in I$

(ii) $n\pi + (-1)^n \frac{\pi}{10}, n \in I$ or $n\pi - (-1)^n \frac{3\pi}{10}, n \in I$

(iii) $x = \frac{n\pi}{3} - \frac{\pi}{9}, n \in I$

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Do yourself-3

1. **(a)** $x = 2n\pi - \frac{\pi}{4}, n \in I$ **(b)** $2m\pi + \frac{\pi}{2}, m \in I$

2. $2n\pi, n \in I$ or $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$ **3.** $x = n\pi + \frac{\pi}{6}, n \in I$, $x = \frac{n\pi}{2} - \frac{\pi}{12}, n \in I$

4. (i) $n\pi + \frac{\pi}{6} + (-1)^n \frac{\pi}{4}, n \in I$ (ii) $2n\pi + \frac{\pi}{2}, n \in I$ or $2n\pi + 2\alpha$ where $\alpha = \tan^{-1}\frac{3}{7}, n \in I$

Do yourself-4

1. $\theta = n\pi$ or $\theta = \frac{m\pi}{3} \pm \frac{\pi}{9}$; $n, m \in I$ **2.** $x = \frac{n\pi}{3}, n \in I$ and $k\pi \pm \frac{\pi}{3}, k \in I$

3. $n\pi - \frac{\pi}{4}, n \in I$ **4.** $n\pi - \frac{\pi}{4}, n \in I$

5. $2n\pi + \frac{\pi}{2}, n \in I$ or $2n\pi, n \in I$ or $n\pi + \frac{\pi}{4}, n \in I$

6. D

7. (i) $2n\pi, n \in I$ or $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$

(ii) $x = (2n+1)\frac{\pi}{4}, n \in I$ or $x = (2n+1)\frac{\pi}{2}, n \in I$ or $x = n\pi \pm \frac{\pi}{6}, n \in I$

(iii) $m\pi, m \in I$ or $\frac{m\pi}{n-1}, m \in I$ or $\left(m + \frac{1}{2}\right)\frac{\pi}{n}, m \in I$

Do yourself-5

1. D **2.** $x = \frac{\pi}{4}$ **3.** $(4p - 3)\frac{\pi}{2}, p \in I$ **4.** $2m\pi + \frac{\pi}{2}, m \in I$

5. A **6.** C

Do yourself - 6

1. $\bigcup_{n \in I}\left[2n\pi - \frac{2\pi}{3}, 2n\pi + \frac{2\pi}{3}\right]$ **2.** $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$ **3.** $x \in \left[n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{4}\right] : n \in I$

4. $\left(2n\pi + \frac{7\pi}{6}, 2n\pi + \frac{11\pi}{6}\right)$ **5.** $\theta \in (n\pi + \pi/3, n\pi + \pi/2]$

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EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	C	C	C	B	A	C	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	C	D	C	C	D	B	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	D	A	C	A	B	D	A	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	B,C	A,B	B,C	A,C	A,B,C,D	B,C	A,C,D	A,C	B,C
Que.	11	12	13	14	15					
Ans.	B,D	B,D	A,B,C,D	A,B,C	A,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	C	C	A	B	B	D	A	D	A
Q.10	A	B	C	D					
	S	P	R	Q					

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5025	24	2	6	8	15.00	50.00	17.00	7.00	0.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	3	4	1	1	1	3	1	1
Que.	11	12	13	14	15					
Ans.	1	2	8.00	4	D					

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8
Ans.	D	8	C	0.50	3	2	1.00	B

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FUNDAMENTALS OF GEOMETRY

SIMILAR TRIANGLES

We call two figures similar if one is simply a blown-up or scaled-up, and possibly rotated and/or flipped, version of the other.

Sufficient Conditions For Similarity of Triangles.

Two triangles are said to be similar if any one of the following conditions is satisfied.

(i) their corresponding angles are equal.

(ii) their corresponding sides are proportional.

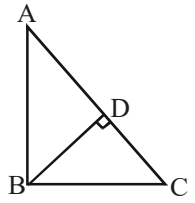
Based on the above, there are three axioms for similarity of two triangles.

1. A.A. (Angle-Angle) Axiom of Similarity.
2. S.A.S (Side-Angle-Side) Axiom of Similarity.
3. S.S.S. (Side-Side-Side) Axiom of Similarity.

Note :

(i) Congruent triangles are necessarily similar but the similar triangles may not be congruent.

(ii) If ABC is a triangle, right-angled at B and $BD \perp AC$, then $\Delta ABC \sim \Delta ADB \sim \Delta BDC$



(iii) If two triangles are similar to a third triangle, then they are similar to each other.

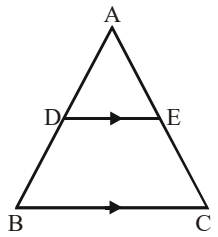
Similar Polygons

Two polygons with the same number of sides are said to be similar if their corresponding angles are equal as well as their corresponding sides are proportional.

Theorem 1 :

Basic Proportionality Theorem (B.P.T.)

If a line is drawn parallel to a side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio. i.e., if in a ΔABC, $DE \| BC$ then, $\frac{AD}{DB} = \frac{AE}{EC}$.



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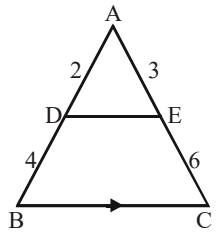
Theorem 2 :

Converse of B.P.T.

If a line divides any two sides of a triangle proportionally, then it must be parallel to the third side.

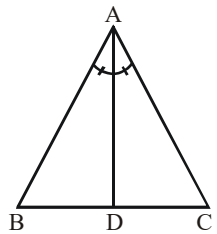


i.e., If in a ΔABC, $\frac{AD}{DB} = \frac{AE}{EC}$ then $DE \| BC$.

Theorem 3 :

Angle Bisector Theorems

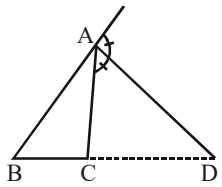
(i) Angle Bisector Theorem (Internal)



The internal bisector of an angle of a triangle divides the opposite side, internally, in the ratio of the sides containing the angle. i.e., If in a ΔABC, AD bisects $\angle A$,

then $\frac{AB}{AC} = \frac{BD}{DC}$.

(ii) Angle Bisector Theorem (External)



The bisector of an exterior angle of a triangle divides the opposite side (provided bisector and opposite side are not parallel) externally, in the ratio of the sides containing the angle.

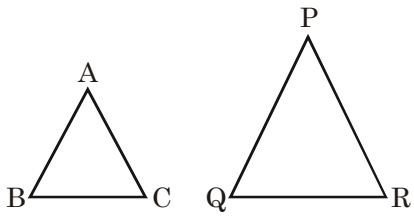
i.e., If in a ΔABC, AD bisects the exterior angle A and intersects side BC produced in D, then $\frac{BD}{CD} = \frac{AB}{AC}$

Converse of both the angle bisector theorems are also true.

Theorem 4 :

Proportion Applied to Area

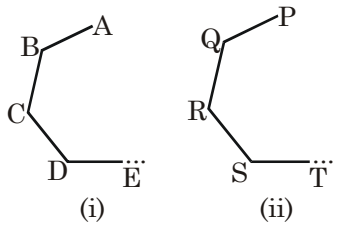
The areas of similar triangles are proportional to the squares of corresponding sides.



$$\frac{\text{Area}(\Delta ABC)}{\text{Area}(\Delta PQR)} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{CA^2}{RP^2} \qquad (\Delta ABC \sim \Delta PQR)$$

Theorem 5 :

The areas of similar polygons are proportional to the squares of corresponding sides.



$$\frac{\text{Area } 1^{st} \text{ poly}}{\text{Area } 2^{nd} \text{ poly}} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{CD^2}{RS^2} = \ldots$$

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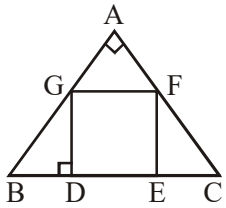
Illustration-1 : In the given figure, DEFG is a square and $\angle BAC = 90°$. Prove that :

(i) $\Delta AGF \sim \Delta DBG$

(ii) $\Delta AGF \sim \Delta EFC$

(iii) $\Delta DBG \sim \Delta EFC$

(iv) $DE^2 = BD \times EC$



Solution : (i) In Δs AGF and DBG, we have :

$\angle GAF = \angle BDG = 90°$ and $\angle AGF = \angle DBG$ [corresponding $\angle$s]

$\therefore \Delta AGF \sim \Delta DBG$ [by A.A. similarity]

(ii) In Δs AGF and EFC, we have :

$\angle FAG = \angle CEF = 90°$ and $\angle AFG = \angle ECF$ [corresponding $\angle$s]

$\therefore \Delta AGF \sim \Delta EFC$ [by A.A. similarity]

(iii) Since, $\Delta AGF \sim \Delta DBG$ and $\Delta AGF \sim \Delta EFC$, it follows that $\Delta DBG \sim \Delta EFC$.

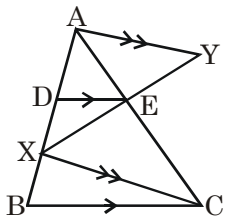
Note : If two triangles are similar to a third triangle, then they are similar to each other.

(iv) Since, $\Delta DBG \sim \Delta EFC$, we have :

$$\frac{BD}{EF} = \frac{DG}{EC} \text{ or } \frac{BD}{DE} = \frac{DE}{EC} \quad [\because EF = DE, DG = DE \text{ (Sides of a sq.)}]$$

Hence, $DE^2 = BD \times EC$.

Illustration-2 : Given the $\overline{DE} \parallel \overline{BC}$ and $\overline{AY} \parallel \overline{XC}$, prove that $\frac{EY}{EX} = \frac{AD}{DB}$.



Solution : The ratios of side lengths in the problem suggest we look for similar triangles.

Since $\overline{AY} \parallel \overline{XC}$, we have $\Delta AYE \sim \Delta CXE$.

$\Rightarrow$ EY/EX = AE/EC. All we have left is to show that AE/EC = AD/DB.

Since $\overline{DE} \parallel \overline{BC}$, we have $\Delta ADE \sim \Delta ABC$. Therefore, AD/AB = AE/AC, which is almost what we want! We break AB and AC into AD + DB and AE + EC, hoping we can do a little algebra to finish :

$$\frac{AD}{AD + DB} = \frac{AE}{AE + EC}$$

If only we could get rid of the AD and AE in the denominators-then we would have AD/DB = AE/EC.

$$\frac{AD + DB}{AD} = \frac{AE + EC}{AE}$$

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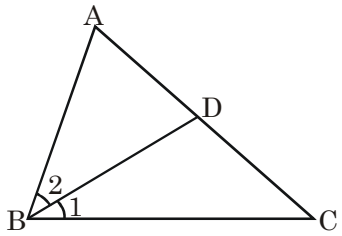
Therefore, $\frac{AD}{AD}+\frac{DB}{AD}=\frac{AE}{AE}+\frac{EC}{AE}$, so $1+\frac{DB}{AD}=1+\frac{EC}{AE}$, which gives us $\frac{DB}{AD}=\frac{EC}{AE}$

Flipping these fractions back over gives us AD/DB = AE/EC. Therefore, we have EY/EX = AE/EC = AD/DB, as desired.

Illustration-3 : In ΔABC, $\angle B = 2\angle C$ and the bisector of $\angle B$ intersects AC at D. Prove that $\frac{BD}{DA}=\frac{BC}{BA}$.

Solution : Given : In ΔABC, $\angle B = 2\angle C$ and BD is bisector of $\angle B$.

To Prove : $\frac{BD}{DA}=\frac{BC}{BA}$



BD is bisector of $\angle CBA$.

$\therefore \frac{BC}{BA}=\frac{CD}{AD}$ ...(1)

$2\angle C = \angle 1 + \angle 2$ [given]

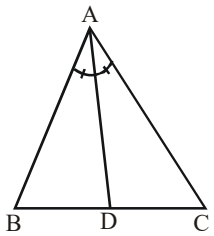
But $\angle 1 = \angle 2$

$\therefore$ $2\angle C = 2\angle 1$

$\Rightarrow$ $\angle C = \angle 1 \Rightarrow BD = CD$ ...(2)

from (1) & (2) $\frac{BC}{BA}=\frac{BD}{DA}$

Illustration-4 : In a ΔABC, AD is the bisector of $\angle BAC$.
If AB = 3.5 cm, AC = 4.2 cm and DC = 2.4 cm
Find BD.



Solution : Since, AD is the bisector $\angle A$, we have $\frac{BD}{DC}=\frac{AB}{AC}$

$\therefore \frac{BD}{2.4}=\frac{3.5}{4.2}\Rightarrow BD=\frac{2.4\times 3.5}{4.2}=2\text{cm}.$

Illustration-5 : O is any point inside a triangle ABC. The bisector of $\angle AOB$, $\angle BOC$ and $\angle COA$ meet the sides AB, BC and CA in points D,E,F respectively. Prove that
AD ×BE × CF = DB × EC × FA.

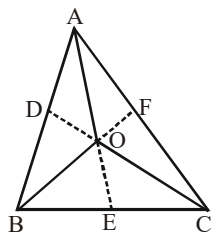
Solution : Given : O is any point inside a ΔABC. The bisectors of $\angle AOB$, $\angle BOC$ and $\angle COA$ meet the sides AB, BC and CA in points D, E, F respectively.
To prove : AD.BE.CF= DB.EC.FA
Proof : In ΔAOB, OD is the bisector of $\angle AOB$.

$\therefore \frac{OA}{OB}=\frac{AD}{DB}$ ...(i)

In ΔBOC, OE is the bisector of $\angle BOC$

$\therefore \frac{OB}{OC}=\frac{BE}{EC}$ ...(ii)

In ΔCOA, OF is the bisector of $\angle COA$.



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$\therefore \dfrac{OC}{OA} = \dfrac{CF}{FA}$ ...(iii)

Multiplying the corresponding sides of (i), (ii) and (iii), we get :

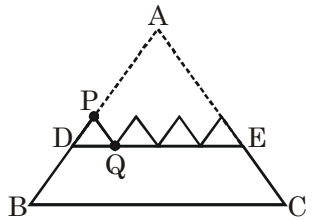
$$\frac{OA}{OB} \times \frac{OB}{OC} \times \frac{OC}{OA} = \frac{AD}{DB} \times \frac{BE}{EC} \times \frac{CF}{FA}$$

$\Rightarrow \quad 1 = \dfrac{AD}{DB} \times \dfrac{BE}{EC} \times \dfrac{CF}{FA}$

$\Rightarrow \quad DB \times EC \times FA = AD \times BE \times CF$

or $\quad AD \times BE \times CF = DB \times EC \times FA.$

Illustration-6 : All the triangles in the diagram below are similar to isosceles triangle ABC in which AB = AC. Each of the smaller triangle has area 1 unit^2 and ΔABC has area 40 unit^2, then area of trapezium DBCE is



(A) 16 (B) 8 (C) 24 (D) 32

Solution : $\Delta PDQ \sim \Delta ABC$

$$\frac{ar.\Delta PDQ}{ar.\Delta ABC} = \frac{DQ^2}{BC^2} = \frac{1}{40}$$

$$\therefore DQ = \frac{BC}{2\sqrt{10}}$$

Also DE = 4.DQ

$$DE = \frac{4.BC}{2\sqrt{10}} \text{ Hence } \frac{DE}{BC} = \frac{2}{\sqrt{10}}$$

$$\frac{ar\Delta ADE}{ar\Delta ABC} = \frac{DE^2}{BC^2} = \frac{4}{10} = \frac{2}{5}$$

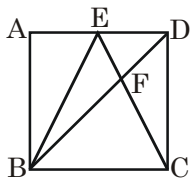
area $\Delta ADE = 16$

area trap DBCE = 40 – 16 = 24

Illustration-7 : ABCD is a square and E is mid point of AD. If area of square is given by A_1. Prove following :

(A) $\dfrac{\text{area of } \Delta EFD}{\text{area of } \Delta BFC} = \dfrac{1}{4}$

(B) $\dfrac{\text{area of } \Delta BEF}{\text{area of } \Delta DFC} = 1$



(C) area of $\Delta EFD = \dfrac{A_1}{12}$

(D) area of $\Delta BEF = \dfrac{A_1}{6}$

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Solution : Let side of square = a

$A_1 = a^2$

area of $\Delta BED = \frac{1}{2} \times a \times \frac{a}{2} = \frac{1}{4}a^2 = \frac{A_1}{4}$

Since ΔEFD is similar to ΔCFB

$\Rightarrow \frac{ED}{BC} = \frac{1}{2} = \frac{EF}{FC} = \frac{BF}{FD}$

$\Rightarrow \frac{\text{area of } \Delta EFD}{\text{area of } \Delta BFC} = \frac{1}{4}$

Let area of $\Delta EFD = \Delta$

$\Rightarrow$ area of $\Delta BFC = 4\Delta$

Also let area of $\Delta BEF =$ x and let area of $\Delta DFC = y$

$\Rightarrow x + \Delta = \frac{A_1}{4}$ and $y + 4\Delta = \frac{A_1}{2}$

Area of ΔBED = Area of ΔCED

($\because$ triangles on the same base ED and between the same parallel lines ED and BC)

$\Rightarrow$ Area of ΔBEF + Area of ΔEFD = Area of ΔCDF + Area of ΔEFD

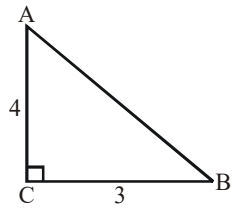
$\Rightarrow x + \Delta = y + \Delta \Rightarrow x = y$

Now x = y therfore $x + \Delta = \frac{A_1}{4}$ and $x + 4\Delta = \frac{A_1}{2}$ on solving these equations

$\Rightarrow \Delta = \frac{A_1}{12} \Rightarrow \qquad x = \frac{A_1}{6}$

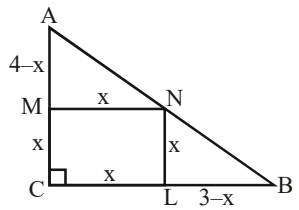
Also $y + \Delta = \frac{A_1}{4} \Rightarrow y = \frac{A_1}{6}$

Illustration-8 : In a right angle triangle ABC. Let k is maximum possible area of a square that can be inscribed when one of its vertices coincide with the vertex of right angle of triangle then find the value of k



Solution : ΔAMN & ΔNLB

are similar.



$\frac{4-x}{x} = \frac{x}{3-x}$

$\Rightarrow 12 - 7x + x^2 = x^2$

$x = \frac{12}{7}$

$k = \frac{144}{49}$

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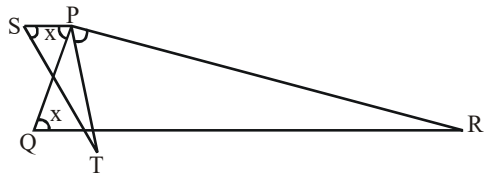
Do yourself-1 :

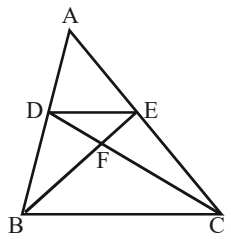
1. In a ΔABC, D and E are points on the sides AB and AC respectively such that DE || BC. If AD = 4x – 3, AE = 8x – 7, BD = 3x – 1 and CE = 5x – 3, find the value of x.

2. In ΔABC, if AD is the bisector of $\angle A$, prove that $\frac{\text{Area }(\Delta ABD)}{\text{Area }(\Delta ACD)} = \frac{AB}{AC}$.

3. In figure, $\angle QPS = \angle RPT$ and $\angle PST = \angle PQR$. Prove that $\Delta PST \sim \Delta PQR$ and hence find the ratio ST : PT, if PR : QR = 4 : 5.

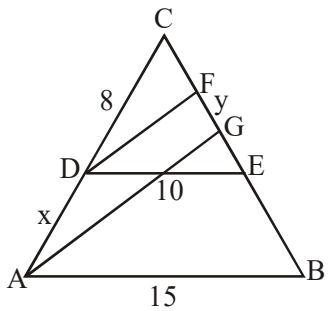


4. In the given figure. DE is parallel to the base BC of triangle ABC and AD : DB = 5 : 3. Find the ratio -

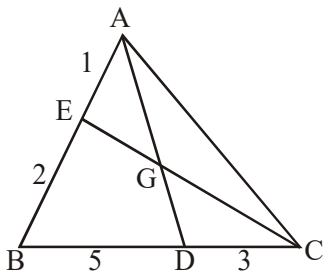


(i) $\frac{AD}{AB}$ (ii) $\frac{\text{Area of } \Delta DEF}{\text{Area of } \Delta CFB}$

5. In the figure DF||AG, DE||AB, AB = 15, CD = 8, AD = x, DE = 10, FG = y and CG = 6.
The ratio x : y equals
(A) 1 : 2
(B) 1 : 3
(C) 2 : 1
(D) 3 : 2



6. In the figure AE : EB = 1 : 2 and BD : DC = 5 : 3.
Find EG : GC.

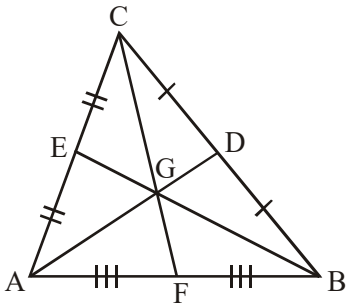


SOME IMPORTANT POINTS AND THEIR MEANING :

MEDIANS AND CENTROID :

A **median** of a triangle is a line segment from a vertex to the midpoint of the opposite side. In the figure below, $\overline{AD}$, $\overline{BE}$ and $\overline{CF}$ are all medians.

- The medians of a triangle are concurrent at a point called the **centroid** of the triangle. The centroid of the triangle is usually labelled by G.



- The medians of a triangle divide the triangle into six little triangles of equal area.
- The centroid of a triangle cuts its medians into 2 : 1 ratio. For example, for the triangle shown, we have $\frac{AG}{GD} = \frac{BG}{GE} = \frac{CG}{GF} = \frac{2}{1}$.
- The median to the hypotenuse of a right triangle is equal in length to half the hypotenuse.

PERPENDICULAR BISECTORS AND CIRCUMCENTRE :

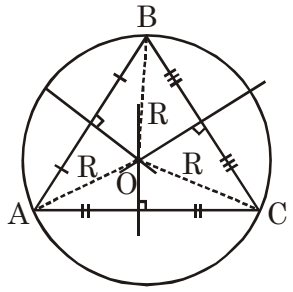
The perpendicular bisector of a segment is the straight line consisting of all the points that are equidistant from the end points of the segment.

Circumcentre :

The perpendicular bisectors of the sides of a triangle are concurrent at a point called the **circumcenter**. The circle centered at the circumcenter that passes through the vertices of the triangle is called the **circumcircle** of the triangle, because it is **circumscribed** about the triangle (meaning it passes through all the vertices of the triangle).

Finally, the radius of this circle is called the **circumradius**, the circumcenter is usually labelled with the letter O, and the circumradius is usually denoted by R.

OA = OB = OC = R



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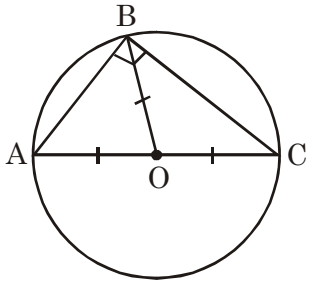
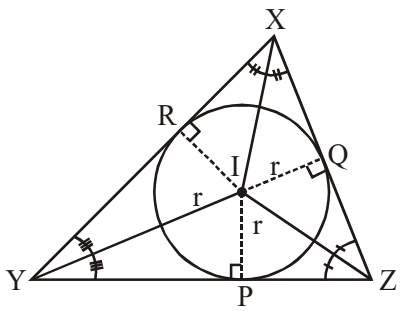
- The circumcenter of a right triangle is the midpoint of the hypotenuse and the circumradius equals half the length of the hypotenuse.



- Just as two points determine a line, we have now shown that three non-collinear points determine a circle. This means that given any three non-collinear points, there is exactly one circle that passes through all three.

ANGLE BISECTORS, INCENTRE & EXCENTRE :

- The angle bisector of an angle consists of all points that are equidistant from the lines forming the angle.
- The angle bisectors of interior angles of a triangle are concurrent at a point called the **incentre**. This point is equidistant from the sides of the triangle. This common distance from the incentre to the sides of a triangle is called the **inradius**. Thus the circle with centre I and this radius is tangent to all three sides of the triangle. This circle is called the **incircle** because it is inscribed in the triangle (meaning it is tangent to all the sides of the triangle). The incentre is usually denoted I, and the inradius is usually written as r.



Escribed Circle : It is a circle touching one side of a triangle internally and the other two externally. A triangle has three escribed circles. The centre of an escribed circle is the point of concurrence of the bisectors of the two exterior angles and the bisector of the third interior angle.

The centres of these escribed circles are called excentres and are labelled with I_1, I_2 and I_3. Also, their corresponding radii called exradii are denoted by r_1, r_2 and r_3.

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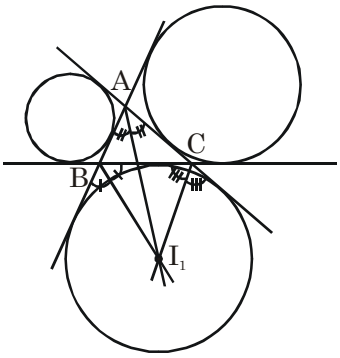
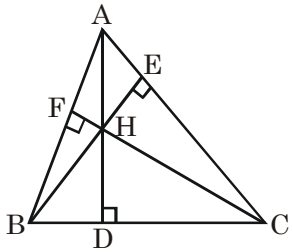


ALTITUDES & ORTHOCENTRE :

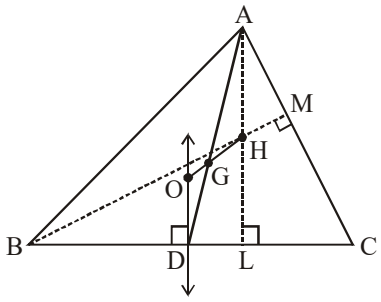
Altitude : It is the perpendicular dropped on any side of a triangle from the opposite vertex.

Orthocentre : The three altitudes of a triangle are concurrent and the point of concurrence is the orthocentre, which is denoted by H.



EULER'S LINE :

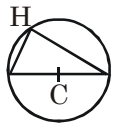
In the figure



- AD is a median and G the centroid.
  AG : GD = 2 : 1.
- BM is the altitude to AC from B.
  Here H is the orthocentre.
- OD is the perpendicular bisector of side BC.
  O is the circumcentre.

Note : O, G and H are collinear. This line is known as Euler's line and HG : GO = 2 : 1.

- In an equilateral triangle these four centres (centroid, orthocentre, incentre, circumcentre) are coincident.
- In an isosceles triangle they are collinear.
- The incentre and centroid are always within the triangle.
- In an acute angled triangle circumcentre and orthocentre also lie inside the triangle
- In a right triangle circumcentre will be the midpoint of the hypotenuse and the orthocentre is the vertex containing the right angle.



- In obtuse angled triangle orthocentre and circumcentre lie outside the triangle.

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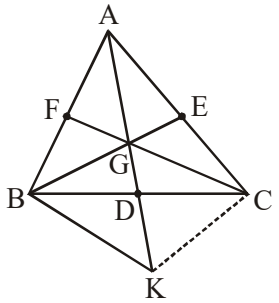
Illustration-9 : For any triangle, the area of triangle formed by medians of the triangle is $\left(\frac{3}{4}\right)^{th}$ the area of given triangle.

Solution : Consider ΔABC, AD, BE and CF are length of median say ℓ_1, ℓ_2 and ℓ_3 respectively
We have to find area of triangle, the length of whose sides are ℓ_1, ℓ_2 and ℓ_3.
Let G be the centroid of ΔABC
Let us produce AG to K
Such that GD = DK
and join CK and BK
We know that
BD = DC and GD = DK
$\therefore$ BKCG is a parallelogram and hence



BK = GC and CK = BG

$BK = \frac{2}{3}CF$ and $CK = \frac{2}{3}BE$

$BK = \frac{2}{3}\ell_3$ and $CK = \frac{2}{3}\ell_2$

and $GK = 2GD = 2.\left(\frac{1}{3}AD\right) = \frac{2}{3}\ell_1$

Now say PQR is triangle formed by length ℓ_1, ℓ_2 and ℓ_3 as sides. $\Delta PQR \sim \Delta GKC$

$$\frac{\text{area } \Delta PQR}{\text{area } \Delta GKC} = \frac{(\ell_1)^2}{\left(\frac{2\ell_1}{3}\right)^2} = \frac{9}{4}$$

$$\text{area } \Delta PQR = \frac{9}{4} \text{ area } \Delta GKC = \frac{9}{4}\times\frac{1}{3} \text{ area } \Delta ABC$$

$$\text{area } \Delta PQR = \frac{3}{4} \text{ area } \Delta ABC$$

Illustration-10 : If I is the incenter of ΔABC then $\angle BIC = 90° + \frac{1}{2}A$.

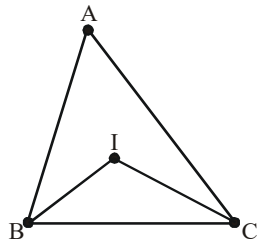
Solution : We have

$\angle BIC = 180° - (\angle IBC + \angle ICB)$

$= 180° - \frac{1}{2}(B + C)$

$= 180° - \frac{1}{2}(180° - A)$

$= 90° + \frac{1}{2}A.$



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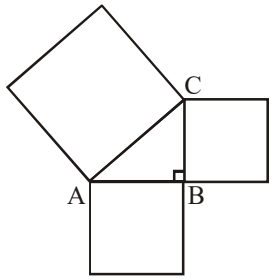
SOME THEOREMS ON SIDE LENGTH OF A TRIANGLE



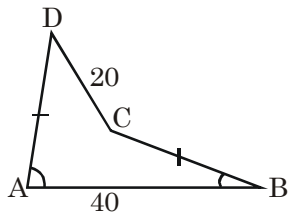
Baudhayana Theorem (Pythagorean/Pythagoras' Theorem) :

In a right-angled triangle the square described on the hypotenuse is equal to the sum of the squares described on the other two sides.

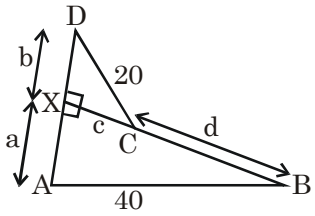
In Right Angle Triangle ABC, right angled at B,

$AC^2 = AB^2 + BC^2$

Illustration-11 : ABCD is a quadrilateral (shown in figure) AD = BC, AB = 40, CD = 20. $\angle DAB + \angle ABC = 90°$. If area of quadrilateral is N then $\frac{N}{100}$ is



Solution : Given : AD = BC, AB = 40



CD = 20, $\angle DAB + \angle ABC = 90°$

Construction :

Extend BC to intersect AD in X.

$\Rightarrow \angle AXB = 180° - (\angle DAB + \angle ABC)$

$\Rightarrow \angle AXB = 180° - 90° = 90°$

Let CX = c, AX = a, XD = b

CB = d

BC = AD

BC = AX + XD

$\boxed{BC = a + b}$

$\therefore \boxed{d = a + b}$

$$\text{area (quad)} = \frac{1}{2}a(c + d) + \frac{1}{2}bc$$

$$= \frac{1}{2}[ac + ad + bc]$$

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$$= \frac{1}{2}[ad + c(a + b)]$$

$$= \frac{1}{2}[ad + cd]$$

In ΔBXA

$$a^2 + (c + d)^2 = 40^2 \qquad ...(1)$$

In ΔCXD

$$c^2 + b^2 = 20^2 \qquad ...(2)$$

On substracting (2) from (1)

$$a^2 + 2cd + d^2 - b^2 = 1200$$

$$a^2 + 2cd + (a + b)^2 - b^2 = 1200 \qquad (d = a + b)$$

$$2a^2 + 2cd + 2ab = 1200$$

$$2a(a + b) + 2cd = 1200$$

$$2ad + 2cd = 1200$$

$$\frac{1}{2}(ad + cd) = 300 \ = \text{Area of quad. ABCD}$$

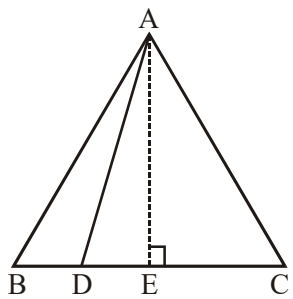
Illustration-12 : In an equilateral triangle ABC, the side BC is trisected at D. Prove that :

$9\ AD^2 = 7\ AB^2$

Solution : Given : An equilateral ΔABC, in which the side BC is trisected at D.

To Prove : $9\ AD^2 = 7AB^2$

Const. : Draw $AE \perp BC$



Proof : We know that in an equilateral triangle perpendicular from a vertex bisects the base.

$\therefore\ BE = EC = \frac{1}{2}BC \quad ...(i)$

Also, the side BC is trisected at D. [given]

$\Rightarrow BD = \frac{1}{3}BC$ [from figure] ...(ii)

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Now, in a right-triangle AED, We have :

$$AD^2 = AE^2 + DE^2$$

$$= AE^2 + (BE - BD)^2 \qquad [\because DE = BE - BD]$$

$$= AE^2 + BE^2 + BD^2 - 2\ BE.BD$$

$$= AB^2 + BD^2 - 2\ BE.BD \qquad [\because AB^2 = AE^2 + BE^2]$$

$$= BC^2 + \left(\frac{1}{3}BC\right)^2 - 2\left(\frac{1}{2}BC\right).\left(\frac{1}{3}BC\right)$$

$[\because AB = BC, BD = \frac{1}{3}BC$ and $BE = \frac{1}{2}BC$ see (i) and (ii)$]$

$$= BC^2 + \frac{BC^2}{9} - \frac{BC^2}{3} = \frac{9BC^2 + BC^2 - 3BC^2}{9}$$

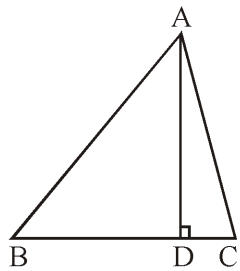
$$\Rightarrow AD^2 = \frac{7BC^2}{9} \text{ or } 9AD^2 = 7BC^2$$

or $9AD^2 = 7AB^2$ $\qquad$ $(AB = BC = CA)$

Illustration-13 : The perpendicular AD on the base BC of a ΔABC intersects BC at D so that DB = 3 CD.

Prove that : $2AB^2 = 2\ AC^2 + BC^2$.

Solution : Given : In figure, $AD \perp BC$ and DB = 3CD.



To Prove : $2AB^2 = 2AC^2 + BC^2$.

Proof : From right triangles ADB and ADC, we have :

$$AB^2 = AD^2 + BD^2$$

and $AC^2 = AD^2 + CD^2$

$$\therefore \quad AB^2 - AC^2 = BD^2 - CD^2$$

$$= (3\ CD)^2 - CD^2 \qquad [\because BD = 3\ CD]$$

$$= 9CD^2 - CD^2 = 8CD^2 = 8 \times \left(\frac{1}{4}BC\right)^2 = 8 \times \frac{BC^2}{16} = \frac{BC^2}{2}$$

$$\left(\because \frac{BD}{CD} = \frac{3}{1} \Rightarrow \frac{BD + CD}{CD} = \frac{3+1}{1} \Rightarrow \frac{BC}{CD} = \frac{4}{1} \Rightarrow CD = \frac{1}{4}BC\right)$$

$$\therefore\ 2AB^2 - 2AC^2 = BC^2$$

Hence, $2AB^2 = 2AC^2 + BC^2$.

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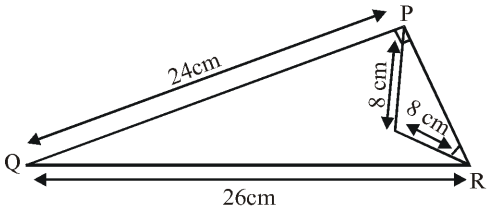
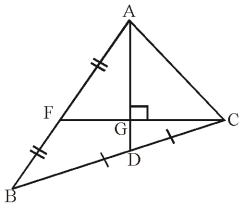
Do yourself-2 :

1. In a triangle, if the square of one side is equal to the sum of the square of the other two sides, prove that the angle opposite to the first side is a right angle. Use the above theorem to find the measure of $\angle PKR$ in figure



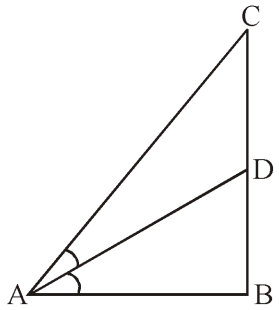
2. ABC is a right-angle triangle, right angled at A. A circle is inscribed in it. The lengths of the two sides containing the right angle are 6 cm and 8 cm, then radius of the circle is :

3. Consider ΔABC, the medians AD & CF intersect at right angles at G. If BC = 3cm and AB = 4 cm, then the length of AC is (in cm) :



4. In triangle ABC, let D be the midpoint of BC. If $\angle ADB = 45^o$ and $\angle ACD = 30^o$, determines $\angle BAD$.

5. If CD = 15, DB = 9, AD bisects $\angle A$ and $\angle ABC = 90^o$, then AB has length :



(A) 32 (B) 18 (C) 7 (D) 24

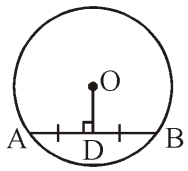
6. Let ΔXOY be a right angled triangle with $\angle XOY = 90^o$. Let M and N be the midpoints of legs OX and OY, respectively. Given that XN = 19 and YM = 22, then length XY is equal to

(A) 24 (B) 26 (C) 28 (D) 34

CIRCLE

Theorem 1 :

If a straight line drawn from the center of a circle bisects a chord, not passing through the centre, then it cuts the chord at right angles. Conversely, if it cuts the chord at right angles, then it bisects the chord.



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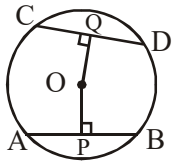
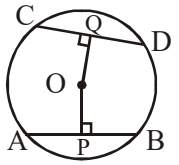
Theorem 2 :



Equal chords of a circle are equidistant from the center.

AB = CD, OP ⊥ AB and OQ ⊥ CD ⇒ OP = OQ

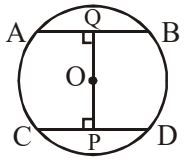
Conversely,



Chords which are equidistant from the centre are equal.

OP = OQ, OP ⊥ AB and OQ ⊥ CD ⇒ AB = CD

Theorem 3 :

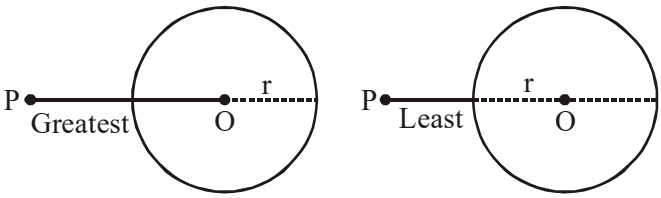


Of any two chords of a circle, that which is nearer to the centre is greater than one lying remote. Conversely, the greater of two chords is nearer to the centre.

OP > OQ ⇒ CD < AB

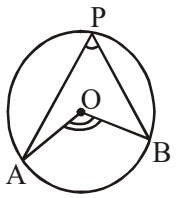
Theorem 4 :

If P be any point (interior, exterior or on the circle) then greatest distance between point P and circumference of the circle is PO +r and the smallest distance between point P and circumference of the circle is |PO –r| (where r is radius and O is centre)



ANGLES IN A CIRCLE

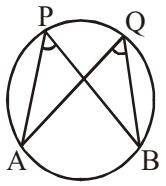
Theorem 1 :



The angle at the centre of a circle is double of an angle at the circumference subtended by the same arc.

∠AOB = 2∠APB

Theorem 2 :



Angles in the same segment of a circle are equal.

∠APB = ∠AQB

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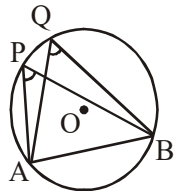
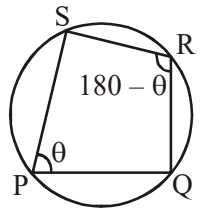
Converse of Theorem 2 :



If a line segment joining two points subtends equal angles at two other points lying on the same side of the line containing the line segment, then four points lie on a circle (i.e. they are concyclic). If $\angle APB = \angle AQB \Rightarrow AB$ is a chord of circle and A, P, Q, B are concyclic points

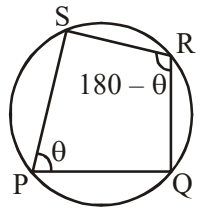
Theorem 3 :



The opposite angles of any quadrilateral inscribed in a circle are together equal to two right angles.

$\angle P + \angle R = 180°$ and $\angle Q + \angle S = 180°$

Converse of Theorem 3 :

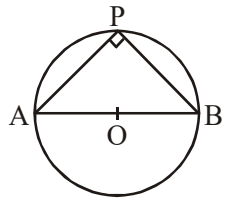


If a pair of opposite angles of a quadrilateral are supplementary, then its vertices are concyclic.

If $\angle P + \angle R = 180°$ or $\angle Q + \angle S = 180°$

$\Rightarrow$ Then points P, Q, R and S are concyclic.

Theorem 4 :



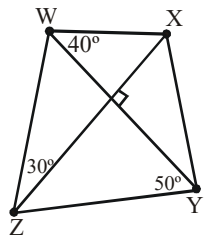
The angle in a semi-circle is a right angle.

$\angle APB = 90°$

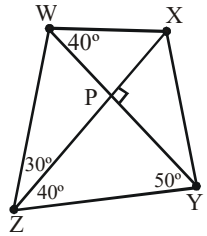
Illustration-14 : In quadrilateral WXYZ with perpendicular diagonals, we are given $\angle WZX = 30º$, $\angle XWY = 40°$, and $\angle WYZ = 50°$.

(a) Compute $\angle Z$.

(b) Compute $\angle X$.



Solution: Let P be the intersection of the diagonals.



In ΔPZY

$\angle PZY = 90° - 50° = 40°$.

Now consider angles.

$\angle XWY$ and $\angle XZY$

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Since $\angle XWY = \angle XZY = 40°$

Using theorem : WXYZ is cyclic quadrilateral.

$\angle Z = \angle WZX + \angle PZY$

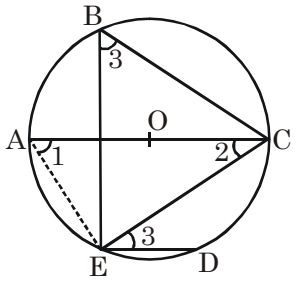
$= 30° + 40° = 70°$

Now $\angle Z + \angle X = 180°$ {opposite angle of cyclic quadrilateral}

$\angle X = 180° - 70° = 110°$

Illustration-15 : In the given figure, the chord ED is parallel to the diameter AC and $\angle CBE = 50°$. Find $\angle CED$.

Solution:



$\angle CBE = \angle CAE$ ($\because$ angles in the same segment)

$\angle CAE = \angle 1 = 50°$ ...(1) ($\because \angle CBE = 50°$)

$\angle AEC = 90°$ ...(2) ($\because$ Angle in a semi circle is a right angle)

Now in ΔAEC

$\angle 1 + \angle AEC + \angle 2 = 180°$ [$\because$ sum of angles of a $\Delta = 180°$]

$\therefore 50° + 90° + \angle 2 = 180°$

$\Rightarrow \angle 2 = 40°$ ...(3)

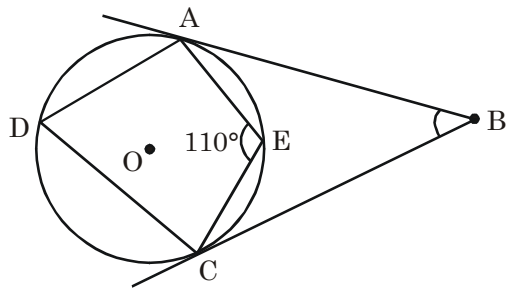
Also, ED $\parallel$ AC (Given)

$\therefore \angle 2 = \angle 3$ (Alternate angles)

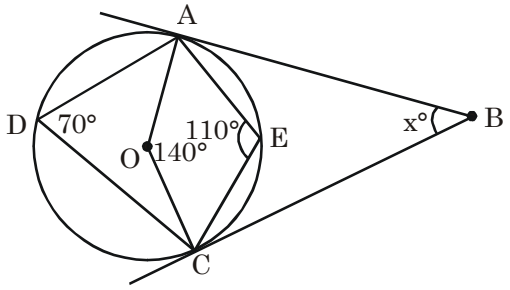
$\therefore 40° = \angle 3$ i.e., $\angle 3 = 40°$

Hence $\angle CED = 40°$

Illustration-16 : In the figure below, BA and BC are tangents to the circle at points A and C, respectively. If $\angle AEC = 110°$. Find $\angle ABC$.



Solution :



Consider the diagram to the right.

• Opposite angles of a cyclic quadrilateral sum up to 180°.

So $\angle ADC = 180° - 110° = 70°$.

• From the inscribed angle theorem, we can see that

$\angle AOC = 2\angle ADC = 2(70°) = 140°$

Since the sum of interior angle of a quadrilateral is 360°, we have

$$\angle ABC = 360° - \angle BCO - \angle BAO - \angle AOC$$
$$= 360° - 90° - 90° - 140°$$
$$= 40°$$

Illustration-17 : An acute isosceles triangle, ABC is inscribed in a circle. Through B and C, tangents to the circle are drawn, meeting at point D. If $\angle ABC = \angle ACB = 2\angle D$, find the measure of $\angle A$.

Solution : Let $\angle D = \theta$

$\Rightarrow \angle ABC = \angle ACB = 2\theta$

In ΔBCD.

$180° - 4\theta + 180° - 4\theta + \theta = 180°$

$\Rightarrow 7\theta = 180°$

$\Rightarrow \theta = \dfrac{180°}{7}$

$\therefore \angle A = 180° - 4\theta$

$= 180° - \dfrac{4 \times 180°}{7}$

$\angle A = \dfrac{3 \times 180°}{7}$

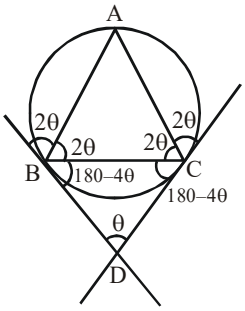
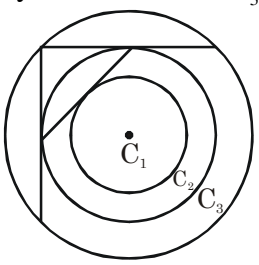


Illustration-18 : In the given figure if C_1, C_2, C_3 are three concentric circles such that radius of C_1 and C_2 is 1 and 3 unit respectively, then radius of C_3 is



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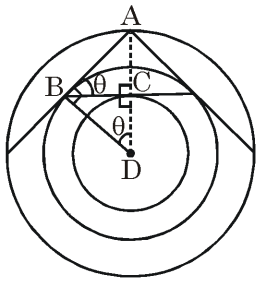
Solution : Let $\angle BDC = \theta$, then



$\angle DBC = \frac{\pi}{2} - \theta$

$\therefore \angle CBA = \theta$, So

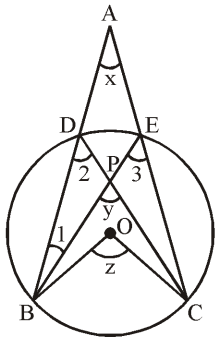
$\Delta ABD \sim \Delta BCD$, so

$\frac{AD}{BD} = \frac{BD}{CD}$

$\Rightarrow R_3 = \frac{(3)^2}{1} = 9$

Illustration-19 : In the given fig., O is the centre of the circle. Prove that $\angle x + \angle y = \angle z$.

Solution : Given : In fig., O is the centre of the circle.



To Prove : $\angle x + \angle y = \angle z$

Proof : $\angle z = 2\angle 2$ [angle at the centre is double to the angle in the remaining part.]

or $\angle z = \angle 2 + \angle 2$

But, $\angle 2 = \angle 3$ [angles in the same segment]

$\Rightarrow \angle z = \angle 2 + \angle 3$ ... (i)

Now, we will determine the values of $\angle 2$ and $\angle 3$ in terms of $\angle x$ and $\angle y$.

$\because$ $\angle 3$ is an exterior angle of ΔAEB,

$\therefore$ $\angle 3 = \angle 1 + \angle x$ ... (ii) [ext. $\angle$ = sum of two int. opp. $\angle$s]

Again, $\angle y$ is an exterior angle of ΔDEP

$\therefore \angle y = \angle 1 + \angle 2 \Rightarrow \angle 2 = \angle y - \angle 1$ ... (iii)

(i), (ii) and (iii)

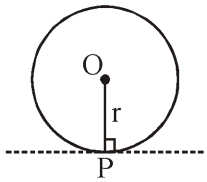
$\Rightarrow \angle z = (\angle y - \angle 1) + (\angle 1 + \angle x) = \angle x + \angle y$

Hence, $\angle x + \angle y = \angle z$.

TANGENT

Theorem 1 :

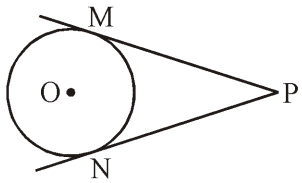
The tangent at any point of a circle is perpendicular to the radius drawn to the point of contact.



Theorem 2 :

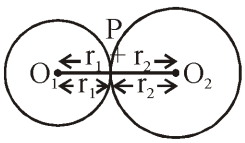
Two tangents can be drawn to a circle from an external point and length of these tangents are equal.

So, PM = PN

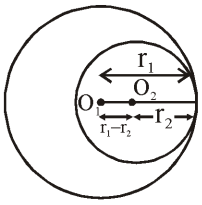


Theorem 3 :

If two circles touch one another, the centres and the point of contact are collinear.



$O_1O_2 = r_1 + r_2$



$O_1O_2 = |r_1 - r_2|$

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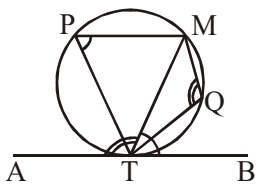
Theorem 4 (Alternate Segments Theorem) :



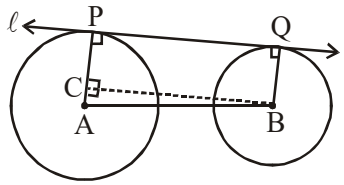
The angles made by a tangent to a circle with a chord drawn from the point of contact are respectively equal to the angles in the alternate segments of the circle.

$\angle MTB = \angle TPM$, $\angle MTA = \angle MQT$

LENGTH OF COMMON TANGENTS

Direct Common Tangent (External Common Tangent) :

Consider two circle with centre A and B and line 'ℓ' touching the two circles at P and Q. Let radius of the circles be r_1 and r_2.



Now draw perpendicular from 'B' to AP meeting at 'C'.

Thus, CPQB is a rectangle and hence PQ = CB; PC = BQ

Now, in ΔACB

$AB^2 = AC^2 + BC^2$

$\Rightarrow BC^2 = AB^2 - AC^2$

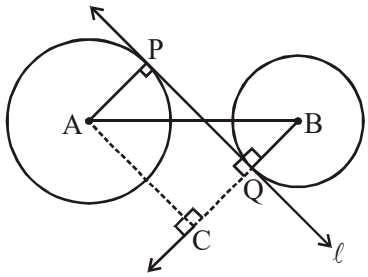
$\Rightarrow PQ^2 = AB^2 - AC^2$ [BC = PQ]

$\Rightarrow PQ^2 = AB^2 - (r_1 - r_2)^2$ [$AC = AP - PC = r_1 - r_2$]

Hence, length of direct common tangent

$PQ = \sqrt{d^2 - (r_1 - r_2)^2}$ [where AB = d = distance between the centres]

Transverse Common Tangent (Internal Common Tangent) :



Drop perpendicular from A to meet BQ produced in 'C'

Now $AP = r_1$; $BQ = r_2$; $AB = d$

AP = QC and PQ = AC

Also BC = BQ + QC

So $BC = r_2 + r_1$

Now in ΔACB

$AB^2 = AC^2 + BC^2$

$\Rightarrow d^2 = AC^2 + (r_1 + r_2)^2$

$\Rightarrow AC^2 = d^2 - (r_1 + r_2)^2$

$\Rightarrow PQ^2 = d^2 - (r_1 + r_2)^2$

$\Rightarrow PQ = \sqrt{d^2 - (r_1 + r_2)^2}$

Length of transverse common tangent is $\sqrt{d^2 - (r_1 + r_2)^2}$

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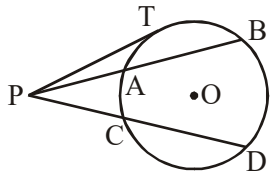
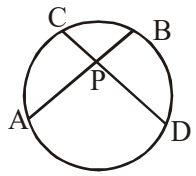
POWER OF A POINT

Suppose a line through a point P intersects a circle in two points, A and B. The **power of a point theorem** states that for all such lines, the product (PA)(PB) is constant. We call this product the power of point P.



P lies outside the circle

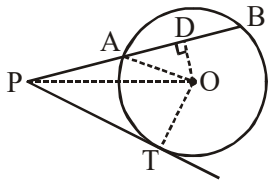


P lies inside the circle

$PT^2 = (PA)(PB) = (PC)(PD)$ $\qquad$ $PA \cdot PB = PC \cdot PD$

$= $ The power of point P

Proof : Given : A secant PAB to a circle C(O, r) intersecting it in points A and B and PT is a tangent segment of the circle.



To Prove : $PA \times PB = PT^2$.

Construction : Draw $OD \perp AB$. Join OP, OT and OA.

Proof : $OD \perp AB \Rightarrow AD = DB$

Now, $PA \times PB = (PD - AD)(PD + DB)$

$= (PD - AD)(PD + AD)$ $\quad [\because DB = AD]$

$= PD^2 - AD^2$

$= (OP^2 - OD^2) - AD^2$ $\quad [OP^2 = OD^2 + PD^2]$

[Pythagoras Theorem]

$= OP^2 - (OD^2 + AD^2)$

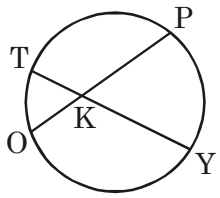
$= OP^2 - OA^2$ $\quad [\because OA^2 = OD^2 + AD^2]$

$= OP^2 - OT^2$ $\quad [\because OA = OT]$ [radii of same circle]

$= PT^2$ $\quad [\because \angle OTP = 90°, OP^2 = OT^2 + PT^2]$

Illustration-20 : Chords $\overline{TY}$ and $\overline{OP}$ meet at point K such that TK = 2, KY = 16 and KP = 2(KO). Find OP.

Solution : A quick sketch suggests how to apply power of a point.



From the power of point K, we have

$$(KP)(KO) = (KT)(KY).$$

Substituting the given information in this equation yields

$$2(KO)(KO) = 2(16),$$

from which we find KO = 4. Therefore, KP = 2KO = 8, so OP = KO + KP = 12.

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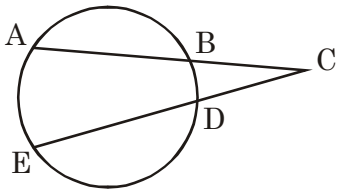
Illustration-21 : In the diagram, CB = 9, BA = 11 and CE = 18. Find DE.

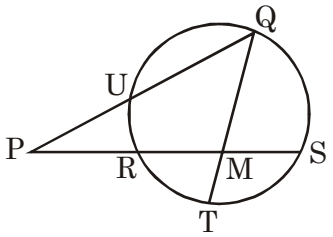


Solution : We have two intersecting secants, so we apply power of a point, which gives

$$(CD)(CE) = (CB)(CA).$$

Therefore, (CD)(18) = 9(20), so CD = 10 and DE = CE – CD = 8.

Illustration-22 : Points R and M trisect $\overline{PS}$, so PR = RM = MS. Point U is the midpoint of $\overline{PQ}$, TM = 2 and MQ = 8. Find PU.



Solution : The power of point M gives us

$$(MR)(MS) = (MT)(MQ).$$

We know that RM = MS, so substitution gives $MR^2 = (2)(8)$, i.e., MR = 4. Therefore, PR = MR = 4 and PS = 3(MR) = 12. Since U is the midpoint of $\overline{PQ}$, we have PQ = 2PU.

Now we can apply the power of point P to find :

$$(PU)(PQ) = (PR)(PS).$$

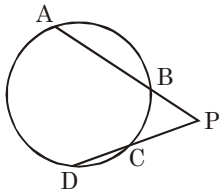
Substitution gives (PU)(2PU) = 4(12), so PU = $2\sqrt{6}$.

Illustration-23 : In the diagram,

we have BP = 8, AB = 10,

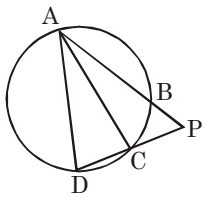
CD = 7 and $\angle APC = 60°$.

Find the area of the circle.



Solution : It's not immediately obvious how we will find the radius, so we start by finding what we can. The power of a point P gives us

(PC)(PD) = (PB)(PA),



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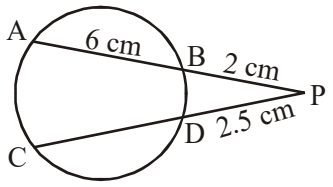
so (PC)(PC + 7) = 144. Therefore, $PC^2 + 7PC - 144 = 0$, so (PC + 16)(PC – 9) = 0. PC must be positive, so PC = 9.

Seeing that $\angle APC = 60°$ makes us wonder if there are any equilateral or 30-60-90 triangle lurking about. Since CP = AP/2 and the angle between these sides is 60°, the sides adjacent to the 60° angle in ΔACP are in the same ratio as the sides adjascent to the 60° angle in a 30-60-90 triangle with right angle at $\angle ACP$! $\Rightarrow \angle ACP = 90° \Rightarrow \angle ACD = 90°$

Since $\angle ACD$ is right and inscribed in $\overset{\frown}{AD}$, we know $\overset{\frown}{AD}$ is a semicircle. Therefore, $\overline{AD}$ is a diameter of the circle. Since AC = $CP\sqrt{3} = 9\sqrt{3}$ from our 30-60-90 triangle, we have $AD = \sqrt{AC^2 + CD^2} = \sqrt{243 + 49} = 2\sqrt{73}$. Finally, the radius of the circle is AD/2 = $\sqrt{73}$, so the area is $\left(\sqrt{73}\right)^2 \pi = 73\pi$.

Illustration-24 : Chords AB and CD of a circle intersect each other at P as shown in fig., if AB = 6 cm, PB = 2 cm, PD = 2.5 cm, find CD.



Solution : Since chords AB and CD of a circle meet in P (when produced),

$\therefore$ PA.PB = PC.PD

$\Rightarrow$ (6 + 2).2 = PC (2.5)

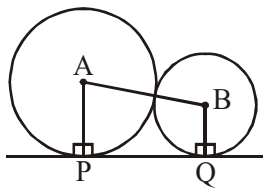
$\Rightarrow$ $PC = \frac{16}{2.5} = \frac{16 \times 2}{5} = \frac{32}{5} = 6.4$ cm

$\therefore$ CD = PC – PD = 6.4 – 2.5 = 3.9 cm.

Illustration-25 : If two circles of radius 1 cm and 4 cm touch each other, then find the length of common tangent.

Solution : Distance between centres AB = $r_1 + r_2 = 5$

Length of tangent



$PQ = \sqrt{d^2 - (r_1 - r_2)^2}$

$= \sqrt{5^2 - (4-1)^2}$

PQ = 4 cm

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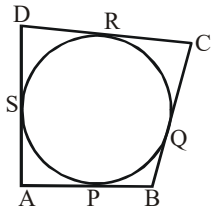
Do yourself-3 :

1. O and C are the orthocenter and the circumcenter of an acute angled triangle PQR respectively. The points P and O are joined and produced to meet the side QR at S. If $\angle QCR = 130°$ and $\angle PQS = 60°$, then find $\angle RPS$.

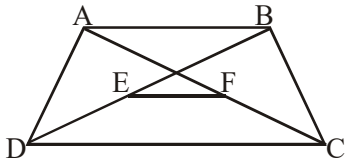
2. In figure, a circle touches all the four sides of a quadrilateral ABCD with AB = 6 cm, BC = 7 cm and CD = 4 cm. Find AD.



3. In a circle of radius 5 cm, AB and AC are two chords, such that AB = AC = 6 cm. Find the length of chord BC.

4. The inradius and the circumradius of a right angled triangle are 9 and 37.5 respectively. Find the perimeter of the triangle.

5. In the trapezium shown, AB||DC and E and F are the midpoints of the two diagonals. If DC = 60 and EF = 5 then the length of AB is equal to



(A) 40 (B) 45 (C) 50 (D) 55

6. ABCD is a trapezium in which AB||CD. If $\angle ADC = 2\angle ABC$, AD = a cm and CD = b cm, then the length (in cm) of AB is :

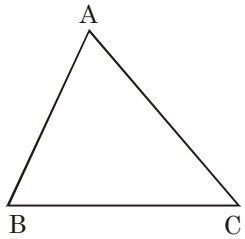
(A) $\frac{a}{2} + 2b$ (B) $a + b$ (C) $\frac{2}{3}a + b$ (D) $a + \frac{2}{3}b$

INEQUALITIES IN TRIANGLE

In any triangle, the longest side is opposite the largest angle and the shortest side is opposite the smallest angle. The middle side, of course, is therefore opposite the middle angle.

In other words, in ΔABC, $AB \geq AC \geq BC$ if and only if $\angle C \geq \angle B \geq \angle A$.

Important :



- $\angle C$ of ΔABC is acute if and only if $AB^2 < AC^2 + BC^2$.

  $\angle C$ of ΔABC is right if and only if $AB^2 = AC^2 + BC^2$.

  $\angle C$ of ΔABC is obtuse if and only if $AB^2 > AC^2 + BC^2$.

- The Triangular Inequality states that for any three points, A, B and C, we have

  $$AB + BC \geq AC,$$

  where equality holds if and only if B lies on $\overline{AC}$. Therefore, for nondegenerate triangles (i.e., those in which the vertices are not collinear), AB + BC > AC.

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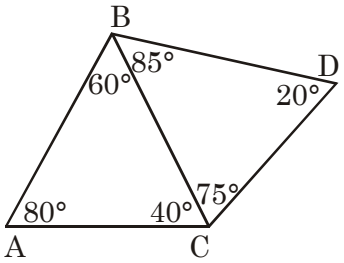
Illustration-26 : Given the angles as shown, order the lengths AB, BD, CD, BC and AC from greatest to least. (Note : the diagram is not drawn to scale!)

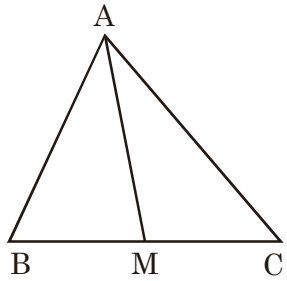


Solution : First we focus on ΔABC. Since $\angle A > \angle B > \angle C$, we have $BC > AC > AB$. Then, we turn to ΔBDC. Since $\angle B > \angle C > \angle D$, we have $CD > BD > BC$. BC is the smallest in one inequality string and the largest in the other, so we can put the inequalities together :

$$CD > BD > BC > AC > AB.$$

Illustration-27 : In ΔABC the median $\overline{AM}$ is longer than BC/2. Prove that $\angle BAC$ is acute.

Solution : We start with a diagram. We'd like to prove something about an angle, but all we are given is an inequality regarding lengths. So, we use the length inequality to get some angle inequalities to work with. Specifically, since $AM > BM$ in ΔABM and $AM > MC$ in ΔACM (because $AM > BC/2$ and $BM = MC = BC/2$), we have



$$\angle B > \angle BAM$$
$$\angle C > \angle CAM.$$

We want to prove something about $\angle BAC$, which equals $\angle BAM + \angle CAM$, so we add these two inequalities to give $\angle B + \angle C > \angle BAC$. Since $\angle B + \angle C + \angle BAC = 180°$, we can write $\angle B + \angle C > \angle BAC$ as

$$180° - \angle BAC > \angle BAC.$$

This gives us $180° > 2\angle BAC$, so $90° > \angle BAC$. Therefore, $\angle BAC$ is acute.

Illustration-28 : In ΔXYZ, we have $XY = 11$ and $YZ = 14$. For how many integer values of XZ is ΔXYZ acute ?

Solution : In order for ΔXYZ to be acute, all three of its angles must be acute. Using the given side lengths and our inequalities above, we see that we must have

$$121 + 196 > XZ^2$$

$$196 + XZ^2 > 121$$

$$121 + XZ^2 > 196$$

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Simplifying these three yields :

$$317 > XZ^2$$

$$XZ^2 > -75$$

$$XZ^2 > 75$$

The middle inequality is clearly always true. (We really didn't even have to include it – clearly 11 couldn't ever be the largest side!) Combining the other two add noting that we seek integer values of XZ, we have $9 \le XZ \le 17$ (since $8^2 < 75 < 9^2$ and $18^2 > 317 > 17^2$). So, there are 9 integer values of XZ such that ΔXYZ is acute.

Illustration-29 : In how many ways can we choose three different numbers from the set {1, 2, 3, 4, 5, 6} such that the three could be the sides of a nondegenerate triangle ?

(Note : The order of the chosen numbers doesn't matter; we consider {3, 4, 5} to be the same as {4, 3, 5}.)

Solution : We first notice that if we have three numbers to consider as possible side lengths of a triangle, we only need to make sure that the sum of the smallest two is greater than the third. (Make sure you see why!) We could just start listing all the ones we see that work, but we should take an organized approach to make sure we don't miss any. We can do so by classifying sets of three numbers by the smallest number.

Case 1 : Smallest side has length 1. No triangles can be made with three different lengths from our set if we include one of length 1.

Case 2 : Smallest side has length 2. The other two sides must be 1 apart, giving the sets {2, 3, 4}, {2, 4, 5}, and {2, 5, 6}.

Case 3 : Smallest side has length 3. There are only three possibilities and they all work : {3, 4, 5}, {3, 4, 6}, {3, 5, 6}.

Case 4 : Smallest side has length 4. The only possibility is {4, 5, 6}, which works.

Adding them all up, we have 7 possibilities.

We sometimes have to use some other tools in addition to triangle inequality.

Illustration-30 : Can the lengths of the altitudes of a triangle be in the ratio 2 : 5 : 6 ? Why or why not ?

Solution : We don't know anything about how the lengths of the altitudes of a triangle are related to each other. We do, however, know a whole lot about how the lengths of the sides of a triangle are related to each other. Therefore, we turn the problem from one involving altitudes into one involving side lengths. We let the area be K, let the side lengths be a, b, c and the lengths of the altitudes to these sides be h_a, h_b, h_c, respectively. Therefore, we have $K = ah_a/2 = bh_b/2 = ch_c/2$, so the sides of the triangle have lengths

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$$\frac{2K}{h_a}, \frac{2K}{h_b}, \frac{2K}{h_c}.$$

If our heights are in the ratio 2 : 5 : 6, then for some x, our heights are 2x, 5x and 6x. Then, our sides are

$$\frac{2K}{2x}, \frac{2K}{5x}, \frac{2K}{6x}.$$

However, the sum of the smallest two sides is then

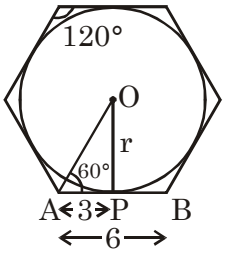
$$\frac{2K}{5x} + \frac{2K}{6x} = \frac{12K}{30x} + \frac{10K}{30x} = \frac{22K}{30x},$$

which is definitely less than the largest side, which is 2K/2x = K/x. Therefore, the sides don't satisfy the triangle inequality, which means it is impossible to have a triangle with heights in the ratio 2 : 5 : 6.

Concept : When facing problems involving lengths of altitudes of a triangle, consider using area as a tool.

Illustration-31 : If a circle is inscribed in regular hexagon of side 6 cm, then area of inscribed circle.

Solution : Since we know that sum of angles of polygon = (n – 2)180°



Each interior angle of polygon = $\frac{(n-2)180°}{n}$

For hexagon n = 6

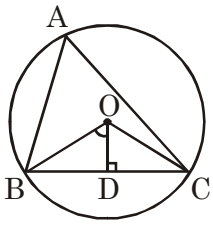
$\therefore$ Each angle = $\frac{(6-2)180°}{6} = 120°$

In ΔOPA

$\frac{r}{3} = \tan 60° \therefore r = 3\sqrt{3}$

Area of circle = $\pi r^2 = \pi\left(3\sqrt{3}\right)^2 = 27\pi$

Illustration-32 : In given circle with centre O a triangle ABC is inscribed, $\angle A = 60°$. OD is perpendicular to BC, then what is $\angle BOD$?



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Solution : In ΔOBD and ΔODC

$OD \perp BC$

$\therefore \angle ODB = \angle ODC = 90°$

OB = OC (radius of circle)

BD = DC [$\perp$ from centre bisects chord]

$\therefore \Delta OBD \cong \Delta OCD$

Hence $\angle BOD = \angle COD = \frac{1}{2}\angle BOC$ .... (1)

Again $\angle BOC = 2\angle BAC$

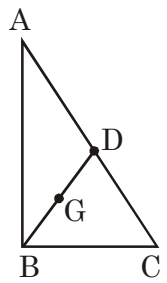
$\angle BOC = 2[\angle A]$ .... (2)

From (1) and (2)

$\angle BOD = \frac{1}{2}(2\angle A) = 60°$

Illustration-33 : In right ΔABC right angled at 'B' (As in figure).
Prove that distance between circumcentre and centroid

$GD = \frac{1}{3}(BD) = \frac{1}{3}\left(\frac{1}{2}AC\right)$. If AC = 10, then find GD

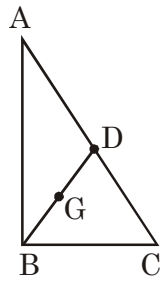


Solution. : **Property :** We know that mid-point of hypotenuse is equidistance from all the three vertices of triangle

$\therefore$ AD = CD = BD

$\therefore BD = \frac{1}{2}AC$

$BD = \frac{1}{2}(10) = 5$

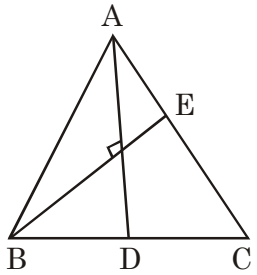


Again centroid divides median in ratio 2 : 1

$\therefore GD = \frac{1}{3}BD = \frac{5}{3}$

[D is circumcentre of ΔABC since mid-point of hypotenuse is circumcentre of triangle]

Illustration-34 : In ΔABC, AD and BE are medians such that they intersect at right angles. If AD = 6 and $BE = \frac{9}{2}$ then area of ΔABC is



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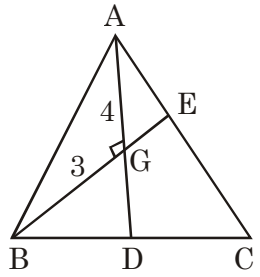
Solution. : Let G be the centroid of ΔABC i.e. point of intersection of medians AD and BE



Now $AG = \frac{2}{3}AD = \frac{2}{3}[6] = 4$

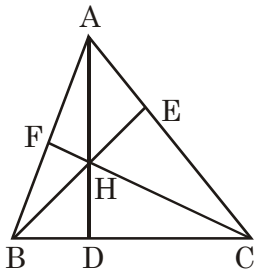
$BG = \frac{2}{3}BE = \frac{2}{3} \times \frac{9}{2} = 3$

Thus Area of ΔAGB is $\frac{1}{2} \times 3 \times 4 = 6$ unit^2

Now we know that area of ΔABC

$= 3 \times$ Area $(\Delta AGB) = 3(6) = 18$ unit^2

Illustration-35 : In, ΔABC, AD, BE and CF are altitudes and H is orthocentre. If $\angle A = 40°$ then find $\angle FDE$.



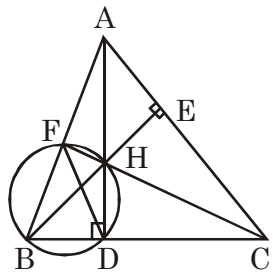
Solution. : It is very clear that $\angle HFB = 90°$ and $\angle HDB = 90°$

(Since CF and AD are altitudes)

Thus for quadrilateral HFBD

sum of opposite angle is 180°

Hence it is cyclic



In ΔAEB

$\angle ABE = 90° - A = \angle FBH$

and using the property of circle

$\angle FBH = \angle FDH$

$\therefore \angle FDH = 90° - A$

Similarly $\angle HDE = 90° - A$

$\therefore \angle FDE = 180° - 2A$

$\angle FDE = 180° - 2(40°) = 100°$

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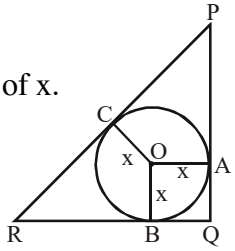
Do yourself-4 :

1. Let P be a point on the circumference of a circle. Perpendiculars PA and PB are drawn to points A and B on two mutually perpendicular diameters. If AB = 36 cm, the diameter of the circle is:

2. PQR is a right angled triangle with PQ = 12 cm and QR = 5 cm.
A circle with centre O and radius x is inscribed in ΔPQR. Find the val ue of x.



3. In triangle ABC with centriod G, if AG = BC, what is the angle BGC in degrees?

4. Let ABC be triangle with AB = AC = 6. If the circumradius of the triangle is 5, then BC equals

(A) 25/3 (B) 9 (C) 48/5 (D) 10

5. The inradius and the circumradius of a right angled triangle are 9 and 37.5 respectively. Find the perimeter of the triangle.

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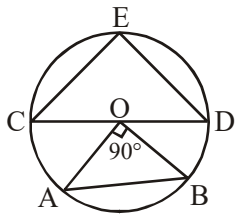
EXERCISE

1. In the diagram 'O' is the centre of the circle. The point E lies on the circumference of the circle such that the area of the ΔECD is maximum. If $\angle AOB$ is a right angle, then the ratio of the area of ΔECD to the area of the ΔAOB is :



(A) 2 : 1 (B) 3 : 2 (C) 4 : 3 (D) 4 : 1

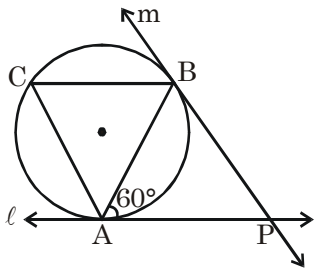
GM0001

2. Let ΔXOY be a right angled triangle with $\angle XOY = 90°$. Let M and N be the midpoints of legs OX and OY, respectively. Given that XN = 19 and YM = 22, the length XY is equal to

(A) 24 (B) 26 (C) 28 (D) 34

GM0002

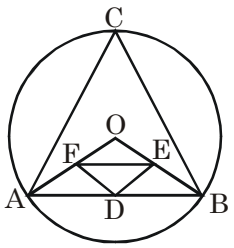
3. In the diagram below, if ℓ and m are two tangents and AB is a chord making an angle of 60° with the tangent ℓ, then the angle between ℓ and m is



(A) 45° (B) 30° (C) 60° (D) 90°

GM0003

4. In the diagram, O is the centre of the circle and D, E and F are mid points of AB, BO and OA respectively. If $\angle DEF = 30°$, then $\angle ACB$ is



(A) 30° (B) 60° (C) 90° (D) 120°

GM0004

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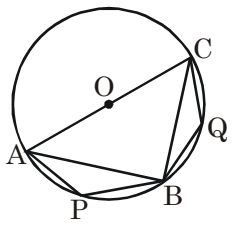
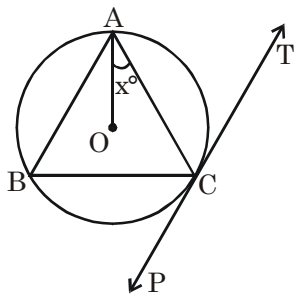
5. In the below diagram, O is the centre of the circle, AC is the diameter and if $\angle APB = 120°$, then $\angle BQC$ is



(A) 30° (B) 150° (C) 90° (D) 120°

GM0005

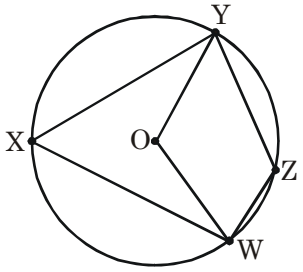
6. In the adjoining figure, PT is a tangent at point C of the circle. O is the circumcentre of ΔABC. If $\angle ACP = 118°$, then the measure of $\angle x$ is



(A) 28° (B) 32° (C) 42° (D) 38°

GM0006

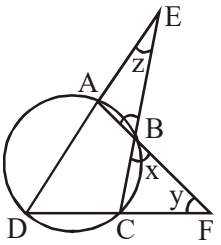
7. In the cyclic quadrilateral WXYZ on the circle centered at O, $\angle ZYW = 10°$ and $\angle YOW = 100°$. What is the measure of $\angle YWZ$?



(A) 30° (B) 40° (C) 50° (D) 60°

GM0007

8. In the adjacent figure, if $\angle y + \angle z = 100°$ then the measure of $\angle x$ is :



(A) 50° (B) 40° (C) 45° (D) Cannot be determined

GM0008

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9. In a ΔABC if P be the point of intersection of the interior angle bisectors and Q be the point of intersection of the exterior angle bisectors of angles B and C respectively, then the figure BPCQ is a :

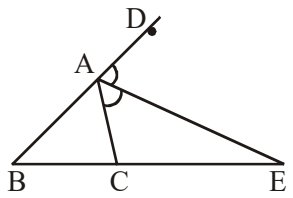
(A) Parallelogram (B) Rhombus (C) Trapezium (D) Cyclic quadrilateral

GM0009

10. D, E and F are points on sides BC, CA and AB respectively of ΔABC such that AD bisects $\angle A$, BE bisects $\angle B$ and CF bisects $\angle C$. If AB = 5 cm, BC = 8 cm and CA = 4 cm, determine AF, CE and BD.

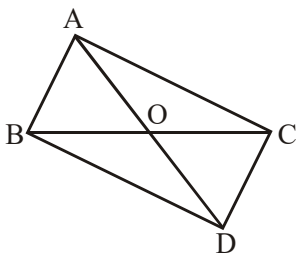
GM0010

11. In the adjoining figure, AE is the bisector of exterior $\angle CAD$ meeting BC produced in E. If AB = 10 cm, AC = 6 cm and BC = 12 cm, find CE.



GM0011

12. In fig. ΔABC and ΔDBC are two triangles on the same base BC. Prove that $\frac{ar(\Delta ABC)}{ar(\Delta DBC)} = \frac{AO}{DO}$.

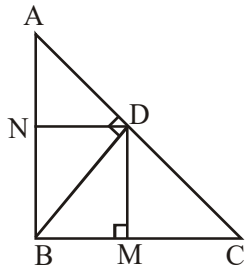


GM0012

13. ABC is a right-triangle with $\angle ABC = 90°$, $BD \perp AC$, $DM \perp BC$ and $DN \perp AB$. Prove that

(i) $DM^2 = DN \times MC$

(ii) $DN^2 = DM \times AN$



GM0013

14. Prove that sum of squares of diagonals of a parallelogram is equal to sum of squares of its sides.

GM0014

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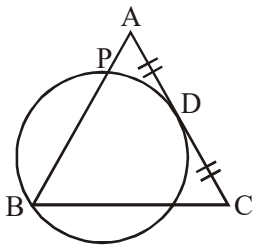
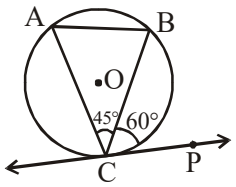
15. In fig., ABC is a triangle in which AB = AC. A circle through B touches AC at D and intersects AB at P. If D is the mid-point of AC, show that 4 AP = AB.



GM0015

16. In fig., CP is a tangent to a circle. If $\angle PCB = 60°$ and $\angle BCA = 45°$, find $\angle ABC$.



GM0016

17. The inscribed circle of ΔABC touches BC, CA and AB at X, Y and Z respectively. If $\angle A = 64°$, $\angle C = 52°$, find $\angle XYZ$ and $\angle XZY$.

GM0017

18. Let $\overline{AM}$ be a median of ΔABC. Prove that AM > (AB + AC – BC)/2.

GM0018

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ANSWER KEY

Do yourself-1

1. 1 **3.** 5 : 4 **4.** (i) $\frac{5}{8}$ (ii) $\frac{25}{64}$ 5. C 6. 5 : 9

Do yourself-2 :

2. 2cm **3.** $\sqrt{5}$ **4.** 30º 5. B 6. B

Do yourself-3 :

1. 35º **2.** 3cm **3.** 9.6 cm **4.** 168 5. C 6. B

Do yourself-4 :

1. 72 **2.** x = 2 cm **3.** 90º 4. C 5. 168

EXERCISE

1. A **2.** B **3.** C **4.** B **5.** B **6.** A

7. B **8.** B **9.** D **10.** AF = $\frac{5}{3}$, CE = $\frac{32}{13}$, BD = $\frac{40}{9}$

11. CE = 18 cm **16.** $\angle ABC = 75°$ **17.** $\angle XYZ = 58°$, $\angle XZY = 64°$

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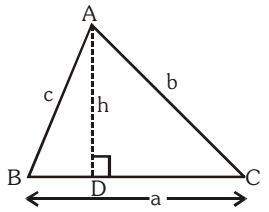
SOLUTIONS OF TRIANGLE

The process of calculating the sides and angles of triangle using given information is called solution of triangle.

In a ΔABC, the angles are denoted by capital letters A, B and C and the length of the sides opposite these angle are denoted by small letter a, b and c respectively.

1. SINE FORMULAE :

In any triangle ABC



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \lambda = \frac{abc}{2\Delta} = 2R$$

where R is circumradius and Δ is area of triangle.

Illustration 1 : Angles of a triangle are in 4 : 1 : 1 ratio. The ratio between its greatest side and perimeter is

(A) $\frac{3}{2+\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2-\sqrt{3}}$ (D) $\frac{1}{2+\sqrt{3}}$

Solution : Angles are in ratio 4 : 1 : 1.

$\Rightarrow$ angles are 120°, 30°, 30°.

If sides opposite to these angles are a, b, c respectively, then a will be the greatest side.

Now from sine formula $\frac{a}{\sin 120°} = \frac{b}{\sin 30°} = \frac{c}{\sin 30°}$

$$\Rightarrow \quad \frac{a}{\sqrt{3}/2} = \frac{b}{1/2} = \frac{c}{1/2}$$

$$\Rightarrow \quad \frac{a}{\sqrt{3}} = \frac{b}{1} = \frac{c}{1} = k \text{ (say)}$$

then $a = \sqrt{3}k$, perimeter = $(2+\sqrt{3})k$

$\therefore$ required ratio = $\frac{\sqrt{3}k}{(2+\sqrt{3})k} = \frac{\sqrt{3}}{2+\sqrt{3}}$ **Ans. (B)**

Illustration 2 : In triangle ABC, if b = 3, c = 4 and $\angle B = \pi/3$, then number of such triangles is -

(A) 1 (B) 2 (C) 0 (D) infinite

Solution : Using sine formulae $\frac{\sin B}{b} = \frac{\sin C}{c}$

$$\Rightarrow \quad \frac{\sin \pi/3}{3} = \frac{\sin C}{4} \Rightarrow \frac{\sqrt{3}}{6} = \frac{\sin C}{4} \Rightarrow \sin C = \frac{2}{\sqrt{3}} > 1 \text{ which is not possible.}$$

Hence there exist no triangle with given elements. **Ans. (C)**

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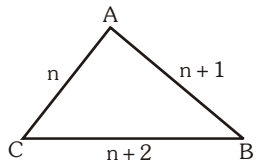
Illustration 3 : The sides of a triangle are three consecutive natural numbers and its largest angle is twice the smallest one. Determine the sides of the triangle.

Solution : Let the sides be n, n + 1, n + 2 cms.



i.e. AC = n, AB = n + 1, BC = n + 2

Smallest angle is B and largest one is A.

Here, $\angle A = 2\angle B$

Also, $\angle A + \angle B + \angle C = 180°$

$\Rightarrow \quad 3\angle B + \angle C = 180° \Rightarrow \angle C = 180° - 3\angle B$

We have, sine law as,

$$\frac{\sin A}{n+2} = \frac{\sin B}{n} = \frac{\sin C}{n+1} \quad \Rightarrow \quad \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin(180-3B)}{n+1}$$

$$\Rightarrow \quad \underset{\text{(i)}}{\frac{\sin 2B}{n+2}} = \underset{\text{(ii)}}{\frac{\sin B}{n}} = \underset{\text{(iii)}}{\frac{\sin 3B}{n+1}}$$

from (i) and (ii);

$$\frac{2\sin B\cos B}{n+2} = \frac{\sin B}{n} \quad \Rightarrow \quad \cos B = \frac{n+2}{2n} \qquad \text{..... (iv)}$$

and from (ii) and (iii);

$$\frac{\sin B}{n} = \frac{3\sin B - 4\sin^3 B}{n+1} \quad \Rightarrow \quad \frac{\sin B}{n} = \frac{\sin B(3-4\sin^2 B)}{n+1}$$

$$\Rightarrow \quad \frac{n+1}{n} = 3 - 4(1-\cos^2 B) \qquad \text{..... (v)}$$

from (iv) and (v), we get

$$\frac{n+1}{n} = -1 + 4\left(\frac{n+2}{2n}\right)^2 \Rightarrow \quad \frac{n+1}{n} + 1 = \left(\frac{n^2+4n+4}{n^2}\right)$$

$$\Rightarrow \quad \frac{2n+1}{n} = \frac{n^2+4n+4}{n^2} \quad \Rightarrow \quad 2n^2 + n = n^2 + 4n + 4$$

$$\Rightarrow \quad n^2 - 3n - 4 = 0 \quad \Rightarrow \quad (n-4)(n+1) = 0$$

$n = 4$ or -1

where $n \neq -1$

$\therefore \quad n = 4$. Hence the sides are 4, 5, 6 **Ans.**

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Do yourself - 1 :

(i) If in a ΔABC, $\angle A = \frac{\pi}{6}$ and $b : c = 2 : \sqrt{3}$, find $\angle B$.

(ii) Show that, in any ΔABC : $a \sin(B - C) + b \sin(C - A) + c \sin(A - B) = 0$.

(iii) If in a ΔABC, $\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$, show that a^2, b^2, c^2 are in A.P.

(iv) Prove that $(b-c)\cos\frac{A}{2} = a\sin\left(\frac{B-C}{2}\right)$.

(v) Prove that $(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C = 0$.

(vi) Prove that $b^2\cos 2A - a^2\cos 2B = b^2 - a^2$.

(vii) The perimeter of a triangle ABC is six times the arithmetic mean of the sines of its angles. If the side b is equal to $\sqrt{3}$, then find angle B.

(viii) In any triangle ABC, if $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A-B)}{\sin(A+B)}$, then prove that the triangle is either right angled or isosceles.

(ix) ABC is a triangle such that $\sin(2A + B) = \sin(C - A) = -\sin(B + 2C) = \frac{1}{2}$. If A, B, C are in A.P. Find A, B, C.

(x) If the angles of ΔABC are in the ratio 1 : 2 : 3, then find the ratio of their corresponding sides

(xi) In a ΔABC prove that $\frac{c}{a-b} = \frac{\tan\frac{A}{2} + \tan\frac{B}{2}}{\tan\frac{A}{2} - \tan\frac{B}{2}}$

(xii) Find the real value of x such that $x^2 + 2x$, $2x + 3$ and $x^2 + 3x + 8$ are lengths of the sides of a triangle.

2. COSINE FORMULAE :

(a) $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ **(b)** $\cos B = \frac{c^2 + a^2 - b^2}{2ca}$ **(c)** $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

or $a^2 = b^2 + c^2 - 2bc\cos A$

Illustration 4 : In a triangle ABC, if $B = 30°$ and $c = \sqrt{3}\,b$, then A can be equal to -

(A) 45° (B) 60° (C) 90° (D) 120°

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Solution : We have $\cos B = \dfrac{c^2+a^2-b^2}{2ca} \Rightarrow \dfrac{\sqrt{3}}{2} = \dfrac{3b^2+a^2-b^2}{2\times\sqrt{3}b\times a}$

$\Rightarrow$ $a^2 - 3ab + 2b^2 = 0 \Rightarrow (a - 2b)(a - b) = 0$

$\Rightarrow$ Either $a = b \Rightarrow A = 30°$

or $a = 2b \Rightarrow a^2 = 4b^2 = b^2 + c^2 \Rightarrow A = 90°$. **Ans. (C)**

Illustration 5 : In a triangle ABC, $(a^2 - b^2 - c^2)\tan A + (a^2 - b^2 + c^2)\tan B$ is equal to -

(A) $(a^2 + b^2 - c^2)\tan C$ (B) $(a^2 + b^2 + c^2)\tan C$

(C) $(b^2 + c^2 - a^2)\tan C$ (D) none of these

Solution : Using cosine law :

The given expression is equal to $-2\,bc\cos A\tan A + 2\,ac\cos B\tan B$

$= 2abc\left(-\dfrac{\sin A}{a} + \dfrac{\sin B}{b}\right) = 0$ **Ans. (D)**

Do yourself - 2 :

(i) If a : b : c = 4 : 5 : 6, then show that $\angle C = 2\angle A$.

(ii) In any ΔABC, prove that :

(a) $\dfrac{\cos A}{a} + \dfrac{\cos B}{b} + \dfrac{\cos C}{c} = \dfrac{a^2+b^2+c^2}{2abc}$

(b) $\dfrac{b^2}{a}\cos A + \dfrac{c^2}{b}\cos B + \dfrac{a^2}{c}\cos C = \dfrac{a^4+b^4+c^4}{2abc}$

(iii) In ΔABC, prove that :

(a) $\dfrac{\cos B}{\cos C} = \dfrac{c - b\cos A}{b - c\cos A}$ (b) $\sum \dfrac{\cos A}{c\cos B + b\cos C} = \dfrac{a^2+b^2+c^2}{2abc}$

(iv) In ΔABC, if A,B,C are in A.P., then prove that $2\cos\dfrac{A-C}{2} = \dfrac{a+c}{\sqrt{a^2+c^2-ac}}$.

(v) If in a triangle ABC, $\angle C = 60°$, then prove that $\dfrac{1}{a+c} + \dfrac{1}{b+c} = \dfrac{3}{a+b+c}$.

(vi) Triangle ABC has BC = 1 and AC = 2. Find the maximum possible value of angle A.

(vii) In a ΔABC, prove that $b^2\sin 2C + c^2\sin 2B = 2bc\sin A$.

(viii) In a ΔABC, if $9a^2 + 9b^2 = 19c^2$, then find the value of $\dfrac{\cot C}{\cot A + \cot B}$.

(ix) In a triangle ABC, the sides a, b, c are roots of $x^3 - 11x^2 + 38x - 40 = 0$. If $\sum\left(\dfrac{\cos A}{a}\right) = \dfrac{p}{q}$, then find the least value of $(p + q)$ where $p,q \in N$.

(x) In a triangle ABC, a : b : c = 4 : 5 : 6. Then 3A + B equals to

(A) 4C (B) 2π (C) $\pi - C$ (D) π

(xi) If in a triangle ABC ; sinA, sinB & sinC are in A.P. ($C > 90°$), then cosA is

(A) $\dfrac{3c-4b}{2b}$ (B) $\dfrac{3c-4b}{2c}$ (C) $\dfrac{4c-3b}{2b}$ (D) $\dfrac{4c-3b}{2c}$

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3. PROJECTION FORMULAE :

(a) $b\cos C + c\cos B = a$ **(b)** $c\cos A + a\cos C = b$ **(c)** $a\cos B + b\cos A = c$

Illustration 6 : In a ΔABC, $c\cos^2\frac{A}{2} + a\cos^2\frac{C}{2} = \frac{3b}{2}$, then show a, b, c are in A.P.

Solution : Here, $\frac{c}{2}(1+\cos A) + \frac{a}{2}(1+\cos C) = \frac{3b}{2}$

$\Rightarrow$ $a + c + (c\cos A + a\cos C) = 3b$

$\Rightarrow$ $a + c + b = 3b$ {using projection formula}

$\Rightarrow$ $a + c = 2b$

which shows a, b, c are in A.P.

Do yourself - 3 :

(i) In a ΔABC, if $\angle A = \frac{\pi}{4}$, $\angle B = \frac{5\pi}{12}$, show that $a + c\sqrt{2} = 2b$.

(ii) In a ΔABC, prove that: (a) $b(a\cos C - c\cos A) = a^2 - c^2$ (b) $2\left(b\cos^2\frac{C}{2} + c\cos^2\frac{B}{2}\right) = a + b + c$.

(iii) In ΔABC, prove that $\frac{\cos A + \cos C}{a+c} + \frac{\cos B}{b} = \frac{1}{b}$.

(iv) In a ΔABC, prove that :

(a) $a(\cos B + \cos C) = 2(b + c)\sin^2 A/2$

(b) $a(\cos C - \cos B) = 2(b - c)\cos^2 A/2$

(v) In ΔABC, prove that $a(b^2 + c^2)\cos A + b(c^2 + a^2)\cos B + c(a^2 + b^2)\cos C = 3abc$.

(vi) In a ΔABC, prove that $\frac{\cos A}{c\cos B + b\cos C} + \frac{\cos B}{a\cos C + c\cos A} + \frac{\cos C}{a\cos B + b\cos A} = \frac{a^2+b^2+c^2}{2abc}$.

(vii) In ΔABC, prove that $(b + c)\cos A + (c + a)\cos B + (a + b)\cos C = a + b + c$.

4. NAPIER'S ANALOGY (TANGENT RULE) :

(a) $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$ **(b)** $\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a}\cot\frac{B}{2}$ **(c)** $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{C}{2}$

Illustration 7 : In a ΔABC, the tangent of half the difference of two angles is one-third the tangent of half the sum of the angles. Determine the ratio of the sides opposite to the angles.

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Solution : Here, $\tan\left(\frac{A-B}{2}\right)=\frac{1}{3}\tan\left(\frac{A+B}{2}\right)$ ..... (i)

using Napier's analogy, $\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right)$ ..... (ii)

from (i) & (ii) ;

$$\frac{1}{3}\tan\left(\frac{A+B}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right) \Rightarrow \frac{1}{3}\cot\left(\frac{C}{2}\right)=\frac{a-b}{a+b}.\cot\left(\frac{C}{2}\right)$$

$$\{\text{as } A+B+C=\pi \quad \therefore \tan\left(\frac{B+C}{2}\right)=\tan\left(\frac{\pi}{2}-\frac{C}{2}\right)=\cot\frac{C}{2}\}$$

$$\Rightarrow \quad \frac{a-b}{a+b}=\frac{1}{3} \quad \text{or} \quad 3a-3b=a+b$$

$$2a=4b \quad \text{or} \quad \frac{a}{b}=\frac{2}{1}\Rightarrow\frac{b}{a}=\frac{1}{2}$$

Thus the ratio of the sides opposite to the angles is b : a = 1 : 2. **Ans.**

Do yourself - 4 :

(i) In any ΔABC, prove that $\frac{b-c}{b+c}=\frac{\tan\left(\frac{B-C}{2}\right)}{\tan\left(\frac{B+C}{2}\right)}$.

(ii) If ΔABC is right angled at C, prove that : (a) $\tan\frac{A}{2}=\sqrt{\frac{c-b}{c+b}}$ (b) $\sin(A-B)=\frac{a^2-b^2}{a^2+b^2}$.

(iii) If in a ΔABC, b = 3c, then find the value of $\cot\frac{A}{2}.\cot\frac{B-C}{2}$.

(iv) In a triangle ABC, $\angle A = 60^\circ$ and $b:c=\sqrt{3}+1:2$, then find the value of $(\angle B-\angle C)$.

(v) ΔABC with $\angle A$ acute, b = 2 and $c=\left(\sqrt{3}-1\right)$ has area $\left(\sqrt{3}-1\right)/2$, then find the measures of angles of triangle.

(vi) In a ΔABC if b = 3, c = 5 and $\cos(B-C)=\frac{7}{25}$, then find the value of $\sin\frac{A}{2}$.

(vii) Find the unknown elements of the ΔABC in which $a=\sqrt{3}+1, b=\sqrt{3}-1, C=90^\circ$.

5. HALF ANGLE FORMULAE :

$s=\frac{a+b+c}{2}$ = semi-perimeter of triangle.

(a) (i) $\sin\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{bc}}$ (ii) $\sin\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{ca}}$ (iii) $\sin\frac{C}{2}=\sqrt{\frac{(s-a)(s-b)}{ab}}$

(b) (i) $\cos\frac{A}{2}=\sqrt{\frac{s(s-a)}{bc}}$ (ii) $\cos\frac{B}{2}=\sqrt{\frac{s(s-b)}{ca}}$ (iii) $\cos\frac{C}{2}=\sqrt{\frac{s(s-c)}{ab}}$

(c) (i) $\tan\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ (ii) $\tan\frac{B}{2}=\sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$ (iii) $\tan\frac{C}{2}=\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$

$=\frac{\Delta}{s(s-a)}$ $=\frac{\Delta}{s(s-b)}$ $=\frac{\Delta}{s(s-c)}$

(d) Area of Triangle

$$\Delta=\sqrt{s(s-a)(s-b)(s-c)}=\frac{1}{2}bc\sin A=\frac{1}{2}ca\sin B=\frac{1}{2}ab\sin C=\frac{1}{2}ap_1=\frac{1}{2}bp_2=\frac{1}{2}cp_3,$$

where p_1, p_2, p_3 are altitudes from vertices A,B,C respectively.

Illustration 8 : If in a triangle ABC, CD is the angle bisector of the angle ACB, then CD is equal to-

(A) $\frac{a+b}{2ab}\cos\frac{C}{2}$ (B) $\frac{2ab}{a+b}\sin\frac{C}{2}$ (C) $\frac{2ab}{a+b}\cos\frac{C}{2}$ (D) $\frac{b\sin\angle DAC}{\sin(B+C/2)}$

Solution : $\Delta CAB = \Delta CAD + \Delta CDB$

$$\Rightarrow \quad \frac{1}{2}ab\sin C=\frac{1}{2}b.CD.\sin\left(\frac{C}{2}\right)+\frac{1}{2}a.CD\sin\left(\frac{C}{2}\right)$$

$$\Rightarrow \quad CD(a+b)\sin\left(\frac{C}{2}\right)=ab\left(2\sin\left(\frac{C}{2}\right)\cos\left(\frac{C}{2}\right)\right)$$

So $CD=\frac{2ab\cos(C/2)}{(a+b)}$

and in ΔCAD, $\frac{CD}{\sin\angle DAC}=\frac{b}{\sin\angle CDA}$ (by sine rule)

$$\Rightarrow \quad CD=\frac{b\sin\angle DAC}{\sin(B+C/2)}$$

Ans. (C,D)

Illustration 9 : If Δ is the area and 2s the sum of the sides of a triangle, then show $\Delta\le\frac{s^2}{3\sqrt{3}}$.

Solution : We have, $2s = a + b + c$, $\Delta^2 = s(s-a)(s-b)(s-c)$

Now, A.M. $\ge$ G.M.

$$\frac{(s-a)+(s-b)+(s-c)}{3}\ge\{(s-a)(s-b)(s-c)\}^{1/3}$$

or $\frac{3s-2s}{3}\ge\left(\frac{\Delta^2}{s}\right)^{1/3}$

or $\frac{s}{3}\ge\left(\frac{\Delta^2}{s}\right)^{1/3}$

or $\frac{\Delta^2}{s}\le\frac{s^3}{27}$ $\Rightarrow$ $\Delta\le\frac{s^2}{3\sqrt{3}}$

Ans.

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Do yourself - 5 :

(i) Given a = 6, b = 8, c = 10. Find

(a) sinA (b) tanA (c) $\sin\frac{A}{2}$ (d) $\cos\frac{A}{2}$ (e) $\tan\frac{A}{2}$ (f) Δ

(ii) Prove that in any ΔABC, (abcs) $\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2} = \Delta^2$.

(iii) If the cotangents of half the angles of a triangle are in A.P., then prove that the sides are in A.P.

(iv) If in a triangle ABC, $\Delta = a^2 - (b - c)^2$, then find the value of tanA.

(v) In any triangle ABC, prove that $(a+b+c)\left(\tan\frac{A}{2}+\tan\frac{B}{2}\right) = 2c\cot\frac{C}{2}$.

(vi) In a triangle ABC, if $\tan\frac{A}{2}.\tan\frac{B}{2} = \frac{1}{3}$ and ab = 4, then find the least value of c.

(vii) In ΔABC, if a,b,c (taken in that order) are in A.P. then find the value of $\cot\frac{A}{2}\ \cot\frac{C}{2}$.

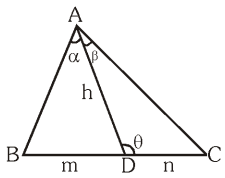
(viii) If A,B,C are the angle of a triangle, then prove that $\cot\frac{A}{2}+\cot\frac{B}{2}+\cot\frac{C}{2} = \frac{s^2}{\Delta}$

(ix) In a ΔABC if b sinC(b cosC + c cosB) = 64, then find the area of the ΔABC.

6. m-n THEOREM :

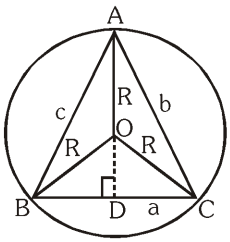
$(m + n)\cot\theta = m\cot\alpha - n\cot\beta$

$(m + n)\cot\theta = n\cot B - m\cot C$.



7. RADIUS OF THE CIRCUMCIRCLE 'R' :

Circumcentre is the point of intersection of perpendicular bisectors of the sides and distance between circumcentre & vertex of triangle is called circumradius 'R'.



$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}.$$

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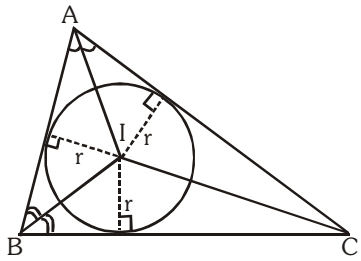
8. RADIUS OF THE INCIRCLE 'r' :

Point of intersection of internal angle bisectors is incentre and perpendicular distance of incentre from any side is called inradius 'r'.



$$r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2} = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$$

$$= a\frac{\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}} = b\frac{\sin\frac{A}{2}\sin\frac{C}{2}}{\cos\frac{B}{2}} = c\frac{\sin\frac{B}{2}\sin\frac{A}{2}}{\cos\frac{C}{2}}$$

Illustration 10 : In a triangle ABC, if a : b : c = 4 : 5 : 6, then ratio between its circumradius and inradius is-

(A) $\frac{16}{7}$ (B) $\frac{16}{9}$ (C) $\frac{7}{16}$ (D) $\frac{11}{7}$

Solution : $\frac{R}{r} = \frac{abc}{4\Delta} \Big/ \frac{\Delta}{s} = \frac{(abc)s}{4\Delta^2} \Rightarrow \frac{R}{r} = \frac{abc}{4(s-a)(s-b)(s-c)}$ ....(i)

$\because$ a : b : c = 4 : 5 : 6 $\Rightarrow \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k$ (say)

$\Rightarrow$ a = 4k, b = 5k, c = 6k

$\therefore$ $s = \frac{a+b+c}{2} = \frac{15k}{2}$, $s-a = \frac{7k}{2}$, $s-b = \frac{5k}{2}$, $s-c = \frac{3k}{2}$

using (i) in these values $\frac{R}{r} = \frac{(4k)(5k)(6k)}{4\left(\frac{7k}{2}\right)\left(\frac{5k}{2}\right)\left(\frac{3k}{2}\right)} = \frac{16}{7}$ **Ans. (A)**

Illustration 11 : If A, B, C are the angles of a triangle, prove that : cosA + cosB + cosC = $1+\frac{r}{R}$.

Solution : $\cos A + \cos B + \cos C = 2\cos\left(\frac{A+B}{2}\right).\cos\left(\frac{A-B}{2}\right) + \cos C$

$$= 2\sin\frac{C}{2}.\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2} = 1 + 2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \sin\left(\frac{C}{2}\right)\right]$$

$$= 1 + 2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right)\right] \qquad \left\{\because \frac{C}{2} = 90° - \left(\frac{A+B}{2}\right)\right\}$$

$$= 1 + 2\sin\frac{C}{2}.2\sin\frac{A}{2}.\sin\frac{B}{2} = 1 + 4\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2}$$

$$= 1 + \frac{r}{R} \qquad \{\text{as, } r = 4R\sin A/2 \,.\, \sin B/2 \,.\, \sin C/2\}$$

$\Rightarrow$ cosA + cosB + cosC = $1+\frac{r}{R}$. Hence proved.

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Do yourself - 6 :

(i) If in ΔABC, a = 3, b = 4 and c = 5, find

(a) Δ (b) R (c) r

(ii) In a ΔABC, show that :

(a) $\dfrac{a^2-b^2}{c}=2R\sin(A-B)$ (b) $r\cos\dfrac{A}{2}\cos\dfrac{B}{2}\cos\dfrac{C}{2}=\dfrac{\Delta}{4R}$ (c) $a+b+c=\dfrac{abc}{2Rr}$

(iii) Let Δ & Δ' denote the areas of a Δ and that of its incircle. Prove that

$$\Delta : \Delta' = \left(\cot\frac{A}{2}.\cot\frac{B}{2}.\cot\frac{C}{2}\right) : \pi$$

(iv) If the median AD of a ΔABC makes an angle $3\pi/4$ with the side BC, then evaluate $|\cot B - \cot C|$.

(v) If in a ΔABC, AB = 4 cm, AC = 8cm and $\cos\left(\dfrac{B-C}{2}\right)=\dfrac{1}{\sqrt{2}}$; then evaluate $|2\cot B - \cot C|$.

(vi) In ΔABC, if circumradius 'R' and inradius 'r' are connected by relation $R^2 - 4Rr + 8r^2 - 12r + 9 = 0$, then find semiperimeter of the ΔABC.

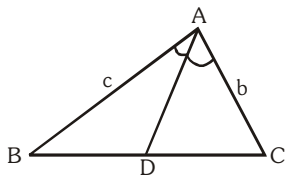
(vii) In a ΔABC, prove that $\sin 2A + \sin 2B + \sin 2C = 2\Delta/R^2$

(viii) In a ΔABC, prove that $\dfrac{b^2-c^2}{2a}=R\sin(B-C)$

(ix) In a ΔABC, prove that $a\cos B\cos C + b\cos C\cos A + c\cos A\cos B = \dfrac{\Delta}{R}$

9. ANGLE BISECTORS & MEDIANS :

An angle bisector divides the base in the ratio of corresponding sides.



$$\frac{BD}{CD}=\frac{c}{b} \Rightarrow BD = \frac{ac}{b+c} \quad \& \quad CD = \frac{ab}{b+c}$$

If m_a and β_a are the lengths of a median and an angle bisector from the angle A then,

$$m_a=\frac{1}{2}\sqrt{2b^2+2c^2-a^2} \text{ and } \beta_a=\frac{2bc\cos\frac{A}{2}}{b+c}$$

Note that $m_a^2+m_b^2+m_c^2=\dfrac{3}{4}(a^2+b^2+c^2)$

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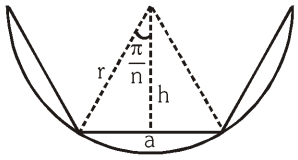
10. REGULAR POLYGON :

A regular polygon has all its sides equal. It may be inscribed or circumscribed.

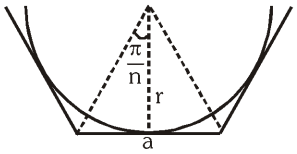


(a) Inscribed in circle of radius r :

(i) $a = 2h\tan\frac{\pi}{n} = 2r\sin\frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides inscribed in a circle of radius r are given by $P = 2nr\sin\frac{\pi}{n}$ and $A = \frac{1}{2}nr^2\sin\frac{2\pi}{n}$

(b) Circumscribed about a circle of radius r :



(i) $a = 2r\tan\frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides circumscribed about a given circle of radius r is given by $P = 2nr\tan\frac{\pi}{n}$ and

$A = nr^2\tan\frac{\pi}{n}$

Do yourself - 7 :

(i) If the perimeter of a circle and a regular polygon of n sides are equal, then

prove that $\frac{\text{area of the circle}}{\text{area of polygon}} = \frac{\tan\frac{\pi}{n}}{\frac{\pi}{n}}$.

(ii) The ratio of the area of n-sided regular polygon, circumscribed about a circle, to the area of the regular polygon of equal number of sides inscribed in the circle is 4 : 3. Find the value of n.

(iii) A regular pentagon and a regular decagon have the same perimeter, prove that their areas are in the ratio $2 : \sqrt{5}$.

(iv) If the area of circle is A_1 and area of regular pentagon inscribed in the circle is A_2, then find the ratio of areas A_1/A_2.

(v) A circle is inscribed in a regular octagon whose area is Δ. If area of a new regular octagon formed by points of contact of polygon and incircle is Δ', then $\frac{\Delta}{\Delta'}$ is

(A) $\cos^2\frac{\pi}{8}$ (B) $\sec\frac{\pi}{8}$ (C) $\sec^2\frac{\pi}{8}$ (D) $\cot^2\frac{\pi}{8}$

(vi) Let A_i, i = 1,2,3,....10 are vertices of regular decagon P. If area of polygon P is $5\sin\left(\frac{\pi}{5}\right)$ square units then the value of $(A_1A_2)(A_1A_3)(A_1A_4)....(A_1A_{10})$ is

(A) 10 (B) 9 (C) $45\sin\left(\frac{\pi}{5}\right)$ (D) $45\cos\left(\frac{\pi}{5}\right)$

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EXERCISE (O-1)

1. In triangle ABC, if $\sin^3 A + \sin^3 B + \sin^3 C = 3\sin A.\sin B.\sin C$, then triangle is

(A) obtuse angled (B) right angled (C) obtuse right angled (D) equilateral

TS0021

2. A triangle has vertices A, B and C, and the respective opposite sides have lengths a, b and c. This triangle is inscribed in a circle of radius R. If b = c = 1 and the altitude from A to side BC has length $\sqrt{\frac{2}{3}}$, then R equals

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2\sqrt{2}}$

TS0022

3. In a triangle ABC, if $\angle C = 105°$, $\angle B = 45°$ and length of side AC = 2 units, then the length of the side AB is equal to

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\sqrt{2}+1$ (D) $\sqrt{3}+1$

TS0023

4. In a triangle ABC, if a = 13, b = 14 and c = 15, then angle A is equal to
(All symbols used have their usual meaning in a triangle.)

(A) $\sin^{-1}\frac{4}{5}$ (B) $\sin^{-1}\frac{3}{5}$ (C) $\sin^{-1}\frac{3}{4}$ (D) $\sin^{-1}\frac{2}{3}$

TS0024

5. In triangle ABC, if AC = 8, BC = 7 and D lies between A and B such that AD = 2, BD = 4, then the length CD equals

(A) $\sqrt{46}$ (B) $\sqrt{48}$ (C) $\sqrt{51}$ (D) $\sqrt{75}$

TS0025

6. In triangle ABC, if 2b = a + c and A – C = 90°, then sin B equals
[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) $\frac{\sqrt{7}}{5}$ (B) $\frac{\sqrt{5}}{8}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{5}}{3}$

TS0026

7. In a triangle ABC, $a^3 + b^3 + c^3 = c^2 (a + b + c)$
(All symbol used have usual meaning in a triangle.)
Statement–1: The value of $\angle C = 60°$.
Statement –2: ΔABC must be equilateral.
(A) Statement–1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

TS0027

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8. A circle is inscribed in a right triangle ABC, right angled at C. The circle is tangent to the segment AB at D and length of segments AD and DB are 7 and 13 respectively. Area of triangle ABC is equal to

(A) 91 (B) 96 (C) 100 (D) 104

TS0028

9. In triangle ABC, If $\frac{1}{a+c}+\frac{1}{b+c}=\frac{3}{a+b+c}$ then angle C is equal to

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) 30° (B) 45° (C) 60° (D) 90°

TS0029

10. In a triangle ABC, if $b = (\sqrt{3}-1)a$ and $\angle C = 30°$, then the value of (A – B) is equal to

(All symbols used have usual meaning in a triangle.)

(A) 30° (B) 45° (C) 60° (D) 75°

TS0030

11. Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c}=\sqrt{\frac{3}{2}}$, then $\angle A$ is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$

TS0001

12. The sides a, b, c (taken in that order) of triangle ABC are in A.P.

If $\cos\alpha=\frac{a}{b+c}, \cos\beta=\frac{b}{c+a}, \cos\gamma=\frac{c}{a+b}$ then $\tan^2\left(\frac{\alpha}{2}\right)+\tan^2\left(\frac{\gamma}{2}\right)$ is equal to

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

TS0032

13. AD and BE are the medians of a triangle ABC. If AD = 4, $\angle DAB = \frac{\pi}{6}, \angle ABE = \frac{\pi}{3}$, then area of triangle ABC equals

(A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{32}{9}\sqrt{3}$

TS0033

14. In a triangle ABC, if (a + b + c) (a + b – c) (b + c – a) (c + a – b) = $\frac{8a^2b^2c^2}{a^2+b^2+c^2}$, then the triangle is

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) isosceles (B) right angled (C) equilateral (D) obtuse angled

TS0034

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15. For right angled isosceles triangle, $\frac{r}{R} =$

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) $\tan\frac{\pi}{12}$ (B) $\cot\frac{\pi}{12}$ (C) $\tan\frac{\pi}{8}$ (D) $\cot\frac{\pi}{8}$

TS0035

16. If K is a point on the side BC of an equilateral triangle ABC and if $\angle BAK = 15°$, then the ratio of lengths $\frac{AK}{AB}$ is

(A) $\frac{3\sqrt{2}\left(3+\sqrt{3}\right)}{2}$ (B) $\frac{\sqrt{2}\left(3+\sqrt{3}\right)}{2}$ (C) $\frac{\sqrt{2}\left(3-\sqrt{3}\right)}{2}$ (D) $\frac{3\sqrt{2}\left(3-\sqrt{3}\right)}{2}$

TS0002

17. Let ABC be a right triangle with length of side AB = 3 and hypotenuse AC = 5.

If D is a point on BC such that $\frac{BD}{DC} = \frac{AB}{AC}$, then AD is equal to

(A) $\frac{4\sqrt{3}}{3}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\frac{4\sqrt{5}}{3}$ (D) $\frac{5\sqrt{3}}{4}$

TS0003

18. In ΔABC, if a = 2b and A = 3B, then the value of $\frac{c}{b}$ is equal to

(A) 3 (B) $\sqrt{2}$ (C) 1 (D) $\sqrt{3}$

TS0004

19. If the angle A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of expression $E = \left(\frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A\right)$, is

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

TS0005

20. The ratio of the sides of a triangle ABC is $1:\sqrt{3}:2$. Then ratio of A : B : C is

(A) 3 : 5 : 2 (B) 1 : $\sqrt{3}$: 2 (C) 3 : 2 : 1 (D) 1 : 2 : 3

TS0006

21. If in a triangle sin A : sin C = sin (A – B) : sin (B – C), then a^2, b^2, c^2

(A) are in A.P. (B) are in G.P. (C) are in H.P. (D) none of these

TS0007

22. In a triangle tan A : tan B : tan C = 1 : 2 : 3, then $a^2 : b^2 : c^2$ equals

(A) 5 : 8 : 9 (B) 5 : 8 : 12 (C) 3 : 5 : 8 (D) 5 : 8 : 10

TS0008

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23. In an acute triangle ABC, $\angle ABC = 45°$, AB = 3 and AC = $\sqrt{6}$. The angle $\angle BAC$, is

(A) 60° (B) 65° (C) 75° (D) 15° or 75°

TS0009

24. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. If a = 1, b = 3 and C = 60°, then $\sin^2 B$ is equal to

(A) $\frac{27}{28}$ (B) $\frac{3}{28}$ (C) $\frac{81}{28}$ (D) $\frac{1}{3}$

TS0010

25. In triangle ABC, If $s = 3+\sqrt{3}+\sqrt{2}$, 3B – C = 30°, A + 2B = 120°, then the length of longest side of triangle is

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) 2 (B) $2\sqrt{2}$ (C) $2(\sqrt{3}+1)$ (D) $\sqrt{3}-1$

TS0011

26. In ΔABC if a = 8, b = 9, c = 10, then the value of $\frac{\tan C}{\sin B}$ is

(A) $\frac{32}{9}$ (B) $\frac{24}{7}$ (C) $\frac{21}{4}$ (D) $\frac{18}{5}$

TS0012

27. If the sides of a triangle are $\sin\alpha$, $\cos\alpha$, $\sqrt{1+\sin\alpha\cos\alpha}$, $0 < \alpha < \frac{\pi}{2}$, the largest angle is

(A) 60° (B) 90° (C) 120° (D) 150°

TS0013

28. In a triangle ABC, $\angle A = 60°$ and b : c = $\left(\sqrt{3}+1\right)$: 2 then ($\angle B - \angle C$) has the value equal to

(A) 15° (B) 30° (C) 22.5 ° (D) 45°

TS0014

29. In triangle ABC, if $\Delta = a^2 - (b-c)^2$, then tan A =

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) $\frac{15}{16}$ (B) $\frac{1}{2}$ (C) $\frac{8}{17}$ (D) $\frac{8}{15}$

TS0015

30. In triangle ABC, if $\cot\frac{A}{2} = \frac{b+c}{a}$, then triangle ABC must be

[Note: All symbols used have usual meaning in ΔABC.]

(A) isosceles (B) equilateral

(C) right angled (D) isoceles right angled

TS0016

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EXERCISE (O-2)

Multiple Correct Answer Type :

1. Given an acute triangle ABC such that $\sin C = \frac{4}{5}$, $\tan A = \frac{24}{7}$ and AB = 50. Then-

(A) centroid, orthocentre and incentre of ΔABC are collinear

(B) $\sin B = \frac{4}{5}$

(C) $\sin B = \frac{4}{7}$

(D) area of $\Delta ABC = 1200$

TS0036

2. In ΔABC, angle A is 120°, BC + CA = 20 and AB + BC = 21, then

(A) AB > AC (B) AB < AC

(C) ΔABC is isosceles (D) area of $\Delta ABC = 14\sqrt{3}$

TS0037

3. In which of the following situations, it is possible to have a triangle ABC?
(All symbols used have usual meaning in a triangle.)

(A) $(a + c - b)(a - c + b) = 4bc$ (B) $b^2 \sin 2C + c^2 \sin 2B = ab$

(C) a = 3, b = 5, c = 7 and $C = \frac{2\pi}{3}$ (D) $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$

TS0038

4. If the lengths of the medians AD,BE and CF of triangle ABC are 6, 8,10 respectively, then-

(A) AD & BE are perpendicular (B) BE and CF are perpendicular

(C) area of $\Delta ABC = 32$ (D) area of $\Delta DEF = 8$

TS0039

5. In ΔABC, angle A, B and C are in the ratio 1 : 2 : 3, then which of the following is (are) correct?
(All symbol used have usual meaning in a triangle.)

(A) Circumradius of $\Delta ABC = c$ (B) $a : b : c = 1 : \sqrt{3} : 2$

(C) Perimeter of $\Delta ABC = 3 + \sqrt{3}$ (D) Area of $\Delta ABC = \frac{\sqrt{3}}{8} c^2$

TS0040

6. In a triangle ABC, let BC = 1, AC = 2 and measure of angle C is 30°. Which of the following statement(s) is (are) correct?

(A) $2 \sin A = \sin B$

(B) Length of side AB equals $5 - 2\sqrt{3}$

(C) Measure of angle A is less than 30°

(D) Circumradius of triangle ABC is equal to length of side AB

TS0041

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7. In a triangle ABC, if cos A cos 2B + sin A sin 2B sin C = 1, then

(A) A,B,C are in A.P. (B) B,A,C are in A.P. (C) $\frac{r}{R} = 2$ (D) $\frac{r}{R} = \sqrt{2}\sin\frac{\pi}{12}$

TS0042

8. Let one angle of a triangle be 60°, the area of triangle is $10\sqrt{3}$ and perimeter is 20 cm. If a > b > c where a, b and c denote lengths of sides opposite to vertices A, B and C respectively, then which of the following is (are) correct?

(A) Inradius of triangle is $\sqrt{3}$

(B) Length of longest side of triangle is 7

(C) Circumradius of triangle is $\frac{7}{\sqrt{3}}$

(D) Radius of largest escribed circle is $\frac{1}{12}$

TS0045

9. In triangle ABC, let b = 10, c = $10\sqrt{2}$ and $R = 5\sqrt{2}$ then which of the following statement(s) is (are) correct?

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) Area of triangle ABC is 50.

(B) Distance between orthocentre and circumcentre is $5\sqrt{2}$

(C) Sum of circumradius and inradius of triangle ABC is equal to 10

(D) Length of internal angle bisector of ∠ACB of triangle ABC is $\frac{5}{2\sqrt{2}}$

TS0046

10. In a triangle ABC, if a = 4, b = 8 and ∠C = 60°, then which of the following relations is (are) correct?

[**Note:** All symbols used have usual meaning in triangle ABC.]

(A) The area of triangle ABC is $8\sqrt{3}$

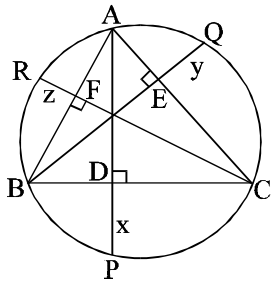
(B) The value of $\sum \sin^2 A = 2$

(C) Inradius of triangle ABC is $\frac{2\sqrt{3}}{3+\sqrt{3}}$

(D) The length of internal angle bisector of angle C is $\frac{4}{\sqrt{3}}$

TS0047

11. As shown in the figure AD is the altitude on BC and AD produced meets the circumcircle of ΔABC at P where DP = x. Similarly EQ = y and FR = z. If a, b, c respectively denotes the sides BC, CA and AB then $\frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z}$ has the value equal to



(A) tanA + tanB + tanC

(B) cotA + cotB + cotC

(C) cosA + cosB + cosC

(D) cosecA + cosecB + cosecC

TS0050

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Paragraph for Question 12 and 13

Let incircle of ΔABC with $\angle B = \angle C$, touches AB, BC, CA at R, P, Q respectively. If inradius 'r' is integer and perimeter of ΔABC is smallest possible integer such that $\frac{2}{AR}+\frac{5}{BP}+\frac{5}{CQ}=\frac{6}{r}$ then

12. The value of 'r' is

(A) 4 (B) 3 (C) 2 (D) 1

TS0113

13. Area of ΔABC is

(A) 15 sq. unit (B) 21 sq. unit (C) 24 sq. unit (D) 27 sq. unit

TS0114

Match the column

14. If symbols have usual notations in triangle ABC, then match the following columns :

	Column-I		Column-II
(A)	$r\, r_1 r_2 r_3$	(P)	rs^2
(B)	$r_1 r_2 r_3$	(Q)	$4R \cos C$
(C)	$r_1 + r_2 - r_3 + r$	(R)	$\frac{1}{2Rr}$
(D)	$\frac{1}{bc}+\frac{1}{ca}+\frac{1}{ab}$	(S)	Δ^2

TS0115

Numerical Grid Type

15. In a ΔABC with usual notations, if 3a + 3b = 4c then $\cot\frac{A}{2}\cot\frac{B}{2}$ is

TS0116

16. In ΔABC (with usual notations), if $r_1 = \Delta$ & $r_1 = r + r_2 + r_3$, then r is equal to

TS0117

17. In triangle ABC, a = 7, b = 6, c = 5. Incircle touches the sides BC, AC and AB at D, E and F respectively and K, L, M are the feet of perpendiculars from circumcentre to sides BC, AC and AB respectively, then the value of DK + EL + FM is

TS0106

18. If sides of a triangle ABC are given by 4,7,7 then length of the tangent drawn from vertex A to the ex-circle opposite to vertex A is equal to

TS0107

19. If f, g and h are the lengths of the perpendiculars from the circumcentre on the sides a, b and c of a triangle ABC respectively then $\frac{a}{f}+\frac{b}{g}+\frac{c}{h}=K\frac{abc}{fgh}$ where K has the value equal to _____.

TS0108

20. In ΔABC with usual notations, if a = 3, b = 2 and $\cos(A-B)=\frac{12}{13}$, then area of ΔABC is

TS0109

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EXERCISE (JM)

1. If 5, 5r, $5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to : **[JEE(Main)-Jan 2019]**

(1) $\frac{3}{2}$ (2) $\frac{3}{4}$ (3) $\frac{5}{4}$ (4) $\frac{7}{4}$ **TS0076**

2. With the usual notation, in ΔABC, if $\angle A + \angle B = 120^\circ$, $a = \sqrt{3}+1$ and $b = \sqrt{3}-1$, then the ratio $\angle A : \angle B$, is : **[JEE(Main)-Jan 2019]**

(1) 7 : 1 (2) 5 : 3 (3) 9 : 7 (4) 3 : 1 **TS0077**

3. In a triangle, the sum of lengths of two sides is x and the product of the lengths of the same two sides is y. If $x^2 - c^2 = y$, where c is the length of the third side of the triangle, then the circumradius of the triangle is : **[JEE(Main)-Jan 2019]**

(1) $\frac{y}{\sqrt{3}}$ (2) $\frac{c}{\sqrt{3}}$ (3) $\frac{c}{3}$ (4) $\frac{3}{2}y$ **TS0078**

4. Given $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$ for a ΔABC with usual notation. If $\frac{\cos A}{\alpha} = \frac{\cos B}{\beta} = \frac{\cos C}{\gamma}$, then the ordered triad (α, β, γ) has a value :- **[JEE(Main)-Jan 2019]**

(1) (3, 4, 5) (2) (19, 7, 25) (3) (7, 19, 25) (4) (5, 12, 13)

TS0079

5. If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is : **[JEE(Main)-Apr 2019]**

(1) 5 : 9 : 13 (2) 5 : 6 : 7 (3) 4 : 5 : 6 (4) 3 : 4 : 5

TS0080

6. The angles A, B and C of a triangle ABC are in A.P. and a : b = 1 : $\sqrt{3}$. If c = 4 cm, then the area (in sq. cm) of this triangle is : **[JEE(Main)-Apr 2019]**

(1) $4\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$ (3) $2\sqrt{3}$ (4) $\frac{4}{\sqrt{3}}$

TS0081

7. If a ΔABC has vertices A(–1, 7), B(–7, 1) and C(5, –5), then its orthocentre has coordinates:

(1) (3, –3) (2) $\left(-\frac{3}{5}, \frac{3}{5}\right)$ (3) (–3, 3) (4) $\left(\frac{3}{5}, -\frac{3}{5}\right)$

[JEE(Main)-2020]

TS0118

8. If in a triangle ABC, AB = 5 units, $\angle B = \cos^{-1}\left(\frac{3}{5}\right)$ and radius of circumcircle of ΔABC is 5 units, then the area (in sq. units) of ΔABC is : **[JEE(Main)-2021]**

(1) $10 + 6\sqrt{2}$ (2) $8 + 2\sqrt{2}$ (3) $6 + 8\sqrt{3}$ (4) $4 + 2\sqrt{3}$

TS0119

9. In ΔABC, the lengths of sides AC and AB are 12 cm and 5 cm, respectively. If the area of ΔABC is 30 cm^2 and R and r are respectively the radii of circumcircle and incircle of ΔABC, then the value of 2R + r (in cm) is equal to _____. **[JEE(Main)-2021]**

TS0120

10. The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for x = k, the area of the triangle is maximum, then $3k^2$ is equal to : **[JEE(Main)-2022]**

(A) 5 (B) 8 (C) 10 (D) 12 **TS0121**

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EXERCISE (JA)

1. Let ABC and ABC′ be two non-congruent triangles with sides AB = 4, AC = AC′ = $2\sqrt{2}$ and angle B = 30°. The absolute value of the difference between the areas of these triangles is **[JEE 2009, 5]**

TS0082

2. (a) If the angle A,B and C of a triangle are in an arithmetic progression and if a,b and c denote the length of the sides opposite to A,B and C respectively, then the value of the expression $\frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A$, is -

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

TS0083

(b) Consider a triangle ABC and let a,b and c denote the length of the sides opposite to vertices A,B and C respectively. Suppose a = 6, b = 10 and the area of the triangle is $15\sqrt{3}$. If ∠ACB is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to

TS0084

(c) Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a,b and c denote the lengths of the sides opposite to A,B and C respectively. The value(s) of x for which a = $x^2 + x + 1$, b = $x^2 - 1$ and c = 2x + 1 is/are **[JEE 2010, 3+3+3]**

(A) $-\left(2+\sqrt{3}\right)$ (B) $1+\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $4\sqrt{3}$

TS0085

3. Let PQR be a triangle of area Δ with a = 2, b = $\frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals **[JEE 2012, 3M, –1M]**

(A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$ (C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$

TS0086

4. In a triangle PQR, P is the largest angle and cos P = $\frac{1}{3}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are)

[JEE(Advanced) 2013, 3, (–1)]

(A) 16 (B) 18 (C) 24 (D) 22

TS0087

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5. In a triangle the sum of two sides is x and the product of the same two sides is y. If $x^2 - c^2 = y$, where c is a third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is -

[JEE(Advanced)-2014, 3(–1)]

(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

TS0088

6. In a triangle XYZ, let x,y,z be the lengths of sides opposite to the angles X,Y,Z, respectively and 2s = x + y + z. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then-

(A) area of the triangle XYZ is $6\sqrt{6}$

(B) the radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$

(C) $\sin\frac{X}{2}\sin\frac{Y}{2}\sin\frac{Z}{2} = \frac{4}{35}$

(D) $\sin^2\left(\frac{X+Y}{2}\right) = \frac{3}{5}$ **[JEE(Advanced)-2016, 4(–2)]**

TS0089

7. In a triangle PQR, let $\angle PQR = 30°$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10, respectively. Then, which of the following statement(s) is (are) TRUE ? **[JEE(Advanced)-2018, 4(–2)]**

(A) $\angle QPR = 45°$

(B) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120°$

(C) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$

(D) The area of the circumcircle of the triangle PQR is 100π.

TS0090

8. In a non-right-angled triangle ΔPQR, let p, q, r denote the lengths of the sides opposite to the angles at P, Q, R respectively. The median from R meets the side PQ at S, the perpendicular from P meets the side QR at E, and RS and PE intersect at O. If $p = \sqrt{3}$, q = 1, and the radius of the circumcircle of the ΔPQR equals 1, then which of the following options is/are correct ?

[JEE(Advanced)-2019, 4(–1)]

(1) Area of $\Delta SOE = \frac{\sqrt{3}}{12}$

(2) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}}{2}(2-\sqrt{3})$

(3) Length of $RS = \frac{\sqrt{7}}{2}$

(4) Length of $OE = \frac{1}{6}$

TS0091

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9. Let x, y and z be positive real numbers. Suppose x, y and z are lengths of the sides of a triangle opposite to its angles X, Y and Z, respectively. If $\tan\frac{X}{2}+\tan\frac{Z}{2}=\frac{2y}{x+y+z}$, then which of the following statements is/are TRUE? **[JEE(Advanced)-2020]**

(A) 2Y = X + Z

(B) Y = X + Z

(C) $\tan\frac{X}{2}=\frac{x}{y+z}$

(D) $x^2+z^2-y^2=xz$

TS0110

10. In a triangle ABC, let $AB=\sqrt{23}$, $BC=3$ and CA= 4. Then the value of $\frac{\cot A+\cot C}{\cot B}$ is_____.

[JEE(Advanced)-2021]

TS0111

11. Consider a triangle PQR having sides of lengths p, q and r opposite to the angles P, Q and R, respectively. Then which of the following statements is (are) TRUE? **[JEE(Advanced)-2021]**

TS0112

(A) $\cos P \geq 1-\frac{p^2}{2qr}$

(B) $\cos R \geq \left(\frac{q-r}{p+q}\right)\cos P+\left(\frac{p-r}{p+q}\right)\cos Q$

(C) $\frac{q+r}{p}<2\frac{\sqrt{\sin Q\sin R}}{\sin P}$

(D) If $p<q$ and $p<r$, then $\cos Q>\frac{p}{r}$ and $\cos R>\frac{p}{q}$

12. Let PQRS be a quadrilateral in a plane, where QR = 1, ∠PQR = ∠QRS = 70º, ∠PQS = 15º and ∠PRS = 40º. If ∠RPS = θº, PQ = α and PS = β, then the interval(s) that contain(s) the value of $4\alpha\beta\sin\theta^\circ$ is/are **[JEE(Advanced)-2022]**

(A) $\left(0,\sqrt{2}\right)$

(B) (1, 2)

(C) $\left(\sqrt{2},3\right)$

(D) $\left(2\sqrt{2},3\sqrt{2}\right)$

TS0122

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ANSWER KEY

Do yourself - 1

1 : **(i)** 90° **(vii)** 60º **(ix)** 45°,60°,75°

(x) a : b : c = sinA : sinB : sinC = $1 : \frac{\sqrt{3}}{2} : \frac{1}{2}$

(xii) $x \in (5, \infty)$

Do yourself - 2

2 : **(viii)** $\frac{5}{9}$ **(ix)** 25 (x) D (xi) D

Do yourself - 4

4 : **(iii)** 2 **(iv)** 30° **(v)** $\angle A = 30º$, $\angle B = 135º$, $\angle C = 15º$

Do yourself - 5

5 : **(i)** **(a)** $\frac{3}{5}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{\sqrt{10}}$ **(d)** $\frac{3}{\sqrt{10}}$ **(e)** $\frac{1}{3}$ **(f)** 24

(iv) $\frac{8}{15}$ **(vi)** 2 **(vii)** 3

Do yourself - 6

6 : **(i)** **(a)** 6 **(b)** $\frac{5}{2}$ **(c)** 1 **(iv)** 2 **(v)** 3 **(vi)** $\frac{9\sqrt{3}}{2}$

Do yourself - 7

7: **(ii)** 6 **(iv)** $\frac{8\pi}{5\sqrt{10+2\sqrt{5}}}$ (v) C (vi) A

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EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	D	D	A	C	C	C	A	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	D	B	C	C	B	D	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	C	A	C	A	C	B	D	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,D	A,D	B,C	A,C,D	B,D	A,C,D	B,D	A,C	A,B,C	A,B
Que.	11	12	13							
Ans.	A	C	D							
Q.14	A	B	C	D						
	S	P	Q	R						
Que.	15	16	17	18	19	20				
Ans.	7	1	2	9	0.25	3				

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	2	3	3	3	3	3	15	C

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	(a) D, (b) 3, (c) B	C	B,D	B	A,C,D	B,C,D	2,3,4	B,C	2
Que.	11	12								
Ans.	A,B	A,B								

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HEIGHT AND DISTANCES

1. INTRODUCTION :

One of the important application of trigonometry is in finding the height and distance of the point which are not directly measurable. This is done with the help of trigonometric ratios.

2. ANGLES OF ELEVATION AND DEPRESSION :

Let OP be a horizontal line in the vertical plane in which an object R is given and let OR be joined.

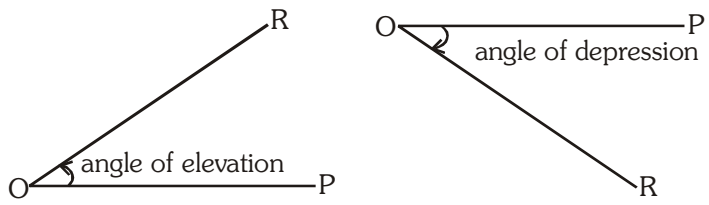


Fig. (a) Fig. (b)

In Fig. (a), where the object R is above the horizontal line OP, the angle POR is called the angle of elevation of the object R as seen from the point O. In Fig. (b) where the object R is below the horizontal line OP, the angle POR is called the angle of depression of the object R as seen from the point O.

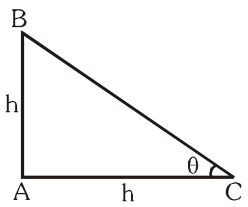
Remark :

Unless stated to the contrary, it is assumed that the height of the observer is neglected, and that the angles of elevation are measured from the ground.

Ex.1 Find the angle of elevation of the sum when the length of shadow of a vertical pole is equal to its height.

Sol. Let height of the pole AB = h and

length of the shadow of the Pole (AC) = h



In ΔABC $\tan\theta = \frac{AB}{AC} = \frac{h}{h} = 1 \Rightarrow \tan\theta = 1$

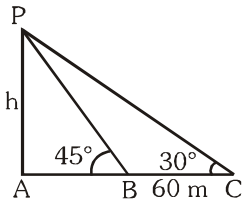
$\Rightarrow \tan\theta = \tan 45° \Rightarrow \theta = 45°$

Ex.2 The shadow of the tower standing on a level ground is found to be 60 metres longer when the sun's altitude is 30° than when it is 45°. The height of the tower is-

(1) 60 m (2) $30(\sqrt{3}-1)$m (3) $60\sqrt{3}$ m (4) $30(\sqrt{3}+1)$ m.

Sol.(4) $AC = h\cot 30° = \sqrt{3}\,h$

$AB = h\cot 45° = h$



$\therefore$ $BC = AC - AB = h(\sqrt{3}-1) \Rightarrow 60 = h(\sqrt{3}-1)$

$\therefore$ $h = \frac{60}{\sqrt{3}-1} = \frac{60(\sqrt{3}+1)}{3-1} = 30(\sqrt{3}+1)$

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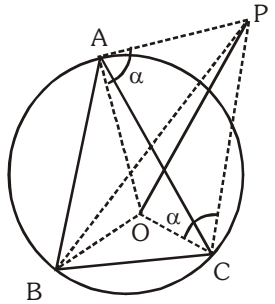
Ex.3 The angle of elevation of the tower observed from each of the three point A,B,C on the ground, forming a triangle is the same angle α. If R is the circum - radius of the triangle ABC, then the height of the tower is -

(1) R sin α (2) R cos α (3) R cot α (4) R tan α

Sol.(4) The tower makes equal angles at the vertices of the triangle, therefore foot of the tower is at the circumcentre.



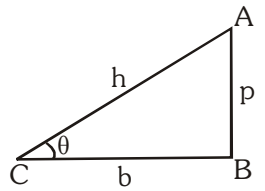
From Δ OCP, OP is perpendicular to OC.

$\angle OCP = \alpha$

so $\tan\alpha = \dfrac{OP}{OA} \Rightarrow OP = OA \tan\alpha$

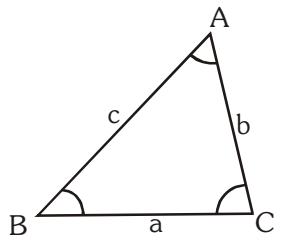
$OP = R \tan\alpha$

3. SOME USEFUL RESULTS :



- In a triangle ABC,

$\sin\theta = \dfrac{p}{h}$, $\cos\theta = \dfrac{b}{h}$, $\tan\theta = \dfrac{P}{b}$

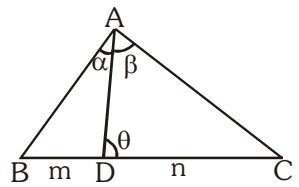


- In any triangle ABC,

$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ [By sine rule]

or cosine formula

i.e. $\cos A = \dfrac{b^2+c^2-a^2}{2bc}$; $\cos B = \dfrac{a^2+c^2-b^2}{2ac}$, $\cos C = \dfrac{a^2+b^2-c^2}{2ab}$

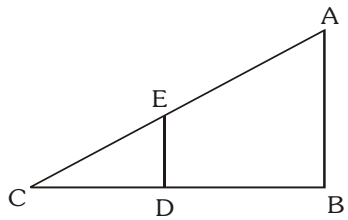


- In any triangle ABC

if BD : DC = m : n and $\angle BAD = \alpha$

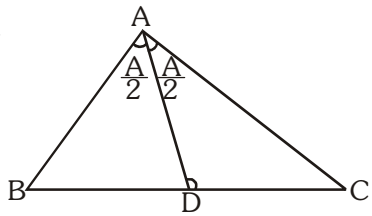
$\angle CAD = \beta$ and $\angle ADC = \theta$,

then $(m+n)\cot\theta = m\cot\alpha - n\cot\beta$



- In a triangle ABC, if DE | | AB

then, $\dfrac{AB}{DE} = \dfrac{BC}{DC}$



- In a triangle the internal bisector of an angle divides the opposite side in the ratio of the arms of the angle

$\therefore \dfrac{BD}{DC} = \dfrac{AB}{AC}$

- In an isosceles triangle the median is perpendicular to the base

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SOLVED EXAMPLES

Ex.1 A tower subtends an angle of 30° at a point on the same level as its foot, and at a second point h m above the first, the depression of the foot of tower is 60°. The height of the tower is.

(1) h m (2) 3h m (3) $\sqrt{3}$ h m (4) $\frac{h}{3}$ m.

Sol.(4) Let OP be the tower of height x.,A the point on the same level as the foot O of the tower and B be the point h m above A (see Fig.) Then $\angle$ AOB = 60° and $\angle$ PAO = 30°. From right-angled triangle AOP, we have

$$OA = x \cot 30°$$

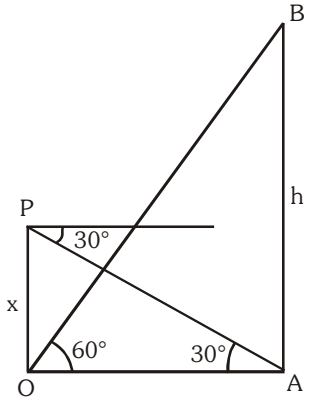
and from right-angled triangle OAB, we have

$$OA = h \cot 60°$$

Therefore, from (1) and (2), we get

$$x \cot 30° = h \cot 60°$$

$$\sqrt{3}\,x = \frac{1}{\sqrt{3}}h \Rightarrow x = \frac{1}{3}h$$



Ex.2 At a point on level ground, the angle of elevation of a vertical tower is found to be such that its tangent is $\frac{5}{12}$. On walking 192 metres towards the tower, the tangent of the angle of elevation is $\frac{3}{4}$. Find the height of the tower.

Sol. Let AB be the tower and let the angle of elevation of its top at C be α. Let D be a point at a distance of 192 metres from C such that the angle of elevation of the top of the tower at D be β.

Let h be the height of the tower and AD = x,

It is given that $\tan\alpha = \frac{5}{12}$ and $\tan\beta = \frac{3}{4}$.

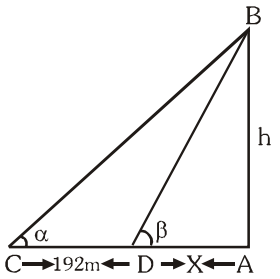
In ΔABC, we have

$$\tan\alpha = \frac{AB}{AC} \Rightarrow \tan\alpha = \frac{h}{192+x} \Rightarrow \frac{5}{12} = \frac{h}{192+x} \quad \text{..... (i)}$$

In ΔABD, we have

$$\tan\beta = \frac{AB}{AD} \Rightarrow \tan\beta = \frac{h}{x} \Rightarrow \frac{3}{4} = \frac{h}{x} \quad \text{..... (ii)}$$

We have to find h. This means that we have to eliminate x from (i) and (ii).



From (ii), we have $3x = 4h \Rightarrow x = \frac{4h}{3}$

Substituting this value of x in (i), we get

$$\frac{5}{12} = \frac{h}{192 + 4h/3} \Rightarrow 5\left(192 + \frac{4h}{3}\right) = 12h$$

$$\Rightarrow 5(576 + 4h) = 36h \Rightarrow 2880 + 20h = 36h$$

$$\Rightarrow 16h = 2880 \Rightarrow h = \frac{2880}{16} = 180$$

Hence, height of tower = 180 metres.

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Ex.3 Let α be the solution of $16^{\sin^2\theta}+16^{\cos^2\theta}=10$ in $(0, \pi/4)$. If the shadow of a vertical pole is $\frac{1}{\sqrt{3}}$ of its height, then then the altitude of the sun is-

(1) α (2) $\frac{\alpha}{2}$ (3) 2α (4) $\frac{\alpha}{3}$

Sol. We have $16^{\sin^2\theta}+16^{\cos^2\theta}=10$

$\Rightarrow\ 16^{\sin^2\theta}+16^{1-\sin^2\theta}=10 \Rightarrow x+\frac{16}{x}=10,$ where $x=16^{\sin^2\theta}$

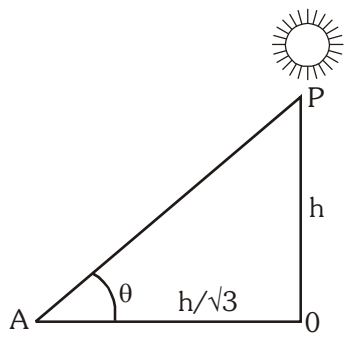
$\Rightarrow\ x=2, 8 \Rightarrow 16^{\sin^2\theta}=2, 8$

$\Rightarrow\ 2^{4\sin^2\theta}=2, 2^3$

$\Rightarrow\ 4\sin^2\theta=2, 3$

$\Rightarrow\ \sin^2\theta=\frac{1}{2}, \left(\frac{\sqrt{3}}{2}\right)^2 \Rightarrow \theta=\frac{\pi}{6}, \frac{\pi}{3}$

$\therefore\ \alpha=\frac{\pi}{6}$



Let the altitude of the sun be θ. Then,

$\tan\theta=\frac{h}{\frac{h}{\sqrt{3}}} \Rightarrow \tan\theta=\sqrt{3} \Rightarrow \theta=\frac{\pi}{3} \Rightarrow \theta=2\alpha$

Ex.4 A vertical lamp-post of height 9 metres stands at the corner of a rectangular field. The angle of elevation of its top from the farthest corner is 30°, while from another corner it is 45°. The area of the field is-

(1) $81\sqrt{2}$ m^2 (2) $9\sqrt{2}$ m^2 (3) $81\sqrt{3}$ m^2 (4) $9\sqrt{3}$ m^2

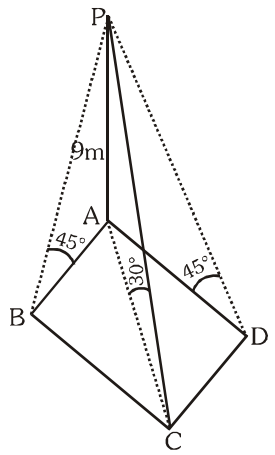
Sol. Let AP be the lamp-post of 9 m standing at corner A of the rectangular field ABCD.

In Δ's BAP and CAP, we have

$\tan 45° = \frac{PA}{BA}$ and $\tan 30° = \frac{PA}{AC}$

$\Rightarrow\ BA = 9$ m and $AC = 9\sqrt{3}$ m

$\therefore\ BC=\sqrt{AC^2-AB^2}=\sqrt{243-81}=\sqrt{162}=9\sqrt{2}$ m



Hence, area of the field = AB $\times$BC = $9\times 9\sqrt{2}$ m^2 = $81\sqrt{2}$ m^2

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Ex.5 A vertical tower stands on a horizontal plane and is surmounted by a vertical flag staff of height h. At a point on the plane, the angle of elevation of the bottom and the top of the flag staff are α and β respectively. Prove that the height of tower is $\frac{h\tan\alpha}{\tan\beta - \tan\alpha}$.

Sol. Let AB be the tower and BC be the flag staff. Let O be a point on the plane containing the foot of the tower such that the angles of elevation of the bottom B and top C of the flagstaff at O are α and β respectively. Let OA = x metres, AB = y metres and BC = h metres.

In ΔOAB, we have

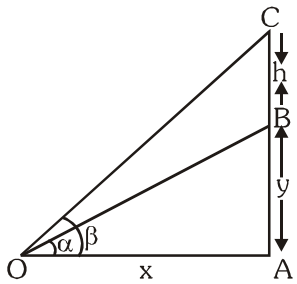
$$\cot\alpha = \frac{OA}{AB} \Rightarrow \cot\alpha = \frac{x}{y}$$

$$\Rightarrow x = y\cot\alpha \quad \text{...(i)}$$

In ΔOAC, we have

$$\cot\beta = \frac{x}{y+h}$$

$$\Rightarrow x = (y + h)\cot\beta \quad \text{...(ii)}$$



Equating the values of x from (i) and (ii), we get

$y\cot\alpha = (y + h)\cot\beta$

$\Rightarrow y\cot\alpha - y\cot\beta = h\cot\beta$

$\Rightarrow y(\cot\alpha - \cot\beta) = h\cot\beta$

$$\Rightarrow y = \frac{h\cot\beta}{\cot\alpha - \cot\beta} \Rightarrow y = \frac{h/\tan\beta}{\frac{1}{\tan\alpha} - \frac{1}{\tan\beta}} = \frac{h\tan\alpha}{\tan\beta - \tan\alpha}$$

Ex.6 A spherical ball of diameter-δ subtends an angle α at the eye of an observer when the elevation of its centre is β. Prove that the height of the centre of the ball is $\frac{1}{2}\delta\sin\beta\,\text{cosec}\left(\frac{\alpha}{2}\right)$.

Sol. O is the position of eye.

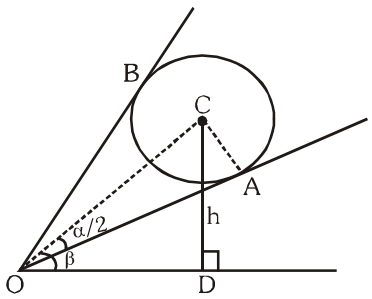
As is clear from figure, from ΔODC,

$$OC = \frac{h}{\sin\beta}$$

From ΔOAC,

$$\sin\frac{\alpha}{2} = \frac{CA}{OC} = \frac{\frac{\delta}{2}}{h/\sin\beta}$$

$$\Rightarrow h = \frac{1}{2}\delta\sin\beta.\,\text{cosec}\,\frac{\alpha}{2}.$$



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CHECK YOUR GRASP — HEIGHTS AND DISTANCE — EXERCISE-I

1. The angle of elevation of the top of a tower from a point 20 metre away from its base is 45°. Then height of the tower is-

(1) 10 m (2) 20 m (3) 40 m (4) $20\sqrt{3}$ m

HD0001

2. At a point 15 metre away from the base of a 15 metre high house, the angle of elevation of the top is-

(1) 45° (2) 30° (3) 60° (4) 90°

HD0002

3. An aeroplane flying at a height 300 metre above the ground passes vertically above another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. Then the height of the lower plane from the ground in metres is-

(1) $100\sqrt{3}$ (2) $100/\sqrt{3}$
(3) 50 (4) $150(\sqrt{3}+1)$.

HD0003

4. If the elevation of the sun is 30° then the length of the shadow cast by a tower of 150 metres height is-

(1) $75\sqrt{3}$m (2) $200\sqrt{3}$m
(3) $150\sqrt{3}$m (4) None of these

HD0004

5. From the top of the cliff 300 metres heigh, the top of a tower was observed at an angle of depression 30° and from the foot of the tower the top of the cliff was observed at an angle of elevation 45°. The height of the tower is -

(1) $50(3-\sqrt{3})$m (2) $200(3-\sqrt{3})$m
(3) $100(3-\sqrt{3})$m (4) None of these

HD0005

6. The angles of elevation of the top of a tower at the top and the foot of a pole of height 10 m are 30° and 60° respectively. The height of the tower is-

(1) 10 m (2) 15 m
(3) 20 m (4) None of these

HD0006

7. A person, standing on the bank of a river, observes that the angle subtended by a tree on the opposite bank is 60°; when he retreates 20 m from the bank, he finds the angle to be 30°. The height of the tree and the breadth of the river are -

(1) $10\sqrt{3}$ m, 10 m (2) 10 m ; $10\sqrt{3}$ m
(3) 20 m, 30m (4) None of these

HD0007

8. From the top of a light house 60 m high with its base at sea level the angle of depression of a boat is 15°. The distance of the boat from the light house is-

(1) $60\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$m (2) $60\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$m

(3) $30\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$m (4) $30\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$m

HD0008

9. The angle of elevation of the sun when the length of the shadow of a pole is $\sqrt{3}$ times the height of the pole will be-

(1) 30° (2) 60° (3) 90° (4) 45°

HD0009

10. If the length of the shadow of a vertical pole on the horizontal ground is equal to its height, find the angle of elevation of the sun -

(1) 60° (2) 30° (3) 45° (4) 90°

HD0010

11. The angle of elevation of the top of a tower from a point 20m away from its base is 60°. The height of the tower is -

(1) 10 m (2) $20/\sqrt{3}$ m (3) 40 m (4) $20\sqrt{3}$ m

HD0011

12. A flag staff on the top of the tower 80 meter high, subtends an angle $\tan^{-1}\left(\frac{1}{9}\right)$ at a point on the ground 100 meters away from the foot of the tower. Find the height of the flag-staff -

(1) 20 m (2) 30 m (3) 25 m (4) 35 m

HD0012

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13. A person walking along a straight road observes that a two points 1 km apart, the angles of elevation of a pole in front of him are 30° and 75°. The height of the pole is -

(1) $250\left(\sqrt{3}+1\right)$ m

(2) $250\left(\sqrt{3}-1\right)$ m

(3) $225\left(\sqrt{2}-1\right)$ m

(4) $225\left(\sqrt{2}+1\right)$ m

HD0013

14. An observer in a boat finds that the angle of elevation of a tower standing on the top of a cliff is 60° and that of the top of cliff is 30°. If the height of the tower be 60 meters, then the height of the cliff is -

(1) 30 m (2) $60\sqrt{3}$ m
(3) $20\sqrt{3}$ m (4) None of these

HD0014

15. ABCD is a square plot. The angle of elevation of the top of a pole standing at D from A and C is 30° and that from B is θ, then tan θ is equal to -

(1) $\sqrt{6}$ (2) $1/\sqrt{6}$
(3) $\sqrt{3}/\sqrt{2}$ (4) $\sqrt{2}/\sqrt{3}$

HD0015

16. The angle of elevation of a ladder against a wall is 58° and the length of foot of the ladder is 9.6 m from the wall. Then the length of the ladder is -
[cos 58° = 0.5299]

(1) 18.11 m (2) 16.11 m

(3) 17.11 m (4) 19.11 m

HD0016

17. From the top of a tower, the angle of depression of a point P on the ground is 30°. If the distance of the point P from the tower be 24 meters then height of the tower is.

(1) 12 m (2) $8\sqrt{3}$ m

(3) $24\sqrt{3}$ m (4) $12\sqrt{3}$ m

HD0017

18. A tower subtends an angle of 30° at a point on the same level as the foot of the tower. At a second point, h metre above first, point the depression of the foot of the tower is 60°, the horizontal distance of the tower from the points is

(1) h cos 60° (2)(h/3) cot 30°
(3) (h/3) cot 60° (4) h cot 30°

HD0018

19. From a point on the ground 100 m away from the base of a building, the angle of elevation of the top of the building is 60°. Which of the following is the best approximation for the height of the building-

(1) 172 m (2) 173 m
(3) 174 m (4) 175 m

HD0019

20. A kite is flying with the string inclined at 75° to the horizon. If the length of the string is 25 m, the height of the kite is-

(1) (25/2) ($\sqrt{3}$ –1)

(2) (25/4) ($\sqrt{3}$ +1)

(3) (25/4) $(\sqrt{3}+1)^2$

(4) (25/4) $(\sqrt{6}+\sqrt{2})$

HD0020

21. If a flagstaff 6 metres high placed on the top of a tower throws a shadow of $2\sqrt{3}$ metres along the ground then the angle (in degrees) that the sun makes with the ground is-

(1) 15° (2) 30°

(3) 60° (4) $\tan^{-1} 2\sqrt{3}$

HD0021

22. A 6-ft tall man finds that the angle of elevation of the top of a 24-ft-high pillar and the angle of depression of its base are complementary angles. The distance of the man from the pillar is-

(1) $2\sqrt{3}$ ft (2) $8\sqrt{3}$ ft

(3) $6\sqrt{3}$ ft (4) None of these

HD0022

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23. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar is-

(1) $\sqrt{3}:1$ (2) $1:3$

(3) $1:\sqrt{3}$ (4) $\sqrt{3}:2$

HD0023

24. The shadow of a tower of height $(1+\sqrt{3})$ metre standing on the ground is found to be 2 metre longer when the sun's elevation is 30°, than when the sun's elevation was -

(1) 30° (2) 45° (3) 60° (4) 75°

HD0024

25. The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance of 50 m from its base is 45°. If the angle of elevation of the top of the complete pillar the same point is to be 60°, then the height of the incomplete pillar is to be increased by-

(1) $50(\sqrt{3}-1)$ m (2) $50(\sqrt{3}+1)$ m

(3) 50 m (4) $25\sqrt{2}$ m.

HD0025

26. A vertical tower stands on a declivity which is inclined at 15° to the horizon. From the foot of the tower a man ascends the declivity for 80 feet and then finds that tower subtends an angle of 30°. The height of the tower is-

(1) $20(\sqrt{6}-\sqrt{2})$

(2) $40(\sqrt{6}-\sqrt{2})$

(3) $40(\sqrt{6}+\sqrt{2})$

(4) None of these

HD0026

27. AB is a vertical pole. The point A of pole AB is on the level ground. C is the middle point of AB. P is a point on the level ground. The portion BC substends an angle β at P. If AP = n AB, then tan β is equal to-

(1) $\frac{n}{2n^2+1}$ (2) $\frac{n}{n^2-1}$

(3) $\frac{n}{n^2+1}$ (4) None of these

HD0027

28. The top of a hill observed from the top and bottom of a building of height h is at angles of elevation p and q respectively. The height of the hill is -

(1) $\frac{h\cot q}{\cot q-\cot p}$ (2) $\frac{h\cot p}{\cot p-\cot q}$

(3) $\frac{h\tan p}{\tan p-\tan q}$ (4) None of these

HD0028

29. A and B are two points 30 m apart in a line on the horizontal plane through the foot of a tower lying on opposite sides of the tower. If the distance of the top of the tower from A and B are 20 m and 15 m respectively, the angle of elevation of the top of the tower at A is-

(1) $\cos^{-1}(43/48)$ (2) $\sin^{-1}(43/48)$

(3) $\cos^{-1}(29/36)$ (4) $\sin^{-1}(29/36)$

HD0029

30. A vertical pole subtends an angle $\tan^{-1}(1/2)$ at a point P on the ground. The angle subtended by the upper half of the pole at the point P is-

(1) $\tan^{-1}(1/4)$ (2) $\tan^{-1}(2/9)$

(3) $\tan^{-1}(1/8)$ (4) $\tan^{-1}(2/3)$

HD0030

31. The angle of elevation of the top of a tower standing on a horizontal plane from a point A is α. After walking a distance d towards the foot of the tower, the angle of elevation is found to be β. The height of the tower is-

(1) $\frac{d\sin\alpha\sin\beta}{\sin(\beta-\alpha)}$ (2) $\frac{d\sin\alpha\sin\beta}{\sin(\alpha-\beta)}$

(3) $\frac{d\sin(\beta-\alpha)}{\sin\alpha\sin\beta}$ (4) $\frac{d\sin(\alpha-\beta)}{\sin\alpha\sin\beta}$

HD0031

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FOG-SOT-H & D\Eng\04_SOT & H & D\10_H & D_Ex(Eng).p65

32. The angle of elevation of the top of two vertical towers as seen from the middle point of the line joining the foot of the towers are 60° and 30° respectively. The ratio of the height of the towers is-

(1) 2 : 1 (2) $\sqrt{3}$:1
(3) 3 : 2 (4) 3 : 1

HD0032

33. A person walking along a st. road towards a hill observes at two points, distance $\sqrt{3}$ km, the angles of elevation of the hill to be 30° and 60°. The height of the hill is-

(1) $\frac{3}{2}$ km (2) $\sqrt{\frac{2}{3}}$ km
(3) $\frac{\sqrt{3}+1}{2}$ km (4) $\sqrt{3}$ km

HD0033

34. The length of the shadow of a vertical pole of height h, thrown by the sun's rays at three different moments are h, 2h and 3h. The sum of the angles of elevation of the rays at these three moments is equal to-

(1) $\frac{\pi}{2}$ (2) $\frac{\pi}{3}$
(3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$

HD0034

35. The upper $\frac{3}{4}$th portion of a vertical pole subtends an an angle $\tan^{-1}\frac{3}{5}$ at a point in the horizontal plane through its foot and at a distance 40 m from the foot. A possible height of the vertical pole is-

[AIEEE-2002]

(1) 80 m (2) 20 m
(3) 40 m (4) 60 m

HD0035

36. A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank of the river is 60° and when he retires 40 meters away from the tree the angle of elevation becomes 30°. The breadth of the river is-

[AIEEE-2004]

(1) 20 m (2) 30 m
(3) 40 m (4) 60 m

HD0036

37. A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that AB (=a) subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30°. The height of the tower is- **[AIEEE-2007]**

(1) $2a/\sqrt{3}$ (2) $2a\sqrt{3}$
(3) $a/\sqrt{3}$ (4) $a\sqrt{3}$

HD0037

38. AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation of the point A from a certain point C on the ground is 60°. He moves away from the pole along the line BC to a point D such that CD = 7 m. From D the angle of elevation of the point A is 45°. Then the height of the pole is- **[AIEEE-2008]**

(1) $\frac{7\sqrt{3}}{2}\frac{1}{\sqrt{3}-1}$m
(2) $\frac{7\sqrt{3}}{2}(\sqrt{3}+1)$m
(3) $\frac{7\sqrt{3}}{2}(\sqrt{3}-1)$m
(4) $\frac{7\sqrt{3}}{2}\frac{1}{\sqrt{3}+1}$m

HD0038

39. A man standing on a horizontal plane, observes the angle of elevation of the top of a tower to be α. After walking a distance equal to double the height of the tower, the angle of elevation becomes 2α, then α is -

(1) $\frac{\pi}{18}$ (2) $\frac{\pi}{12}$
(3) $\frac{\pi}{6}$ (4) $\frac{\pi}{2}$

HD0039

40. ABCD is a trapezium such that AB and CD are parallel and BC ⊥ CD. If ∠ADB = θ, BC = p and CD = q, then AB is equal to **[JEE-MAINS-2013]**

(1) $\frac{(p^2+q^2)\sin\theta}{p\cos\theta+q\sin\theta}$ (2) $\frac{p^2+q^2\cos\theta}{p\cos\theta+q\sin\theta}$

(3) $\frac{p^2+q^2}{p^2\cos\theta+q^2\sin\theta}$ (4) $\frac{(p^2+q^2)\sin\theta}{(p\cos\theta+q\sin\theta)^2}$

HD0040

41. If the angles of elevation of the top of a tower from three collinear points A, B and C, on a line leading to the foot of the tower, are 30°, 45° and 60° respectively, then the ratio, AB : BC, is : **[JEE(Main)-2015]**

(1) $1:\sqrt{3}$ (2) 2 : 3

(3) $\sqrt{3}:1$ (4) $\sqrt{3}:\sqrt{2}$

HD0041

42. A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30°. After walking for 10 minutes from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60°. Then the time taken (in minutes) by him, form B to reach the pillar, is : **[JEE(Main)-2016]**

(1) 5 (2) 6 (3) 10 (4) 20

HD0042

43. Let a vertical tower AB have its end A on the level ground. Let C be the mid-point of AB and P be a point on the ground such that AP = 2AB. If ∠BPC = β, then tanβ is equal to :- **[JEE(Main)-2017]**

(1) $\frac{4}{9}$ (2) $\frac{6}{7}$ (3) $\frac{1}{4}$ (4) $\frac{2}{9}$

HD0043

44. PQR is a triangular park with PQ = PR = 200 m. A T.V. tower stands at the mid-point of QR. If the angles of elevation of the top of the tower at P,Q and R are respectively 45°, 30° and 30°, then the height of the tower (in m) is- **[JEE(Main)-2018]**

(1) 50 (2) $100\sqrt{3}$

(3) $50\sqrt{2}$ (4) 100

HD0044

45. Consider a triangular plot ABC with sides AB=7m, BC=5m and CA=6m. A vertical lamp-post at the mid point D of AC subtends an angle 30° at B. The height (in m) of the lamp-post is : **[JEE(Main) 2019]**

(1) $7\sqrt{3}$ (2) $\frac{2}{3}\sqrt{21}$

(3) $\frac{3}{2}\sqrt{21}$ (4) $2\sqrt{21}$

HD0045

46. If the angle of elevation of a cloud from a point P which is 25 m above a lake be 30° and the angle of depression of reflection of the cloud in the lake from P be 60°, then the height of the cloud (in meters) from the surface of the lake is : **[JEE(Main) 2019]**

(1) 42 (2) 50 (3) 45 (4) 60

HD0046

47. Two vertical poles of heights, 20m and 80m stand a part on a horizontal plane. The height (in meters) of the point of intersection of the lines joining the top of each pole to the foot of the other, from this horizontal plane is : **[JEE(Main) Apr-19]**

(1) 12 (2) 15 (3) 16 (4) 18

HD0047

48. Two poles standing on a horizontal ground are of heights 5m and 10 m respectively. The line joining their tops makes an angle of 15° with ground. Then the distance (in m) between the poles, is :- **[JEE(Main) 2019]**

(1) $\frac{5}{2}(2+\sqrt{3})$ (2) $5(\sqrt{3}+1)$

(3) $5(2+\sqrt{3})$ (4) $10(\sqrt{3}-1)$

HD0048

49. ABC is a triangular park with AB = AC = 100 metres. A vertical tower is situated at the mid-point of BC. If the angles of elevation of the top of the tower at A and B are $\cot^{-1}\left(3\sqrt{2}\right)$ and $\text{cosec}^{-1}\left(2\sqrt{2}\right)$ respectively, then the height of the tower (in metres) is : **[JEE(Main) 2019]**

(1) $10\sqrt{5}$ (2) $\frac{100}{3\sqrt{3}}$

(3) 20 (4) 25

HD0049

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\3-CA(TR)-TE-FOG-SOT-H & D\Eng\04_SOT & H & D\10_H & D_Ex(Eng).p65

50. A 2 m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/ sec., then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is : **[JEE(Main) 2019]**

(1) $25\sqrt{3}$ (2) 25

(3) $\frac{25}{\sqrt{3}}$ (4) $\frac{25}{3}$

HD0050

51. The angle of elevation of the top of vertical tower standing on a horizontal plane is observed to be 45° from a point A on the plane. Let B be the point 30 m vertically above the point A. If the angle of elevation of the top of the tower from B be 30°, then the distance (in m) of the foot of the tower from the point A is: **[JEE(Main) 2019]**

(1) $15\left(3-\sqrt{3}\right)$ (2) $15\left(3+\sqrt{3}\right)$

(3) $15\left(1+\sqrt{3}\right)$ (4) $15\left(5-\sqrt{3}\right)$

HD0051

52. Two vertical poles AB = 15 m and CD = 10 m are standing apart on a horizontal ground with points A and C on the ground. If P is the point of intersection of BC and AD, then the height of P (in m) above the line AC is : **[JEE(Main) 2020]**

(1) 20/3 (2) 5

(3) 10/3 (4) 6

HD0052

53. The angle of elevation of a cloud C from a point P, 200 m above a still lake is 30°. If the angle of depression of the image of C in the lake from the point P is 60°, then PC (in m) is equal to : **[JEE(Main) 2020]**

(1) 400 (2) $400\sqrt{3}$

(3) 100 (4) $200\sqrt{3}$

HD0053

54. The angle of elevation of the top of a hill from a point on the horizontal plane passing through the foot of the hill is found to be 45°. After walking a distance of 80 meters towards the top, up a slope inclined at an angle of 30° to the horizontal plane, the angle of elevation of the top of the hill becomes 75°. Then the height of the hill (in meters) is_. **[JEE(Main) 2020]**

HD0054

55. The angle of elevation of the summit of a mountain from a point on the ground is 45°. After climding up one km towards the summit at an inclination of 30° from the ground, the angle of elevation of the summit is found to be 60°. Then the height (in km) of the summit from the ground is : **[JEE(Main) 2020]**

(1) $\frac{1}{\sqrt{3}-1}$ (2) $\frac{1}{\sqrt{3}+1}$

(3) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (4) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

HD0055

56. A spherical gas balloon of radius 16 meter subtends an angle 60° at the eye of the observer A while the angle of elevation of its center from the eye of A is 75°. Then the height (in meter) of the top most point of the balloon from the level of the observer's eye is : **[JEE(Main) 2021]**

HD0056

(1) $8(2+2\sqrt{3}+\sqrt{2})$ (2) $8(\sqrt{6}+\sqrt{2}+2)$

(3) $8(\sqrt{2}+2+\sqrt{3})$ (4) $8(\sqrt{6}-\sqrt{2}+2)$

57. Two poles, AB of length a metres and CD of length a + b (b ≠ a) metres are erected at the same horizontal level with bases at B and D. If BD = x and $\tan\angle ACB = \frac{1}{2}$, then: **[JEE(Main) 2021]**

(1) $x^2 + 2(a + 2b)x - b(a + b) = 0$

(2) $x^2 + 2(a + 2b)x + a(a + b) = 0$

(3) $x^2 - 2ax + b(a + b) = 0$

(4) $x^2 - 2ax + a(a + b) = 0$

HD0057

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58. A man is observing, from the top of a tower, a boat speeding towards the tower from a certain point A, with uniform speed. At that point, angle of depression of the boat with the man's eye is 30° (Ignore man's height). After sailing for 20 seconds, towards the base of the tower (which is at the level of water), the boat has reached a point B, where the angle of depression is 45°. Then the time taken (in seconds) by the boat from B to reach the base of the tower is : **[JEE(Main) 2021]**

(1) 10 (2) $10\sqrt{3}$

(3) $10(\sqrt{3}+1)$ (4) $10(\sqrt{3}-1)$

HD0058

59. Two vertical poles are 150 m apart and the height of one is three times that of the other. If from the middle point of the line joining their feet, an observer finds the angles of elevation of their tops to be complementary, then the height of the shorter pole (in meters) is : **[JEE(Main) 2021]**

(1) $20\sqrt{3}$ (2) $25\sqrt{3}$ (3) 30 (4) 25

HD0059

60. Let AB and PQ be two vertical poles, 160 m apart from each other. Let C be the middle point of B and Q, which are feet of these two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A, respectively. If the height of pole PQ is twice the height of pole AB, then $\tan^2\theta$ is equal to **[JEE(Main) 2022]**

(A) $\frac{3-2\sqrt{2}}{2}$ (B) $\frac{3+\sqrt{2}}{2}$

(C) $\frac{3-2\sqrt{2}}{4}$ (D) $\frac{3-\sqrt{2}}{4}$

HD0060

61. A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower in two parts such that QR = 15 m. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is : **[JEE(Main) 2022]**

HD0061

(A) $5(2\sqrt{3}+3)$ m (B) $5(\sqrt{3}+3)$ m

(C) $10(\sqrt{3}+1)$ m (D) $10(2\sqrt{3}+1)$ m

62. The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45°. Let R be a point on AQ and from a point B, vertically above R, the angle of elevation of P is 60°. If $\angle BAQ = 30°$, AB = d and the area of the trapezium PQRB is α, then the ordered pair (d, α) is : **[JEE(Main) 2022]**

(A) $\left(10\left(\sqrt{3}-1\right),25\right)$ (B) $\left(10\left(\sqrt{3}-1\right),\frac{25}{2}\right)$

(C) $\left(10\left(\sqrt{3}+1\right),25\right)$ (D) $\left(10\left(\sqrt{3}+1\right),\frac{25}{2}\right)$

HD0062

63. The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is $\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then $\cot\alpha$ is equal to : **[JEE(Main) 2022]**

(A) $\frac{6}{5}$ (B) $\frac{9}{5}$ (C) $\frac{4}{3}$ (D) $\frac{7}{3}$

HD0063

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**1**	**1**	**3**	**3**	**2**	**1**	**2**	**1**	**3**	**4**	**1**	**1**	**1**	**2**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**1**	**2**	**2**	**2**	**4**	**3**	**3**	**3**	**2**	**1**	**2**	**1**	**2**	**1**	**2**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**1**	**4**	**1**	**1**	**3**	**1**	**3**	**2**	**2**	**1**	**3**	**1**	**4**	**4**	**2**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**2**	**3**	**3**	**3**	**3**	**2**	**4**	**1**	**80.00**	**1**	**2**	**3**	**3**	**2**	**C**
Que.	61	62	63												
Ans.	**A**	**A**	**A**												

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Important Notes

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